
Foreword

Thank you for use BL3-U series elevator in-built controller. BL3-U series elevator in-built controller is the new generation of intelligent elevator in-built controller. This system includes intelligent logical controller and excellent performance inverter speed-driven controller. This product has advanced technology, excellent performance, safety and reliability, use convenient, reasonable price. The characteristic of BL3-U series elevator controller in-built are as follow:

- ◆ This system includes intelligent logical controller and excellent performance inverter speed-driven controller.
- ◆ Use advanced vector control, make electric motor precise and running much comfortable.
- ◆ Electric energy's using efficiency is higher than traditional PWM.
- ◆ It carry forward all the function and performance of BL2000 and BL3000 control systems, can suit for all kinds of control requirements easily.
- ◆ BL3-U elevator controller in built is used together with Blue light gearless permanent magnetic synchronous traction machine, the controller can select the function as per the traction machine model automatically, Thus motor specification written and motor specification self-learning can be avoid, which make elevator commissioning easier. The setting of motor specification with this kind of method can make the motor with best performance.
- ◆ Fuzziness control unload compensation start-up, even if the elevator haven't set weighing device, it also can start-up comfortably by encoder.
- ◆ Rotate and stop self-study of electric motor parameter and beginner angle easily.
- ◆ It is fit for gearless permanent magnetic synchronous traction machine and induction motor.
- ◆ Internal driven unit can reduce external component cost.
- ◆ Internal encoder and dividing frequency interface can match different PG signal.
- ◆ Advanced double 32 bit CPU+ can complete all control by CPLD.
- ◆ Redundancy design, software and hardware protection can keep running safety and reliability.
- ◆ EMC lab professional test, improve the ability of prevent to electromagnetic interference and self electromagnetic interference.
- ◆ Monitoring current interdiction circs of controller goes to main machine every time when the elevator stops.
- ◆ Create best travel curve through aim floor, make direct stop, improve running efficiency.
- ◆ Use CAN BUS serial communication, datum transmit fast and reliable, simple wiring.
- ◆ Internet wireless remote monitoring system interface make remote debug, maintenance and monitor the elevator running easily.
- ◆ It has upper monitoring and software device, easy to set parameter, fast debug and diagnose trouble.
- ◆ It has the function of upload, download and copy of Parameter, debug conveniently.
- ◆ It sustains emergency running by storage battery when power supply is failed.
- ◆ It complies with GB7588-2003 safety standard.

This user' manual has introduced how to use BL3-U series elevator controller in-built. Please read it carefully and understand safety items before use (install, running, maintenance).

This description is providing to elevator designer, installer and maintenance people. People who had professional trained can install, debug and maintenance.

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Preface

This manual is designed to ensure correct and suitable application of MU Inverters. Read this manual before attempting to install, operate, maintain, or inspect an Inverter and keep it in a safe, convenient location for future reference. Be sure you understand all precautions and safety information before attempting application.

General Precautions
• The diagrams in this manual may be indicated without covers or safety shields to show details. Be sure to restore covers or shields before operating the Units and run the Units according to the instructions described in this manual. • Any illustrations, photographs, or examples used in this manual are provided as examples only and may not apply to all products to which this manual is applicable. • The products and specifications described in this manual or the content and presentation of the manual may be changed without notice to improve the product and/or the manual. • When ordering a new copy of the manual due to damage or loss, contact us on www.ydelevator.com and provide the manual number shown on the front cover. • If nameplates become worn or damaged, order new ones from YIDA elevator.

Safety Information

The following conventions are used to indicate precautions in this manual. Failure to heed precautions provided in this manual can result in serious or possibly even fatal injury or damage to the products or to related equipment and systems.



Indicates precautions that if not heeded could possibly result in loss of life or serious injury.



Indicates precautions that if not heeded could result in relatively serious or minor injury, damage to the product, or faulty operation.



Indicate important information that should be memorized.

Safety Precautions

■ Confirmations upon Delivery



- Never install an Inverter that is damaged or missing components.
Doing so can result in injury.

■ Installation



- Always hold the case when carrying the Inverter.
If the Inverter is held by the front cover, the main body of the Inverter may fall, possibly resulting in injury.
- Attach the Inverter to a metal or other noncombustible material.
Fire can result if the Inverter is attached to a combustible material.
- Install a cooling fan or other cooling device when installing more than one Inverter in the same enclosure so that the temperature of the air entering the Inverters is below 45°C.
Overheating can result in fires or other accidents.

■ Wiring



- Always turn OFF the input power supply before wiring terminals.
Otherwise, an electric shock or fire can occur.
- Wiring must be performed by an authorized person qualified in electrical work.
Otherwise, an electric shock or fire can occur.
- Be sure to ground the ground terminal. (200 V Class: Ground to 100 Ω or less, 400 V Class: Ground to 10 Ω or less)
Otherwise, an electric shock or fire can occur.
- Always check the operation of any fast stop circuits after they are wired.
Otherwise, there is the possibility of injury. (Wiring is the responsibility of the user.)
- Never touch the output terminals directly with your hands or allow the output lines to come into contact with the Inverter case. Never short the output circuits.
Otherwise, an electric shock or ground short can occur.
- Do not use the Inverter for any load other than a three-phase AC motor.
- A permanent magnet motor is a type of permanent magnet motor with a rotor in which a magnet is integrated. Unlike an induction motor, the permanent magnet motor terminal generates high voltage when the motor is running, even when the Inverter power is shut off. Be sure to completely stop the motor before wiring, maintenance and inspection.
Failure to do so may result in electric shock.



- Check to be sure that the voltage of the main AC power supply satisfies the rated voltage of the Inverter.
Injury or fire can occur if the voltage is not correct.
- Do not perform voltage withstand tests on the Inverter.
Otherwise, semiconductor elements and other devices can be damaged.
- Connect braking resistors, Braking Resistor Units, and Braking Units as shown in the I/O wiring examples.
Otherwise, a fire can occur and the Inverter, braking resistors, Braking Resistor Units, and Braking Units can be damaged.
- Tighten all terminal screws to the specified tightening torque.
Otherwise, a fire may occur.
- Do not connect AC power to output terminals U, V, and W.
The interior parts of the Inverter will be damaged if voltage is applied to the output terminals.
- Do not connect phase-advancing capacitors or LC/RC noise filters to the output circuits.
The Inverter can be damaged or interior parts burnt if these devices are connected.
- When a magnetic contactor is connected to the output circuits, do not switch it ON and OFF while the Inverter is running.
Surge current will cause the over current protection circuit inside the Inverter to operate.
- Do not make terminal (+) / P1 and (-) short link.
Otherwise, a fire or explode may occur.

■ Trial Operation



- Check to be sure that the front cover is attached before turning ON the power supply.
Otherwise, an electric shock may occur.
- Provide a separate fast stop switch; the Digital Operator STOP Key is valid only when its function is set.
Otherwise, an Injury may occur.
- Reset alarms only after confirming that the RUN signal is OFF.
Otherwise, an Injury may occur.



- Do not touch the radiation fins (heat sink), braking resistor, or Braking Resistor Unit. These can become very hot.
Otherwise, a burn injury may occur.
- Be sure that the motor and machine is within the applicable ranges before starting operation.
Otherwise, an injury may occur.
- Provide a separate holding brake if necessary.
Always construct the external sequence to confirm that the holding brake is activated in the event of an emergency, a power failure, or an abnormality in the Inverter.
Failure to observe this caution can result in injury.
- If using an Inverter with a elevator, take safety measures on the elevator to prevent the elevator from dropping.
Failure to observe this caution can result in injury.
- Do not check signals while the Inverter is running.

- Otherwise, the equipment may be damaged.
- Be careful when changing Inverter settings. The Inverter is factory set to suitable settings. Otherwise, the equipment may be damaged.

■ Maintenance and Inspection



- Do not touch the Inverter terminals. Some of the terminals carry high voltages and are extremely dangerous.
Doing so can result in electric shock.
- Always have the protective cover in place when power is being supplied to the Inverter. When attaching the cover, always turn OFF power to the Inverter through the MCCB.
Doing so can result in electric shock.
- After turning OFF the main circuit power supply, wait for the time indicated on the front cover, and make sure the CHARGE indicator light has gone out, and then perform maintenance and inspection.
The capacitor will remain charged and is dangerous.
- Maintenance, inspection, and replacement of parts must be performed only by authorized personnel. Remove all metal objects, such as watches and rings, before starting work. Always use grounded tools.
Failure to heed these warning can result in electric shock.
- Be sure to completely stop the permanent magnet motor before maintenance and inspection.
Failure to do so may result in electric shock.



- A CMOS IC is used in the control board. Handle the control board and CMOS IC carefully.
The CMOS IC can be destroyed by static electricity if touched directly.
- Do not change the wiring, or remove connectors or the Digital Operator, during operation.
Doing so can result in personal injury.

■ Other



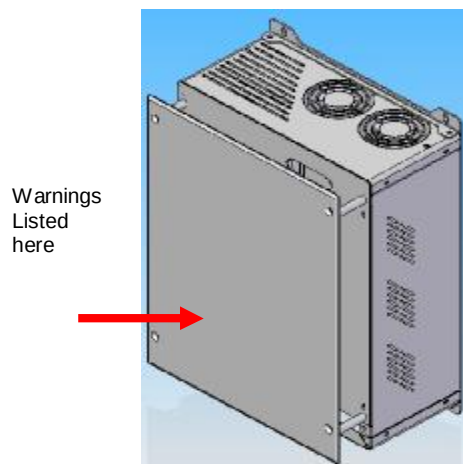
- Do not attempt to modify or alter the Inverter.
Doing so can result in electrical shock or injury.



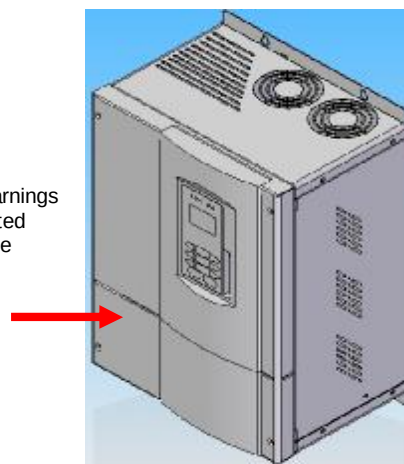
- Do not subject the Inverter to halogen gases, such as fluorine, chlorine, bromine, and iodine, at anytime even during transportation or installation.
Otherwise, the Inverter can be damaged or interior parts burnt.

Warning Labels on Inverter

Be sure to read and follow all warning labels on the Inverter before installation.



Opening Style



Close Style

Text on Warning Labels



WARNING



Risk of electric shock

Read the manual before installing.
Wait 5 minutes for capacitor discharge
after disconnecting power supply.
After switch off the manual switch
between the power supplier and driver,
please wait 5 minutes before inspecting,
performing maintenance or wiring the
drive.

1.1 Inverter Specifications

The Inverter specifications are listed in the following tables.

Table 1.1 400 V Class Inverters

Standard Specification List

Model BL3-U□40□□		4005	4007	4011	4015	4018	4022	4030	4037	4045
Max. applicable motor output (KW)		5.5	7.5	11	15	18.5	22	30	37	45
Output Rating	Rated output capacity (kVA)	9	12	18	22	27	32	43	53	63
	Rated output current (A)	14	18	27	34	41	48	65	80	96
	Max. output voltage (V)	3-phase; 380 VAC (Proportional to input voltage.)								
	Rated output frequency (Hz)	50Hz								
	Max. input frequency (HZ)	120								
	Rated voltage (V)	3-phase; 380 VAC								
	Rated frequency (Hz)	50Hz								
	Rated input current (A)	17	22	32	41	49	58	78	96	115
	Allowable voltage fluctuation	±15%								
	Allowable frequency fluctuation	±5%								
Model BL-U□20□□		2003		2005		2007		2011		2015
Max. applicable motor output (KW)		3.7		5.5		7.5		11		15
Output Rating	Rated output capacity (kVA)	7		10		14		20		27
	Rated output current (A)	17.5		25		33		49		66
	Max. output voltage (V)	3-phase; 220 VAC (Proportional to input voltage.)								
	Rated output frequency (Hz)	50								
	Max. input frequency (HZ)	120								
Power supply characteristics	Rated voltage (V)	3-phase; 220 VAC								
	Rated frequency (Hz)	50								
	Rated input current (A)	21		27		40		52		68
	Allowable voltage fluctuation	±15%								
	Allowable frequency fluctuation	±5%								
Control Method	Simplex Collective, Duplex Collective, Triplex Collective, Up to 8 Group									

		Control
	Speed Range	0.5m/s ~ 4.0m/s
	Max. Landing	64
	Applicable elevator	Passenger, Hospital, Panoramic, Goods, Villa Elevator
	Applicable Traction motor	Gear Traction Machine, Gearless Traction Machine
	Communication BUS	CAN BUS
	Leveling Precision	$\leq 3\text{mm}$
Control Characteristics	Control method	space vector PWM (SVPWM) close loop vector control
	Carrier Frequency	10KHz (5~15KHz adjustable)
	Speed control range	1:1000
	Speed control accuracy10	$\pm 0.05\%$ ($25^{\circ}\text{C} \pm 10^{\circ}\text{C}$)
	Speed response	30Hz
	Torque limit	Available
	Torque accuracy	$\pm 5\%$
	Frequency control range	0~120Hz
	Frequency setting resolution	Digital given: $\pm 0.01\%$ ($-10^{\circ}\text{C} \sim +40^{\circ}\text{C}$)
	Output frequency resolution	Digital given: 0.01Hz
	Overload capacity150%,	150% rating current 60S; 180% rating current 10S
	Starting Torque	180% rating current 0Hz
	Accel/decel time	0.001~600s
I/O control signal	Main control function	Non load compensation starting; Battery-driven running; Motor Auto tuning; load compensation; cooling fan control; basic blocking; torque control; CAN communication given; Normal Acc/Dec; s-curve speed; if main machine's electric current can interdiction monitoring effectively when the car stops; interior braking unit; PG dividing frequency output, fault retry automatic; fault reset automatic; parameter copy
	Photo coupler power supply	Isolated external 24VDC
	Relay power supply	Isolated external 24VDC
	Low voltage Photocopier input signal	26 inputs, 7mA/24VDC, 100Hz (Max.)
	High voltage Photo coupler input signal	6 inputs, 8mA/110VAC, 100Hz (Max.)
	Multifunction contact output signals	12 outputs. Relay contacts, Contact capacity: 5A max. at 30VAC, 5 A max. at 250VDC
	CAN BUS	Two port, used for communicate with car and hall station, duplex control or group control function if has
	Serials communication	1 port, used for digital operator
	RS 232 communication ports	Two port, used by the host for monitoring and commissioning
Display	Digital operator	Chinese/English LCD. Displays parameters such as: elevator speed, motor speed, output frequency, output voltage, output current, output power
	Monitor software	Displays parameters, elevator running state, digital wave
f 3	Instantaneous over	Approx. 200% of rated output current

	current	
	Fuse Open	When The fuse in the main circuit is open. Warning:
	Overload	150% of rated output current , 60 seconds; or 180% of rated output current , 10 seconds
	Overvoltage	Stops at when main DC is 780V or greater.
	Under voltage	Stops at when main DC is about 380V or less.
	Heat sink Over temperature	Via Electronic thermal on heat sink
	IPM protect	IPM over-current, overheated, short circuit, under-voltage protection
	Motor protection	Electronic thermal (Thermistor : Optional)
	IMPACT RESTRAIN LOOP FAULT PROTECTION	Protect via auxiliary contactor feedback of contactor
	Over speed Detection	The motor speed feedback exceeded the setting time
	Excessive Speed Deviation	Available
	PG Open	PG Disconnection
	Auto-tuning Fault protect	When auto-tuning motor, protect motor is there is a fault
	Input Phase Loss	An unusual big ripple on the DC bus voltage has been detected.
	Door interlock fault	Protect, door interlock open while elevator is running
	Safety circuit fault	Protect when safety circuit open
	Brake fault	Protect when command output but not the brake open signal feedback
	Door zone fault	Protect when door zone signal fault
	Output contactor fault	Protect when the output contactor cannot work properly
	Running time limit	Protect when motor running time over the setting in one trip
	Floor counting error	Protect when floor counting error
	Communication error	Communication interference error
	Hoistway learning error	Protect when hoistway leaning failure
Structure	Protection class	IP20
	cooling	Forced air cooling
	Installation	Wall mounting (within cabinet)
Environment	Ambient temperature	-10°C ~ +40°C
	Humidity	5% ~ 95%rh, non-condensing
	Storage temperature	-20°C ~ +60°C
	Location	In doors, without direct sunlight, dust, corrosive/explosive gases, oil fog, vapor, water dripping, or salty substances
	Altitude	< 1,000m (de-rate when above 1,000m)
	Vibration	10~20Hz, <9.8m/S ² ; 20~50Hz, <2m/S ² ;

1.2 Component Names

BL3-U series elevator driver has open classis and enclosed classis style. The open classis has wall-mounted and insert wall-mounted style.

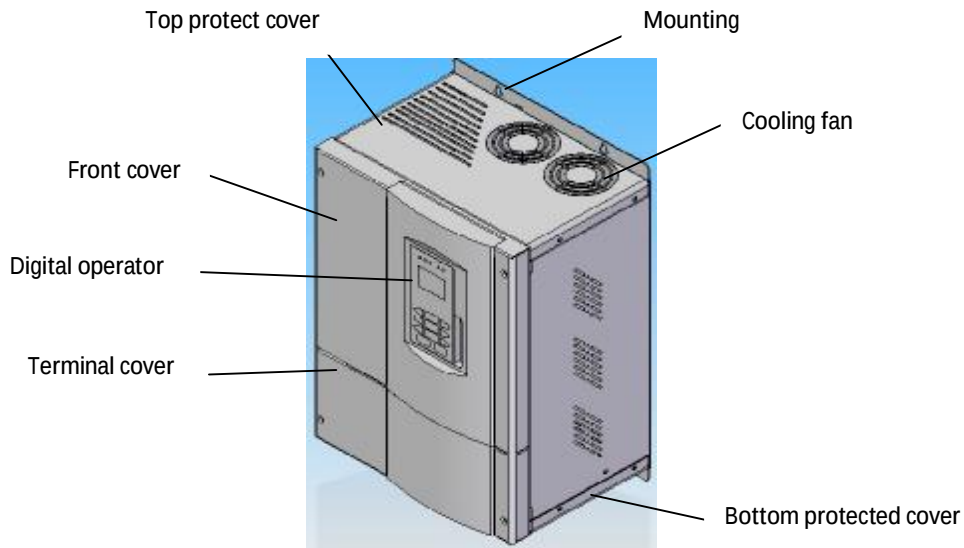


Fig. 1.3 enclosed classis appearance

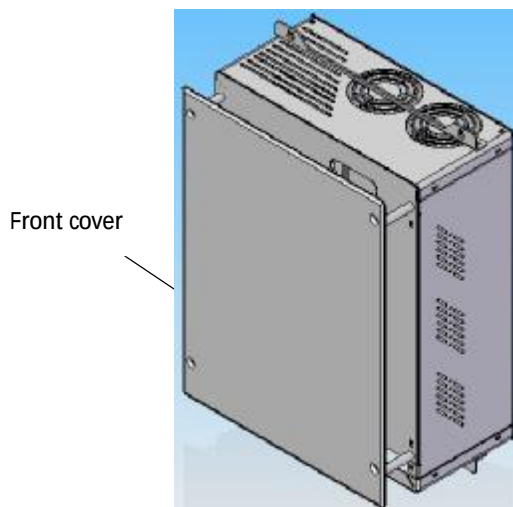


Fig. 1.4 open classes insert wall-mounted appearance

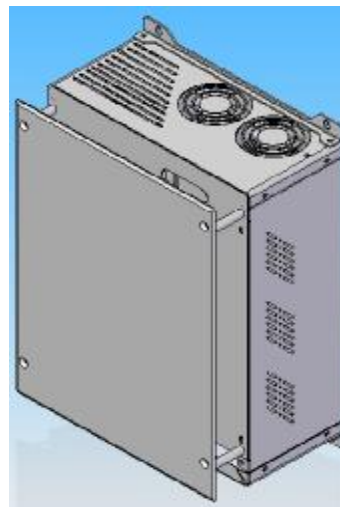


Fig. 1.5 open classis wall-mounted appearance

1.3 Exterior and Mounting Dimensions

■ Open Chassis Inverters

1.3.1 Open classis wall-mounted

Exterior diagrams of the Open Chassis Inverters are shown below.

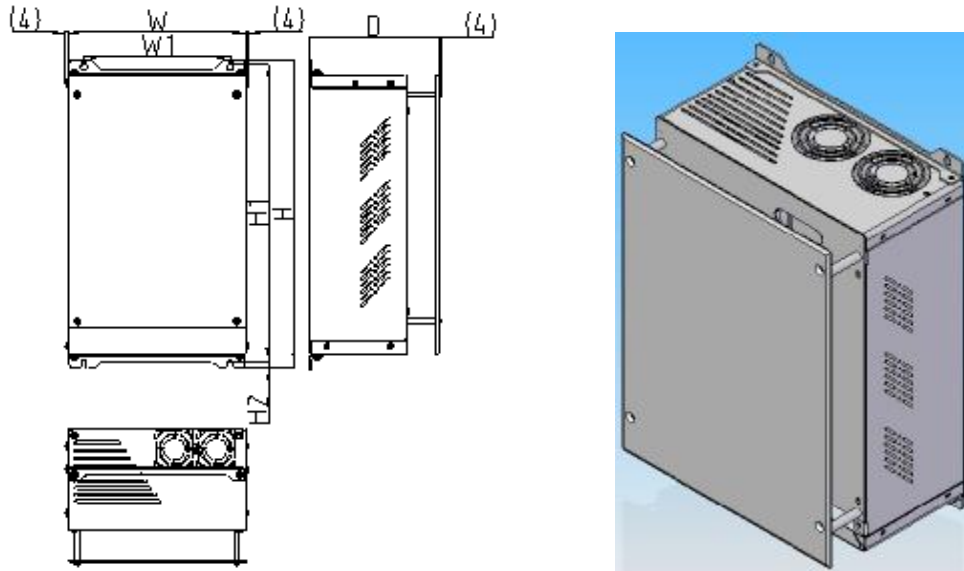


Fig 1.6 Exterior Diagrams of Open Chassis wall-mounted Inverters

1.3.2 Open classes insert wall-mounted

Exterior diagrams of the Open Chassis Inverters are shown below.

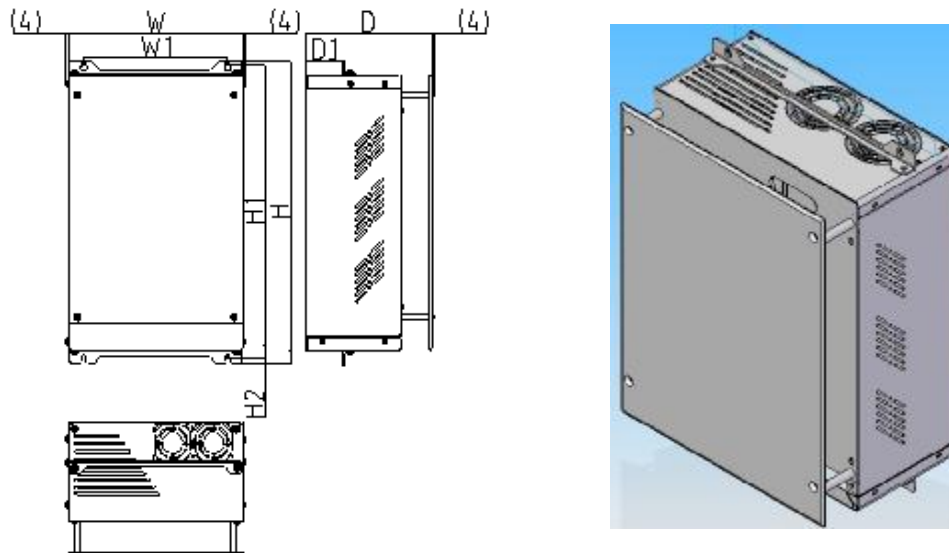


Fig 1.7 Exterior Diagrams of Open Chassis insert wall-mounted Inverters

Table 1.2

Voltage class	Mounting method	Max. Applicable Motor Output [kW]	model	figure	Shape size							Weight Kg						
					W	H	D	W1	H1	H2	D1							
400V (3phase)	Open Chassis wall-mounted	5.5	BL3-UO4005	Fig. 2.6	280	463	203	230	448	8.5	0	16						
		7.5	BL3-UO4007									16						
		11	BL3-UO4011									16						
		15	BL3-UO4015		320			270				16						
		18.5	BL3-UO4018									18						
		22	BL3-UO4022									18						
		30	BL3-UO4030		235		20											
		37	BL3-UO4037		390	650	300	340	600			35						
		45	BL3-UO4045									35						
	Open Chassis insert wall-mounted	5.5	BL3-UO4005	Fig. 2.8	280	463	203	230	448		57.5	16						
		7.5	BL3-UO4007									16						
		11	BL3-UO4011									16						
		15	BL3-UO4015		320			270				16						
		18.5	BL3-UO4018								59	18						
		22	BL3-UO4022									18						
		30	BL3-UO4030		235		20											
		37	BL3-UO4037		390	650	300	340	600		136	35						
		45	BL3-UO4045									35						
200V (3phase)	Chassis wall-mounte	3.7	BL3-UO2003	Fig. 2.6	260	373	203	200	358	8.5	0	16						
		5.5	BL3-UO2005					270	448			16						
		7.5	BL3-UO2007		320	463						16						
		11	BL3-UO2011									18						
		15	BL3-UO2015									18						
	Open Chassis insert wall-mounted	3.7	BL3-UO2003	Fig. 2.8	260	373	203	200	358		57.5	16						
		5.5	BL3-UO2005					270	448			16						
		7.5	BL3-UO2007		320	463						16						
		11	BL3-UO2011									59	18					
		15	BL3-UO2015										18					

■ Enclosed classis

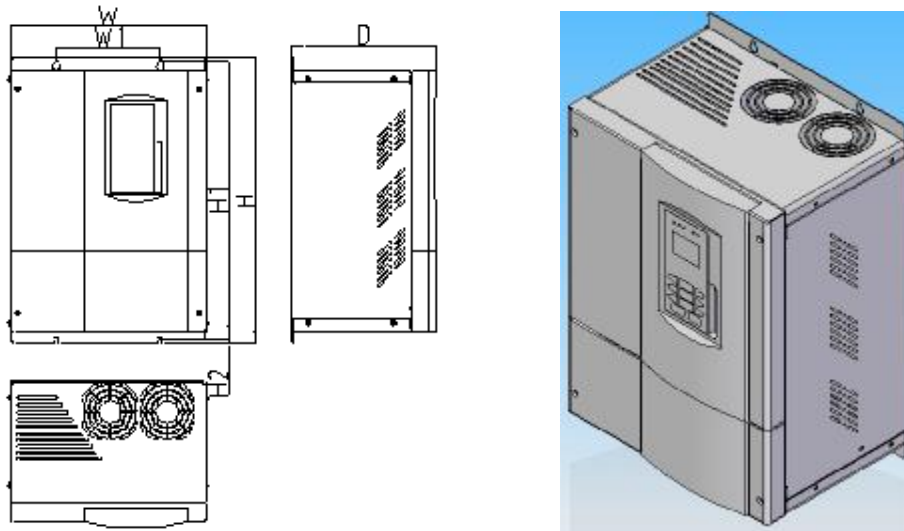


Fig 1.8 Exterior Diagrams of enclosed classis Inverters

Table 1.3

Voltage class	Max. Applicable Motor Output [kW]	model	Dimensions (mm)						Weight Kg
			W	H	D	W1	H1	H2	
400V (3phase)	5.5	BL3-UC4005	280	460	236	130	445	8.5	16
	7.5	BL3-UC4007							16
	11	BL3-UC4011							16
	15	BL3-UC4015							16
	18.5	BL3-UC4018	320	460	236	170	445	8.5	18
	22	BL3-UC4022							18
	30	BL3-UC4030							20
	37	BL3-UC4037	390	650	300	340	600	8.5	35
	45	BL3-UC4045							35

■ Confirmations upon Delivery

Table 1.4 Checks

Item	Method
Has the correct model of Inverter been delivered?	Check the model number on the nameplate on the side of the Inverter.
Is the Inverter damaged in any way?	Inspect the entire exterior of the Inverter to see if there are any scratches or other damage resulting from shipping.
Are any screws or other components loose?	Use a screwdriver or other tools to check for tightness.
Open terminal cover, is the control board loose?	Use a screwdriver or other tools to check for tightness.

If you find any irregularities in the above items, contact the agency from which you purchased the Inverter.

Checking and Controlling the Installation Site

Install the Inverter in the installation site described below and maintain optimum conditions.

■ Installation Site

Install the In-built under the following conditions in pollution degree 2 environments.

Table 1.5 Installation Site

Type	Ambient Operating Temperature	Humidity
Open chassis	-10 to + 40 °C	95% RH or less (no condensation)
Enclosed wall-mounted	-10 to + 45 °C	95% RH or less (no condensation)

Observe the following precautions when mounting the Inverter.

- Install the Inverter in a clean location which is free from oil mist and dust. It can be installed in a totally enclosed panel that is completely shielded from floating dust.
- When installing or operating the Inverter, always take special care so that metal powder, oil, water, or other foreign matter does not get into the Inverter.
- Do not install the Inverter on combustible material, such as wood.
- Install the Inverter in a location free from radioactive materials and combustible materials.
- Install the Inverter in a location free from harmful gasses and liquids.
- Install the Inverter in a location without excessive oscillation.
- Install the Inverter in a location free from chlorides.
- Install the Inverter in a location not in direct sunlight.

■ Controlling the Ambient Temperature

To enhance the reliability of operation, the Inverter should be installed in an environment free from extreme temperature increases. If the Inverter is installed in an enclosed environment, such as a cabinet, use a cooling fan or air conditioner to maintain the internal air temperature below 45°C.

■ Protecting the Inverter from dust

Place a cover over the Inverter during installation to shield it from metal power produced by drilling. Always remove the cover from the Inverter after the completion of the installation. Otherwise, ventilation will be reduced, causing the Inverter to overheat.

Removing and Attaching the Terminal Cover

■ Remove the terminal cover to wire cables to the control circuit and main circuit terminals.

Loosen the screw at the bottom of the terminal cover; press in on the sides of the terminal cover in the directions of arrows

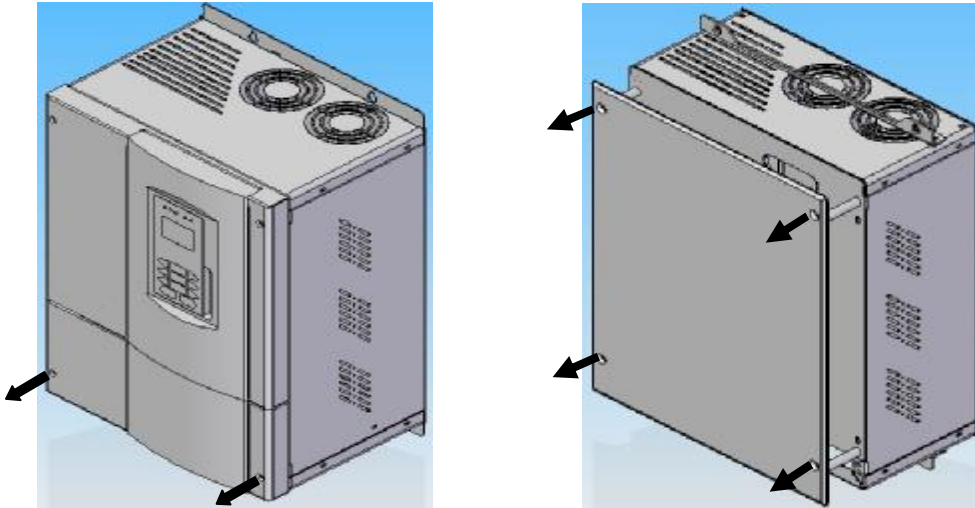


Fig 1.9 Removing the Terminal Cover

■ Attaching the Terminal Cover

When the terminal block wiring has been completed, attach the terminal cover by reversing the removal procedure.

For Inverters with an output of 18.5 kW or less, insert the tab on the top of the terminal cover into the groove on the Inverter and press in on the bottom of the terminal cover until it clicks into place.

Removing/Attaching the Digital Operator/LED Monitor and Front Cover

To attach optional boards or change the control circuit terminal board connector, remove the Digital Operator/LED Monitor and front cover in addition to the terminal cover. Always remove the Digital Operator/LED Monitor from the front cover before removing the front cover.

■ Removing the Digital Operator/LED Monitor

Press the lever on the side of the Digital Operator/LED Monitor in the direction of arrow 1 to unlock the Digital Operator/LED Monitor and lift the Digital Operator/LED Monitor in the direction of arrow 2 to remove the Digital Operator/LED Monitor as shown in the following illustration.

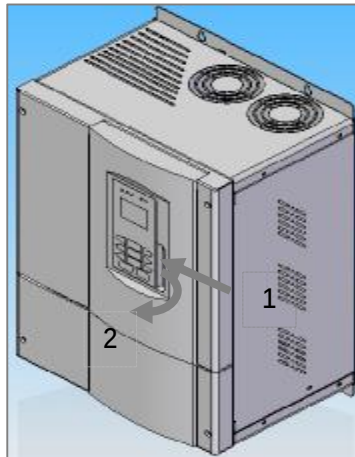


Fig 1.10 Removing the Digital Operator/LED Monitor

■ Mounting the Front Cover

After wiring the terminals, mount the front cover to the Inverter by performing the steps to remove the front cover in reverse order.

1. Do not mount the front cover with the Digital Operator/LED Monitor attached to the front cover; otherwise, Digital Operator/LED Monitor may malfunction due to imperfect contact.
2. Insert the tab of the upper part of the front cover into the groove of the Inverter and press the lower part of the front cover onto the Inverter until the front cover snaps shut.

■ Mounting the Digital Operator/LED Monitor

After attaching the terminal cover, mount the Digital Operator/LED Monitor onto the Inverter using the following procedure.

1. Hook the Digital Operator/LED Monitor at A (two locations) on the front cover.
2. Press the Digital Operator/LED Monitor until it snaps in place

Installation Orientation and Space

Install the Inverter vertically so as not to reduce the cooling effect. When installing the Inverter, always provide the following installation space to allow normal heat dissipation.

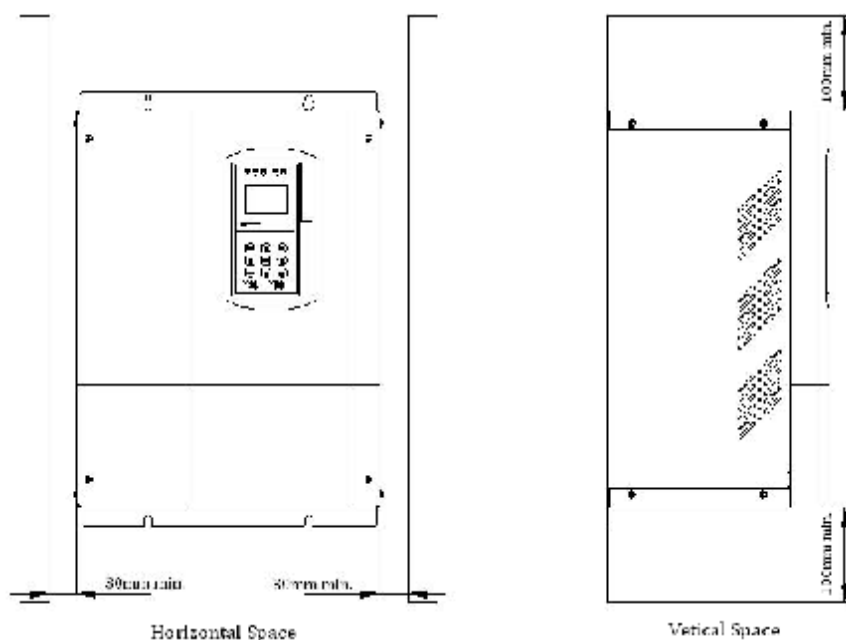


Fig 1.11 Driver Installation Orientation and Space



1. The same space is required horizontally and vertically for both Open Chassis and Enclosed Wall-mounted drivers.
2. Always remove the protection covers before installing driver with an output of in a panel. Always provide enough space for suspension eye bolts and the main circuit lines when installing driver with an output of in a panel.

Braking resistance

The BLU-3 drives have built-in braking units. You only need to mount a braking resistor.

Elevator is typically belonging to potential energy load. When the elevator is brake running, the electricity feedback of motor will make the DC bus voltage increase. Energy can release by allocate suitable brake component. If not, it comes to overload protection. BL3-U series elevator controller in-built have internal brake unit, it only need suitable brake resistance. Allocation resistance is varying with the degree of power.

The detail disposition of brake resistance can see on table 2.6.

Table 1.6 list of braking resistance

Model	Applicable Motor Power (KW)	Applicable Synchronous Motor(Ω /W)	Applicable Asynchronies Motor(Ω /W)
BL3-U□4005	5.5	75 Ω /2000W	75 Ω /1000W
BL3-U□4007	7.5	65 Ω /2500W	65 Ω /1000W
BL3-U□4011	11	40 Ω /3500W	40 Ω /2000W
BL3-U□4015	15	30 Ω / 6000W	30 Ω / 4000W
BL3-U□4018	18.5	25 Ω /6000W	25 Ω /4000W
BL3-U□4022	22	20 Ω /9000W	20 Ω /6000W
BL3-U□4030	30	16 Ω /11000W	16 Ω /8000W
BL3-U□4037	37	12 Ω /15000W	12 Ω /15000W
BL3-U□4045	45	10 Ω /15000W	10 Ω /15000W
BL3-U□2003	3.7	30 Ω /2000W	30 Ω /2000W
BL3-U□2005	5.5	24 Ω /2000W	24 Ω /2000W
BL3-U□2007	7.5	20 Ω /4000W	20 Ω /2000W
BL3-U□2011	11	12 Ω /4000W	12 Ω /4000W
BL3-U□2015	15	10 Ω /6000W	10 Ω /4000W

2 Function Descriptions

Table 1.4 function list

No.	name	purpose	description	remark
1	automation running		1. After leveling, it automatically open door. 2. Automatic close door in delay time. 3. If close door button is press down, door will be closed ahead of delay time. 4. car call register and wrong operation unregister automatically 5. Hall call automatic stop the car at same running direction. 6. Elevator services the opposite direction at the highest/lowest floor calling.	Turn all inspection switch to normally state
2	Attendant running		1. After leveling it automatically open door. 2. It close door after close door button be press down. 3. car call register and wrong operation unregister automatically 4. If there is hall call, elevator services the same direction call.	Turn all inspection switch to normally state
3	Inspection running	Used for commissioning and maintenance	When push down up-run/down-run button in inspection state, elevator runs up/down at inspection speed. It will stop, if you release button.	The switch of normal/overhaul is set on car top 、 in car box and controller box, their priority deceases one by one.
4	Automatically open door after power on	Automatic open door	if car box at leveling zone when elevator is powered on, door is opened automatically.	
5	Automatically close door time set	door keep opening	After the door complete open, it will keep opening, and automatically close after delay time.	when elevator stops for car call, delay time is T, when elevator stops for landing call, delay time is T-2s. when elevator stops for both car call and landing call, delay time is T+2s
6	Open door for current floor landing call	Open for landing call	If you push down landing call button at current floor when door is closing or closed but elevator doesn't start, door will be opened automatically.	Delay time is set by open door holding time.
7	Safe board or photoelectric board protection	Close door safety	If safe board or photoelectric board acts when door is closing, closing action stops at once, door is opened and closed again.	After safe board or photoelectric board action cleared up, the door closed again.

8	overload don't close door	Waiting for cut the weight	if elevator is over load, door is opened, overload light is turned on, buzzer sounds, 'CZ' is displayed on COP and close door button is valid.	It restores normally after overload is clear up.
9	Full-load by pass	Run with same direction to nearest registered floor.	If the elevator is full-load, it responds to the car call and not to landing call.	It restores normally after full-load is clear up.
10	By pass in attendant control	VIP running	In attendant state, elevator responds to car call and not to landing call if pass over button is pushed down.	
11	running state display	Adjust and repair	Elevator running state, direction, floor, door state, load and fault information may be displayed on LCD	

No.	name	purpose	description	Remark
12	Automation control light	Energy saving	Car box light is turned off if elevator isn't used in 15 minutes. It will be turned on after receive any call.	
13	fireman		when fire switch is closed, system enters into fire state: (1) it clears up all calls; (2) return to fire floor; (3) open door (4) When it arrives at fire floor, it output fire signal. (5) If elevator direction is opposite, it stops at nearest floor without open door and returns to fire floor open door.	There are two kinds of fire modes for user to choose: 1.mode 1 Elevator returns to fire floor and stops service. 2.mode 2 a. landing call is invalid; b. Door is open at fire floor. c. When elevator is need to start service, fireman pushes down car call button and presses close door button until door closed. If close door button is released before door closed, door is opened at once. d. When elevator arrives at destination floor, door is closed. Fireman presses open door button until door opened. If open button is released before door open, door is closed at once. e. One destination floor is set every running.

14	Automatically stops for fault	save the passengers	When elevator doesn't stop at leveling zone for fault, it should be automatically crawling to leveling zone and open door.	If safe circuit and inverter are both normal.
15	Parking	Stop running	when electronic lock is closed, elevator enters into parking state. 1.elevator doesn't respond to landing call. If there is car call, it finishes all car call 2.service and returns to parking floor(set). 3.otherwise, it returns to parking floor directly. after it arrives at parking floor, a. landing call box and COP display "ZT". b. it doesn't respond to landing call and car call. c. Door is automatically closed, light is turned off and display is cleared in 10 seconds. d. Press car call button or close/open door button, light is turned on. e. By push down close/open Dutton, door is automatically closed and light is turned off again in 10s.	(1) when electronic lock is closed in inspection state, elevator can't return to parking floor, the other is the same as above. (2) elevator always works in parking state, elevator exits parking state and normally runs as soon as electronic lock is opened.
16	Twins control	two sets of elevators control	1.when there is landing call, two elevators respond to it at one time. One of them finishes the service by the rule of speediness and economy from their position and direction. Elevator efficiency is improved. 2.when two elevators both haven't service, one returns to homing floor, another stays at there.	When parallel connection of two elevators is connected by attached cables and set parameters, parallel running is realized.
17	Group control	Optimize control	It can controls 8elevators in the group control mode.	

Special function list

No.	name	purpose	description	remark
1	Hoistway learning	measure、keep datum of hoistway	System can get and save the position of leveling zone and hoistway switch by hoistway learning.	if control system is unusual in learning, it will stop learn and display fault signal. fault code is on < trouble-shooting of hoistway learning—list 8.2>: ▲caution: if stop self-study, only when the operator show “Success”, learning finished.
2	Cancel mistake operating	Cancel car call	When passenger mistake pushing down car call button and this is registered (light on), it can be canceled by push the button again if elevator doesn't start.	When the elevator doesn't start.
3	Prevent making trouble	cancel car call	1.when elevator arrives at farthest floor and changes direction, it clears all car call registers. 2.if it has load device, when it is light load, the number of car call is 3 at most.	
4	landing call button inset diagnose	Repair indicate	When landing call button is press for 20s, this call is invalid. The corresponding light twinkles to alarm.	When the button is released, system restore normally.
5	Repeat close door		After perform close door command, door inter-lock circuit doesn't put through, elevator opens door and closes it again.	If the action repeats 5 times, door inter-lock circuit still doesn't put through, system stops service and display fault code.
6	floor selection by machine room	adjustment	You can input car call command by LCD operation	
7	open / close door of the machine room	adjustment	You can input close/open door command by LCD operation.	
8	Don't stop floor set	set up a no-stop floor	User may set any floor not to stop.	
9	Homing floor set	Wait at homing floor.	In automation state, if there are no car call and landing call in some time, elevator returns to homing floor.	homing floor only one.
10	Displaying character set	change the display of one floor	Displaying character may set by elevator controller in-built operation, which is letter or figure.	

List 2.8special function list (continue)

No.	name	purpose	description	Remark
11	Attendant select direction	VIP running	Attendant has priority to choose running direction by up-run/down-run button.	in case of attendant state.
12	Automatically start/stop service	automatic start/stop service	User may set start/stop service time by practical requirement.	1.time is set by 24 hours. 2.if set automatic start/stop time of the elevator as 00, this function is canceled. 3.this function is valid in electronic lock is switched on, if it is switched off, elevator works at parking state. 4.if you want elevator exit from automatically stop service, you may perform the following operation: a. you switch electronic lock from on to off for 1s, then you switch it on again, elevator will normally run when elevator finishes using. b. If you want elevator exit normal running, you switch electronic lock from on to off for 1s, then you switch it on again, elevator will enter into automatically start/stop service state.
13	special running	special passenger service	Landing call is invalid. Elevator is controlled by attendant. Door control mode is the attendant mode.	If elevator has special button.
14	Automatically close door delay time set	door-automatic closing time delay	In automation state, when this button is pushed down, open door holding time is door delay time.	1.in case that elevator has a close door delay button; 2.in automation state 3.the function is used in hospital elevator.
15	Both side door		It can control two doors act at corresponding floor.	definition and presetting of penetration-door mode can refer on <6.8- control of going-through door>
16	trouble-shooting	automatic find and record fault information	1.when faults happen, the system diagnose fault and display its code on LCD. 2.the time, type and floor of the latest 30 faults is stored,	trouble code is on <trouble-shooting of elevator-list 8.1>
17	Interference evaluation	automatic inspect immunity to controller in-built	It not only takes measure to improve its resistor interference, but also evaluates system connection with ground. It conducts maintenance to remove hidden trouble and provides more safe and reliable circumstance.	this function is used after finishing debugging.

List 2.8special function list (continue)

No.	name	purpose	description	remark
18	Encoder evaluation	automatic inspect quality of pulse signal from encoder	Encoder is a component detecting elevator speed and position. Many faults are generated from it. It evaluates encoder pulse and conducts maintenance to get rid of trouble of tingling and leveling.	basis of encoder quality and disturb.
19	Input port evaluation	automatically inspect the affect of input signal	Because traveling cable, hoistway cable and MCU unit is parallel connected, the interference can't be neglected, which may lead to display error and leveling fault. This system evaluates input signal and display on LCD. it conducts maintenance lay out cables and solves above problem completely.	direct show of input disturb.
20	emergency automatic landing running	after power cut, the elevator running by emergency automatic leveling control device: leveling at nearest floor	After power cut, the elevator running by emergency automatic leveling control device, leveling at nearest floor, so the passengers can leave safety. emergency automatic leveling running needs to meet these conditions: 1.after power cut, the power of the elevator running is supplied by emergency automatic leveling control device(controller in-built I/O board emergency automatic leveling running input terminal X18 is in effect); 2.no overhaul operation 3.not at the floor. 4.elevator can running. 5.signal of safety locking-door loop is usual. the elevator is not at door emergency automatic leveling running: 6.automatic choose running direction by load status(need to add weight device). 7.the elevator described above will stop and open the door at nearest floor, and keeping door-opening. After a certain time, emergency automatic leveling controller stop power supply. 8.Power recover, the system will record if the elevator had automatic leveling running, and adjust the number when back to bottom floor. The power of the elevator running is supplied by emergency automatic leveling control device (controller in-built I/O board emergency automatic leveling running input terminal X18 is in effect), door opening automatically.	Because the power of emergency automatic leveling running is supplied by elevator emergency automatic leveling controller device. Drawings controller system is different from standard drawing. When you need to this function, ask our technique department for this reference drawing. Use SYT-YY series elevator emergency automatic leveling controller of our company, reference to: <use description of 11.6—STJ-YY series elevator emergency automatic leveling controller>.

List 2.8 special function list(continue)

No.	name	purpose	description	remark
21	Elevator for the handicapped	Use handicapped cop、hop	1、start elevator for handicapped function through parameter; 2、use cop: it have instruction and open-close input button in it. 3、use hop: it will recognize if it is handicapped or not through the different floor; 4、use principle: when leveling, opening time will keep longer if handicapped ion this floor(the opening time can be set); When press cop opening button, opening time will keep longer.	u set address hop: When use elevator for the handicapped (function choose FU14=ON) , hop need to set like this: 1~32is normal floor,1is the lowest floor, 2 is the second floor, it totally can set 32 floors. 33~64 are handicapped floor. 33 is the lowest floor, 34, is the second floor, the highest is 64, it totally has 32 floors. If the building only has one hop, the other hop does not exist. u Choose the link button in car when use optional function. 1~N floor internal select button has been linked to car box. 1~N floor internal select joint is normal internal select. N+1~N+N is handicapped cop 1~N floor internal select. cop opening 2、closing 2 input is handicapped input(caution: the function of elevator for the handicapped can't use together with double-door.)
22	Door-opening in advance	when running slowly, it will open the door in advance to improve running efficiency	When the elevator's speed slow down and attend to landing door, if it can meet the condition of door-opening in advance, it will open in advance. the condition of opening in advance: 1、 running speed need to change; 2、 two of the sensors are useful; 3、 the speed is slower than the set speed.(slower than protective speed); 4、 controller low speed is useful; 5、 safety electric board is useful;	when special parameter's function select parameter FU20=ON, opening in advance is useful; Opening in advance/re-leveling' s link principle picture and describe is on appendix 2- opening in advance/re-leveling function description.
23	re-leveling function	Apply re-leveling	when the elevator stop at the floor, if passengers are too much, the steel rope expand or contract, and the car will move and get to leveling in a low speed. Condition of re-level: 1、 when elevator stopped and leave leveling; 2、 two of the sensors are useful; 3、 the speed is slower than the set speed.; 4、 controller low speed is useful; 5、 safety electric board is useful;	when special parameter's function select parameter FU19=ON, re-leveling is in effect. Opening in advance/re-leveling' s link principle picture and describe is on appendix 2- opening in advance/re-leveling function description.

List 2.9 main safety protect function list

number	name	elevator description
1	Safe loop protection	Once the safety circuit breaks, the elevator immediately stops.
2	Door interlock protection	Only when all doors are interlocked and closed may the elevator be able to run, otherwise it will stop.
3	Running contactor protection	The system will at anytime check the contactors of running circuit for keeping reliable, and once abnormality is found, it will stop immediately.
4	Brake checking protection	The detection switch of brake arm makes a real-time inspection of opening/closing of the brake. If the brake fails to open as per requirements, the system will prevent the elevator from start.
5	Terminal station speed-change and correct the floor display	If the elevator detects a terminal switch during running, it will be forced to decelerate and meanwhile automatically correct the floor display.
6	Position-limit protection	If the elevator detects a limit switch, the entire system will immediately stop.
7	Limit protection	If the elevator runs to a trigger a limit switch, the entire system will be immediately de-energized.
8	Instantaneous over-current protection	Over 200% rating output current stop
9	Fuse blowing protection	when fuse blow, it will stop
10	Overload protection	150% rated current 80% rated current stop after 10S
11	Over-voltage protection	when main loop DC bus voltage higher than 780V, it will stop (400V)
12	Under-voltage protection	when main loop DC bus voltage lower than 380V, it will stop (400V)
13	Radiator overheated protection	thermal resistor protection
14	IPM interior protection	IPM over current、overheated、short circuit、under-voltage protection
15	Electric motor protection	electric thermal protection
16	Impact restrain loop failure protection	through contactor joint to feedback protection
17	Over speed protection	protect when the speed is faster than allowed
18	Over deviation protection	Protect when speed deviation is bigger than allowed.
19	PG fault protection	Protect when PG braking or phase stagger.
20	Self-study protection	Protect when self-study of motor parameter unusual.
21	Phase lack protection	protect when lack of in/output phase
22	Door signal trouble	Protect when door signal unusual.

23	Running time protection	Protect when once run time is longer than limited time.
24	Counter of layer trouble	Protect when layer counter is wrong.
25	Trouble of communication immunity	Protect when communicate interrupt.
26	Hoistway learning error	protect when have trouble in hoistway learning error

List 2.10 optional function list

number	name	elevator description
1	Remote monitoring	You may monitor elevator running in monitor center by wire or wireless module.
2	Stop gang	Clock announce.
3	Voice stop-information	Voice announce.
4	ID identify	ID/IC card control
5	Weight device	Allocated Weight device

This chapter is the instruction of Main Circuit terminals, Control Circuit terminals, PG card terminals and their wiring specification.

3.1 Connection Diagram of elevator controller in-built

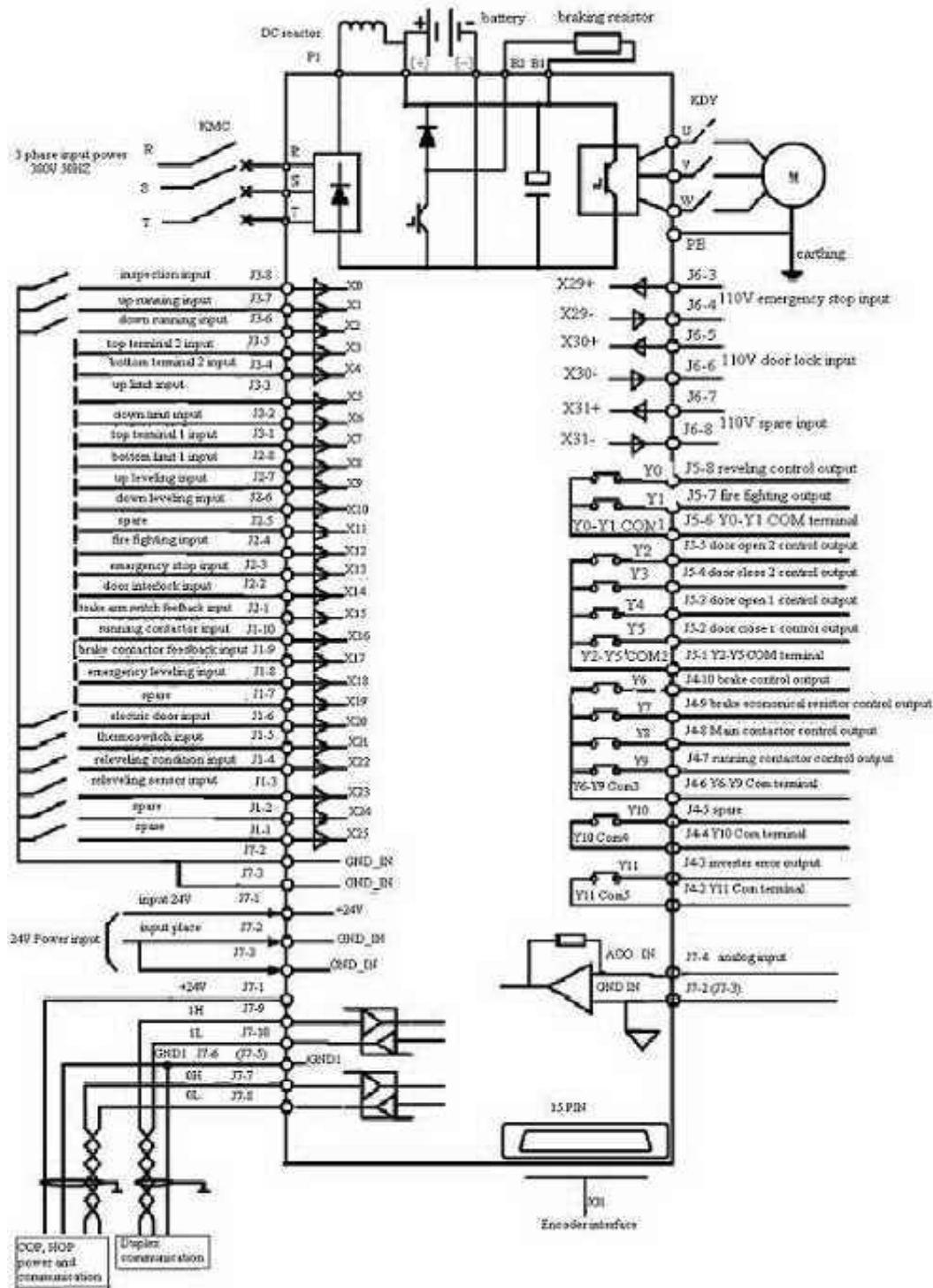


Fig. 3.1 connection diagram of elevator controller in-built 3.1

3.2 Wiring Main Circuit Terminals

3.2.1 Main Circuit Structure

Main Circuit structure is shown as Fig 3.2

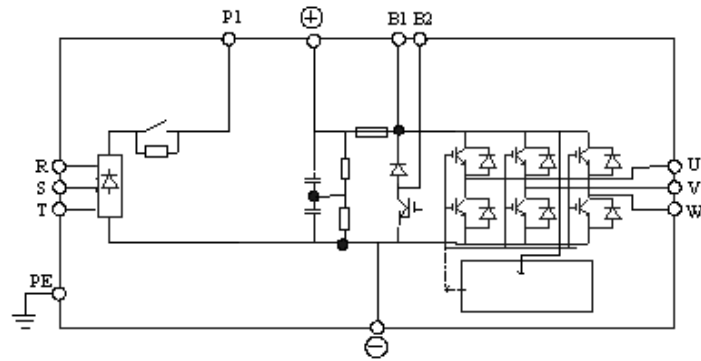


Fig 3.2 Main Circuit Structure

3.2.2 Terminal arrangement for Main Circuit

The terminal arrangement for main circuit is shown as Fig 3.3

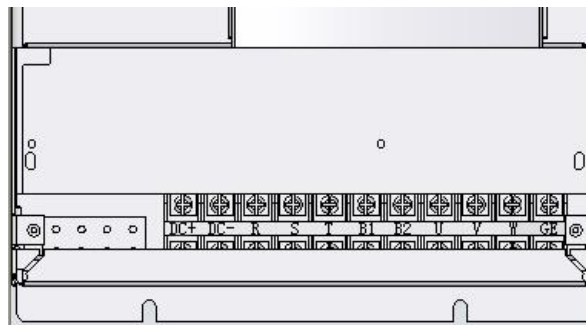


Fig 3.3 Terminal arrangement for main circuit

3.2.3 Main circuit terminal summary and function instruction.

Main circuit terminal summary and function instructions refer to Table 3.1

Table 3.1

Terminal symbol	Function instruction
R, S, T	Main circuit power input (380V/50Hz, 3 ϕ
DC+	DC BUS output +
DC-	DC BUS output +
B1、B2	Braking resistor unit connection
U, V, W	Controller output, connect with motor
DC+ DC-	Connect with battery device when system has butterfly running function
GE	Earthing

3.2.4 Wire size for wiring main circuit

Wire size and clapping torque are shown in Table 3.2

Table 3.2

Controller model	Terminal symbol	Terminal Screws	Claping Torque N•m	Acceptabl e wire size mm2	Recomme nded wire size mm2	Wire type
BL3-U□4005	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M4	1.2~1.5	4	4	Power cables , e.g. 600V vinyl power cables
	PE			2.5~4	2.5	
BL3-U□4007	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M4	1.2~1.5	4	4	
	PE			4	4	
BL3-U□4011	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M6	4~5	6~10	6	
	PE			6~10	6	
BL3-U□4015	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M6	4~5	6~10	6	
	PE			6~10	6	
BL3-U□4018	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M6	4~5	8~10	10	
	PE			8~10	10	
BL3-U□4022	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M6	4~5	10~16	10	
	PE			10~16	16	
BL3-U□4030	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M6	4~5	10~16	16	
	PE			10~16	16	
BL3-U□4037	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M8	9~10	25~50	35	
	PE			25~50	35	
BL3-U□4045	R, S, T, P1, (+), (-), B1、B2 , U, V, W	M8	9~10	35~50	35	
	PE			35~50	35	

3.2.5 Main circuit wiring

3.2.5.1 Wiring main circuit inputs

Observe the following precautions when wiring the main circuit.

1. Installing a Molded-case Circuit Breaker (MCCB).

Connect the power input terminals(R、S、T)and power supply via a molded-case circuit breaker(MCCB) suitable for the controller. Choose an MCCB with a capacity of1.8 to 2 times the controller's rated current. For MCCB's time characteristics, be sure to consider the controller's overload protection (60S at 150% of the rated output current, 10S at 180% of the rated output current).

2. Installing a Earthing Fault Interrupter

Controller outputs use high-speed switching, so high-frequency leakage current is generated. Therefore, at the controller input side, use an earthing fault interrupter to detect only the leakage current in the frequency range that is hazardous to humans and exclude high-frequency leakage current. For a special-purpose earthing fault interrupt, choose an earthing fault interrupter with sensitivity amperage of at least 30mA per controller. When using a general earthing fault interrupts,

choose an earthing fault interrupter with sensitivity amperage of 200mA or more per controller and with an operating time of 0.1s or more.

3. Installing a Magnetic Contactor.

Power input terminals(R、S、T) and power can be connected or be shut off through magnetic contactor (KMC). Choose of magnetic contactor is according with the rated current of the controller. Make sure the contact capacity of magnetic contactor is greater than the rated current of the controller.

4. Wiring the terminals

If there is no phase sequence requirements, the input power supply can be connected with any one terminal (R、S、T, the phase sequence of input power supply is irrelevant to the phase sequence.

5. Installing a DC reactor

If necessary, take out the jumper in P1、(+) terminal of the controller, install the DC reactor. Never touch the P1、(+) terminal or the jumper with bare hand directly.

6. Installing Surge Absorber

Make sure using a surge absorber for inductive loads near the controller. These inductive loads include magnetic contactor, electromagnetic relays and magnetic brakes, and so on.

3.5.2 Wiring the output side of main circuit

1. Connect the controller and Motor

Please connect the motor(input) wires to the output terminals U、V、W. Check that the motor rotates anticlockwise with the forward run command while running, otherwise, exchange the output terminal V and W.

2. Never connect a power supply to output terminals

Never connect a power supply to output terminals U、V、W. If voltage is applied to the output terminals, the internal circuit of the controller will be damaged.

3. Never short or earthing output terminals

Never earthing or short the output terminals. Never contact the controller with bare hands.

4. Never use capacity, surge absorber, power factor regulator and noise filter.

Never connect capacity, surge absorber, power factor regulator and noise filter to output circuit. The high-frequency components of the controller output may result in overheating or damage to these parts or may result in damage to the controller

5. The use of magnetic contactor

Don't connect a magnetic contactor between the controller and the motor and turn it ON or OFF during operation. Otherwise, a large inrush current will be created and the over current protection in the controller will operate. Before shut off the contactor, please stop the controller output first, after a while delay, then shut off.

3.2.5.3 Wiring earthing terminals

- a) Make sure connect the earthing protection terminal PE (earthing resistance of less than 10Ω.
- b) Please do not share the earthing wire with other devices, such as welding machines or power tools.
- c) Minimize the length and dimension of earthing wire.
- d) Please earthing at one point.

3.2.5.4 Connection of brake resistance

-
- a) There is in-built brake unit in the controller, but the energy that the motor feedback when the resistance releases the brake should be equipped. The configuration of the brake resistance refers to the table 2.6.
 - b) connect brake resistor terminal B1、 B2.
 - c) It is better to use heatproof wire with suitable specification to connect with brake resistance. And minimize the length
 - d) The installation of the brake resistance should consider the ventilated condition, so the ventilated fan and protection cover are necessary to make sure ventilate well, to avoid burn and electric shock and other danger once someone touch it un-carefully.
 - e) Never contact the Terminals B1 and B2 with bare hands

3.3wiring Control circuit terminals

3.3.1 Terminal arrangement for control circuit

The terminal arrangement for control circuit is shown as Fig 3.4.

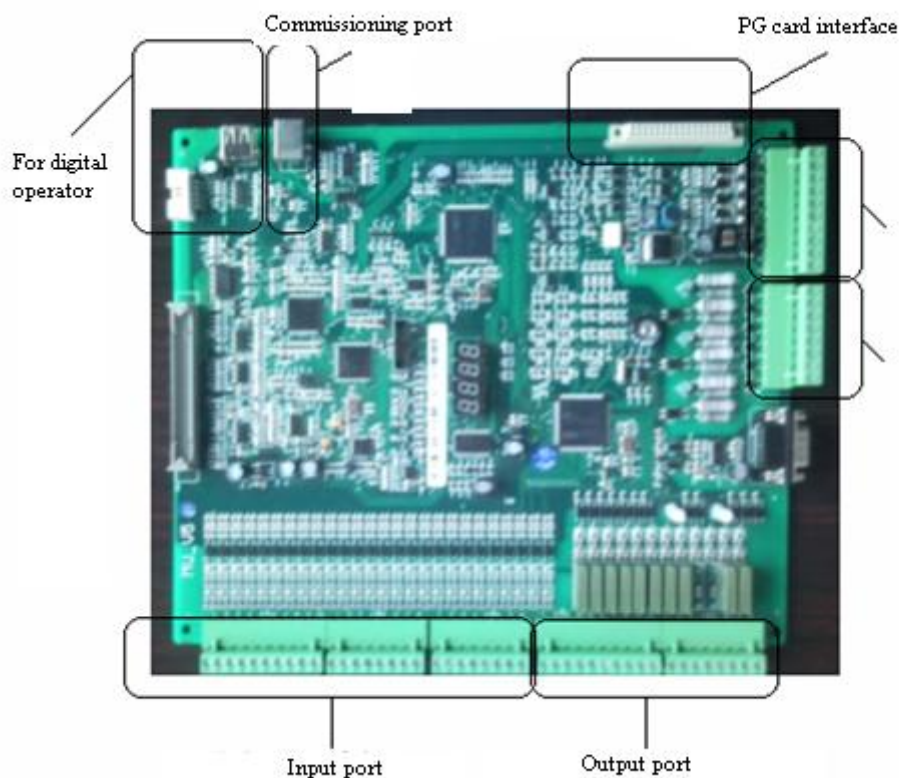


Fig 3.4 terminal arrangement for control circuit

Control circuit includes in-built main control board, PG card and digital operator. PG card and digital operator will be introduced separately in Chapter 4.

The change range of the elevator floor is big, and the input, output of the controller and the CAN communication interface is separate with the interior of the controller, so the I/O interface part of the controller should be connected to DC 24V power supply. Besides, DC 24V power supply should also be used by HOP and COP. So the power supply should be equipped with suitable capacity according with the floor.

3.3.2 Terminal connection for control circuit

The terminal connection for control circuit is shown in Table 3.1.

3.3.3 Control circuit terminal define and function instruction

The control circuit terminals define and function instruction is shown in Table 3.3.

Table 3.3

Terminal name	Terminal symbol	position	Define	Use	Interface technical specification			
					Interface type	Rated capacity	On/off time	Max Speed
J1	X25	J1-1	Spare	input	opto-coupler	DC24V 7mA	10mS	100Hz
	X24	J1-2	Spare					
	X23	J1-3	Re-leveling sensor input					
	X22	J1-4	Re-leveling condition input					
	X21	J1-5	Thermo switch input					
	X20	J1-6	electric lock input					
	X19	J1-7	Spare					
	X18	J1-8	Emergency leveling running input					
	X17	J1-9	Brake contactor fead back input					
	X16	J1-10	Running contactor input					
J2	X15	J2-1	Brake feedback input	input	opto-coupler	DC24V 7mA	10mS	100Hz
	X14	J2-2	Door inter-lock input					
	X13	J2-3	Emergency stop input					
	X12	J2-4	Fireman input					
	X11	J2-5	Spare					
	X10	J2-6	Down-leveling input					
	X9	J2-7	Up-leveling input					
	X8	J2-8	Bottom terminal 1 input					
J3	X7	J3-1	Top terminal 1 input	input	opto-coupler	DC24V 7mA	10mS	100Hz
	X6	J3-2	Down limit input					
	X5	J3-3	Up limit input					
	X4	J3-4	Bottom terminal 2 input					
	X3	J3-5	Top terminal 2 input					
	X2	J3-6	Down running input					
	X1	J3-7	Up running input					
	X0	J3-8	Inspection input					
J4		J4-1	Stand by	output	Relay	DC 10A30V AC 10A250V	5/10mS	20cpm
	COM5	J4-2	Y11 common terminal					
	Y11	J4-3	Inverter fault output					
	COM4	J4-4	Y10 common terminal					
	Y10	J4-5	Spare					
	COM3	J4-6	Y6~Y9 common terminal					
	Y9	J4-7	Running contactor control output					
	Y8	J4-8	Main contactor control output					
	Y7	J4-9	Brake economy resistance control output					
	Y6	J4-10	Brake control output					

*The function of X11 and Y8 need to enable by special function setting F4-06-29.

Terminal name	Terminal symbol	position	Define	Use	Interface technical specification			
					Interface type	Rated capacity	On/off time	Max speed
J5	COM2	J5-1	Y2~Y5 common terminal	Output	Relay	DC 10A30V AC 10A250V	5/10mS	20cpm
	Y5	J5-2	Door close 1 control output					
	Y4	J5-3	Door open 1 control output					
	Y3	J5-4	Door close 2 control output					
	Y2	J5-5	Door open 2 control output					
	COM1	J5-6	Y0~Y1 common terminal					
	Y1	J5-7	Fire fighting output					
	Y0	J5-8	Re-leveling control output					
J6		J6-1	Standby	--	--	--	--	--
		J6-2	standby					
	X29+	J6-3	Emergency stop input+	Input	opto-coupler	AC110V 8mA	10mS	100Hz
	X29-	J6-4	Emergency stop input-					
	X30+	J6-5	Door inter-lock input +					
	X30-	J6-6	Door inter-lock input -					
	X31+	J6-7	Spare					
	X31-	J6-8	Spare					
J7	+24V	J7-1	Input power supply	power supply	power supply	DC 24V 10A		
	GND IN	J7-2	Input					
	GND IN	J7-3						
	AG0 IN	J7-4	analog input	Input	Analog	-10V~+10V		
	GND1	J7-5	output	Communication interface				
	GND1	J7-6						
	0H	J7-7	HOP/COP communications+	Communication interface	CAN	80mA		25KH
	0L	J7-8	HOP/COP communications -					
	1H	J7-9	Duplex/Group control communications+	Communication interface	CAN	80mA		25KH
	1L	J7-10	Duplex/Group control communications -					

3.3.4 Wire size for control circuit terminals

600V plastic insulated wire should be used when connect the control circuit terminals. Choose suitable wire size according with the function of terminals. Refers to Table 3.4

Table 3.4

Terminal function	Acceptable wire size mm ²	Recommended wire size mm ²	Claping Torque N.m	Special requirement
input、output	0.5~1	0.75	0.5~0.6	
Pgcard input/output	0.15~0.5	0.3	0.5~0.6	TWISTED-PAIR SHIELD CABLE
CAN communications	0.75~1.5	0.75(≤10 floors), 1.5(>10 floors)	0.5~0.6	TWISTED-PAIR SHIELD CABLE

3.3.5 Control circuit input and output interface and wiring

3.3.5.1 Digital value input interface

the common terminal of board for digital value input interface is +24V, so it should be contactor input or common emitter input (0V common terminal), its input and connection as per Fig 3.5 and Fig 3.6.

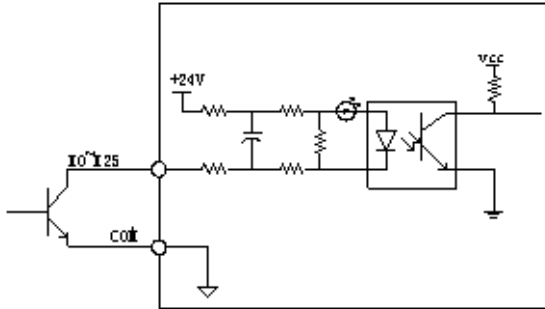


Fig. 3.5 common emitter input

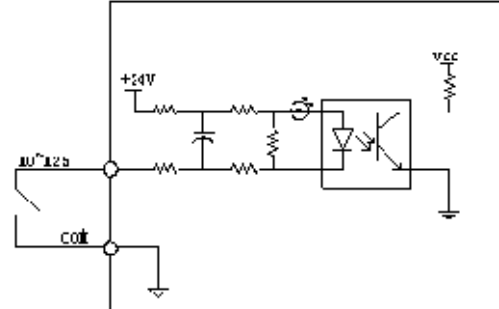


Fig 3.6 contactor switch input

3.3.5.2 CAN Communication interface

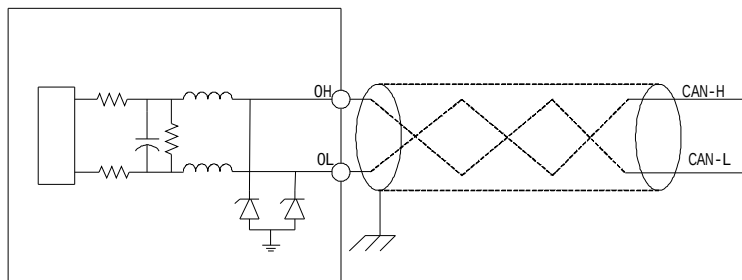


Fig 3.7 CAN Interface and connection

The controller, the COP and the HOP is connected through the CAN wire. There is directly a relation among the reliability of the controller, the communication wire and the wiring. It is required that the shaft cable and the traveling cable should be twisted-pair shield cable. The twisted-pair cable can stop the differential mode interference of the long wire and the electromagnetic interference caused during the elevator running effectively. The function and connection of the CAN communication interface is shown in the Fig 3.7.

3.3.5.3 Analog input interface

The voltage range of analog input interface is -10~+10V, which is for selecting the load information provided by weighing device. For analog inter put is easy to be interrupted. The wire of analog input terminal should be TWISTED-PAIR SHIELD CABLE. Its input and connection is shown in Fig 3.8.

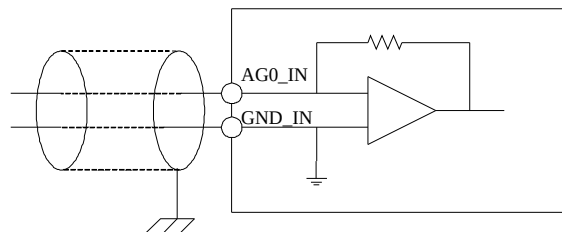


Fig 3.8 analog input and connection

3.3.5.4 AC 110V input interface

AC 110V input interface is for the on-off testing of emergency stop and door lock. Its interface and connection is shown in Fig 3.9

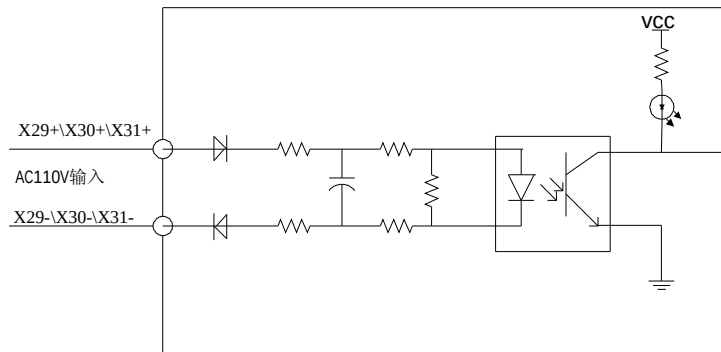


Fig 3.9 AC110Vinput interface and connection

3.3.5.5 Power supply input interface

Power supply interface and its connection is shown in Fig 3.10

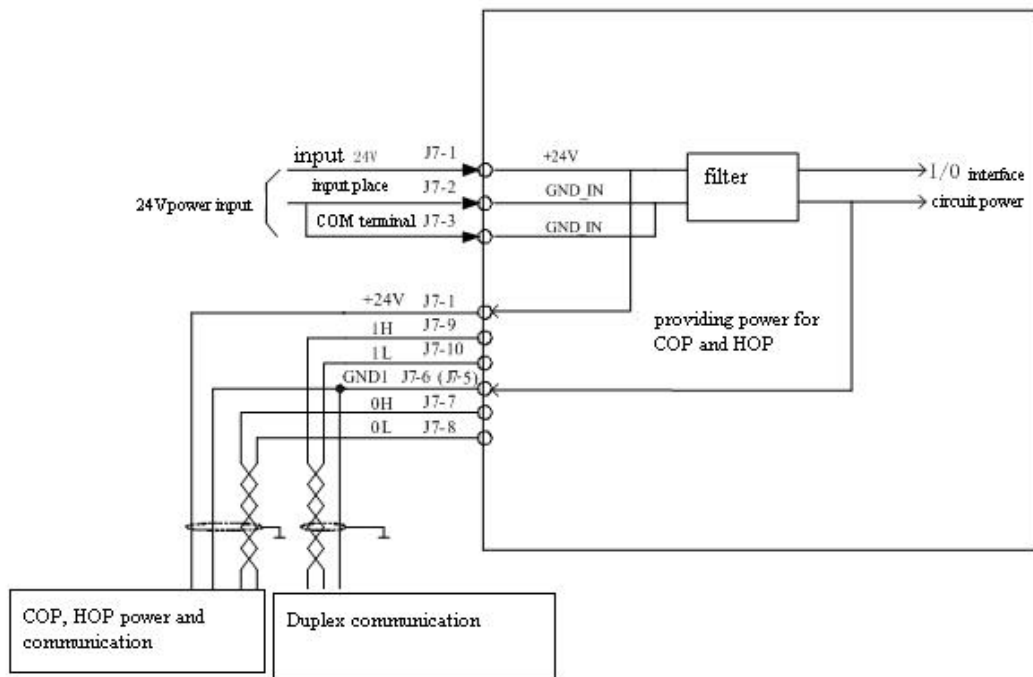


Fig 3.10 Power supply interface and its connection

3.4 PG card installation and wiring

3.4.1 PG—V31 interface card

PG-V31 interface card is used for pulse encoder speed feedback and frequency output card of synchronous/asynchronous motor.

The card is compatible for 12V OC output and push-pull output and 5V long line drive output pulse encode, encoder for synchronous motor (A/B) and encoder for asynchronous motor (A/B/Z/U/V/W)

It is shown in Fig 3.11



Fig 3.11 PG-V31 card

3.4.2 Installation and unload of PG-V31 card

while installing the PG-V31 card, first shut off the controller power supply, confirm that the charge indicator lamp of the controller is off. Then unload the controller and the front cover and install the PG-V31 card.

While installing, first aim the connector of the PG-V31 card (welding parts) towards the JPGR connector of the main control board. Aim the other two installation hole toward the related two nylon tube. Then keep the PG-V31 card horizontal and press the card equably until the connector and the nylon tube are installed well.

While unloading, first shut off the controller power supply, confirm that the charge indicator lamp of the controller is off. Then unload the controller and the front cover, and then unload the terminal connection of PG-V31 card. Put the unloaded wire well, don't contact with other electrics, then unload the D type connector, hold the embossed part of the nylon tube with needle point pliers, raise the PG card slowly until the embossed part is taken out. Deal with the other nylon tube in the same way. Then take out the PG card.

Precautions while installing and unloading

1. Don't touch the chip with bare hand directly, to avoid that the static damage the chip.
2. Choose suitable tool, such as screwdriver, needle point pliers, and so on.
3. Never damage the components of the card.
4. Never damage the components of the main control board.
5. While recovering the connection, please connect as per the wire No. and the requirements, make sure the connection correctly.
6. Please check whether the jumper wire of the new card and the old card is same.

3.4.3 Define and function of the PG-V31 card terminal

The definition and the function of the PG-V31 card is shown in the Table 3.5.

Table 3.5

Terminal name	Terminal mark	Position	Define	Usage	Interface technical specification.		
					Interface type	Rated capacity	Max speed
JG0	+5V	JG0-1	Power supply	5V Power supply	Power output	+200mA/5V±5%	
	A-	JG0-2	Frequency signal difference output A-	synchronization frequency division	Difference output	±50mA	500KHz
	A+	JG0-3	Frequency signal difference output A+	synchronization frequency division	Difference output	±50mA	500KHz
	B-	JG0-4	Frequency signal difference output B-	synchronization frequency division	Difference output	±50mA	500KHz

	B+	JG0-5	Frequency signal difference output B+	synchronization frequency division	Difference output	±50mA	500KHz
	OV	JG0-6	Power supply earthing	Power supply earthing	Power supply earthing	—	
	IB	JG0-7	OC/ push-pull type B phase input	Input signal B	OC/ push-pull input	-10mA/12V-15V	500KHz
	IA	JG0-8	OC/ push-pull type A phase input	Input signal A	OC/ push-pull input	-10mA/12V-15V	500KHz
	+12V	JG0-9	OC/push-pull type power supply	12V power supply	Power output	+150mA/12V±5%	
	GE	JG0-10	Shield	Shield		—	
JG1	+5V	JG1-1	+5V	5V power supply	Power output	+400mA/5V±5%	
	U+	JG1-2	U+	Difference signal U+	Difference input	±20mA/3.1-5V	500KHz
	Z+	JG1-3	Z+	Difference signal Z+	Difference input	±20mA/3.1-5V	500KHz
	B+	JG1-4	B+	Difference signal B+	Difference input	±20mA/3.1-5V	500KHz
	A+	JG1-5	A+	Difference signal A+	Difference input	±20mA/3.1-5V	500KHz
	GND	JG1-6	GND	5V earthing	Power supply earthing	—	
	U-	JG1-7	U-	Difference signal U-	Difference input	±20mA/3.1-5V	500KHz
	Z-	JG1-8	Z-	Difference signal Z-	Difference input	±20mA/3.1-5V	500KHz
	B-	JG1-9	B-	Difference signal B-	Difference input	±20mA/3.1-5V	500KHz
	A-	JG1-10	A-	Difference signal A-	Difference input	±20mA/3.1-5V	500KHz
	GND	JG1-11	GND	5V earthing	Power supply earthing	—	
	V+	JG1-12	V+	Difference signal V+	Difference input	±20mA/3.1-5V	500KHz
	V-	JG1-13	V-	Difference signal V-	Difference input	±20mA/3.1-5V	500KHz
	W+	JG1-14	W+	Difference signal W+	Difference input	±20mA/3.1-5V	500KHz
	W-	JG1-15	W-	Difference signal W-	Difference input	±20mA/3.1-5V	500KHz



the parameter in the Table is given when PG-V3w interface card work in the environment temperature of 0℃-70℃, if the temperature is higher than 70℃, PG interface card will work abnormally or be damaged.

3.4.4 Connection PG-V31 card and 12V push-pull/OC output encoder (for asynchronous motor)

The connection PG-V3 card and 12Vpush-pull/OC output encoder (for asynchronous motor) is shown in Fig 3.12



JA and JB should be shorted, connect the12V/B/A/OV of encoder to the +12V/IB/IA/OV (the terminal 7/8/9/10 of JG0) one-by-one.

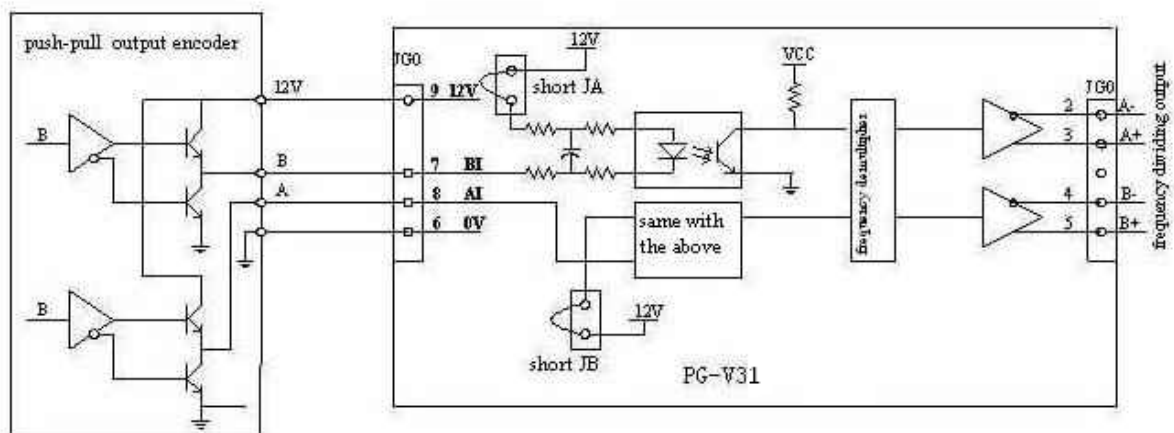


Fig 3.12 connection PG-V31 card and 12V push-pull/OC output encoder

3.4.5 Connection of PG-V31 card and 5V wire output encoder (or asynchronous motor)

The connection of PG-V31 card and 5V wire output encoder (for asynchronous motor) is shown in Fig 3.13



JA and JB should be shorted (The status of factory setting for JA and JB wire jumper is off), then connect the 5V/B+/A+/B-/A-/0V of encoder to the +5V/B+/A+/B-/A-/GND of the D-type connection part JG1 (the related pin of 1/4/5/9/10/6 of JG1 for D-type connection part). After confirming connection well, please connect plug and lock well.

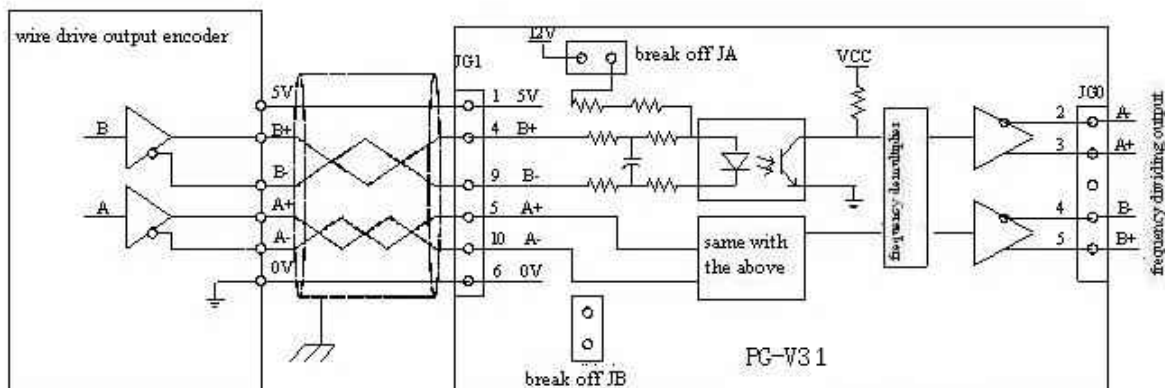


Fig 3.13 connection of PG-V31 card and 5V wire output encoder (for asynchronous motor)

3.4.6 Connection of PG-V3 card and 5V wire drive output encoder (for synchronous motor)

The connection of PG-V31 card and 5V wire drive output encoder (for synchronous motor) is shown in Fig 3.14



JA and JB should be disconnected (The status of factory setting for JA and JB wire jumper is off), then connect the 5V/0V/A+/A-/B+/B-/Z+/Z-/U+/U-/V+/V-/W+/W-/ of encoder to the +5V/GND/A+/A-/B+/B-/Z+/Z-/U+/U-/V+/V-/W+/W-/ of the D-type connection part JG1 (the related pin of 1/6/5/10/4/9/3/8/2/7/12/13/14/15 of JG1 for D-type connection part). After confirming connection well, please connect plug and lock well.

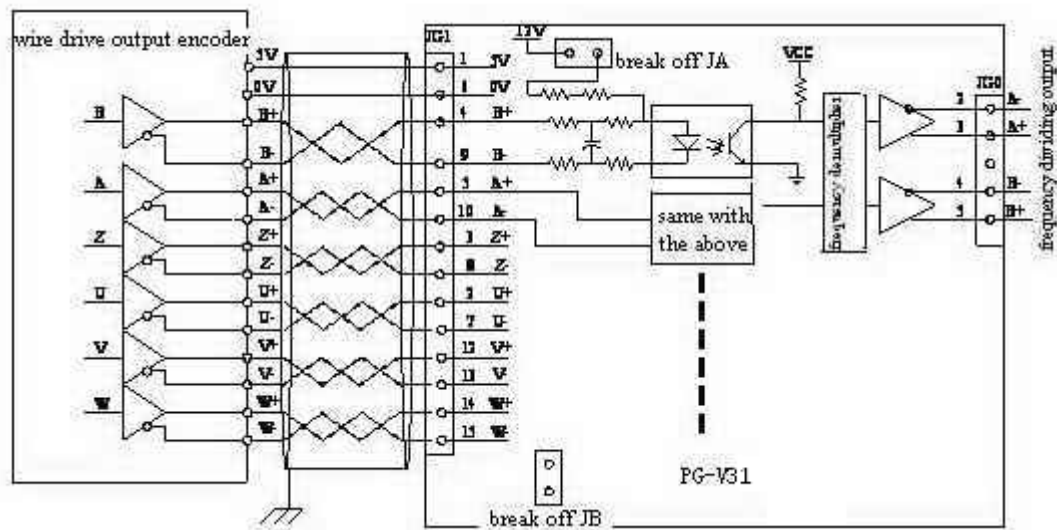


Fig 3.14 connection of PG-V31 card and 5V wire drive output encoder (for synchronous motor)

3.4.7 Precautions for using PG-V31 card

For the 15 pin differential output encoder interface of the PG-V31 card and the OC/push-pull output encoder interface, the input pulse signal frequency is required to be not more than 500KHz, the parameter of the input and output signal current and voltage can't be out of the range given in the above table.

If the parameter of the input and output signal current and voltage is out of the given range, the PG interface card may work abnormally or be damaged; even the encoder may be damaged.

The manufacturer suggests that once the circuit working condition is not very well, to avoid that the PG card is damaged or be interfered, except the encoder, the PG-V31 interface card can't provide +5V power supply for other circuits.

Never connect the PG-V31 interface card to OC/push-pull output encoder interface and the differential output encoder interface simultaneously. Otherwise, the encoder and the PG—V31 card may be damaged.

While using the differential output encoder interface, it is necessary to make one D type plug with 3 line 15 pin corresponding with the "JG1" connector of the PG-V31 interface card. Simultaneously, never connect the jumper "JA" and "JB". If the OC/push-pull output encoder interface is used, the signal interface in the "JG0" terminal can be connected directly, and the jumper "JA" and "JB" should be connected. The position of "JA" and "JB" doesn't affect the use of the frequency output function.

Pay more attention to that the user shall take responsibility for the consequence caused for the above reason.

3.4.8 SPG-V33 interface card

SPG-V33 interface card is shown in the Fig 3.15



Fig 3.15 SPG-V33 card

3.4.9 SPG-V33 interface terminal define and function

SPG-V33 interface terminal define and function is shown in Table 3.6

Table 3.6

Name of Terminal	Terminal mark	Location	Define	Function	Interface technology specifications			
					Inlet type	Rated burden	off/on time	Max speed
JENDAT	+5V	JENDAT -1	Power supply	5Vpower supply	Power output	+500mA/5V±2.5% Voltage ripple less than 50mV		
	B-	JENDAT -2	The B- signal output terminal of difference frequency output	synchronization frequency division	Differential output	5V/±50mA		250KHZ
	B+	JENDAT -3	The B+ signal output terminal of difference frequency output	synchronization frequency division	Differential output	5V/±50mA		250KHZ
	A-	JENDAT -4	The A- signal output terminal of difference frequency output	synchronization frequency division	Differential output	5V/±50mA		250KHZ
	A+	JENDAT -5	The A+ signal output terminal of difference frequency output	synchronization frequency division	Differential output	5V/±50mA		250KHZ
	GND	JENDAT -6	PGND	PGND	PGND			
	GND_E	JENDAT -7	Earthing terminal of encoder shield wire					
	GND_E	JENDAT -8	Earthing terminal of encoder shield wire					
JG1	B-	JG1-1	B-	Differential signal B-	Differential input			40KHz
	*	JG1-2	—	—	—			
	R+	JG1-3	R+	Differential signal R+	Differential input			40KHz
	R-	JG1-4	R-	Differential signal R-	Differential input			40KHz
	A+	JG1-5	A+	Differential signal A+	Differential input			40KHz
	A-	JG1-6	A-	Differential signal A-	Differential input			40KHz
	0V	JG1-7	GND	5V Ground	PGND			
	B+	JG1-8	B+-	Differential signal B+	Differential input			40KHz
	5V	JG1-9	+5V	5Vpower	Power output	+500mA/5V±2.5% Voltage ripple less than 50mV		
	C-	JG1-10	C-	Differential signal C-	Differential input			40KHz
	C+	JG1-11	C+	Differential signal C+	Differential input			40KHz
	D+	JG1-12	D+	Differential signal D+	Differential input			40KHz
	D-	JG1-13	D-	Differential signal D-	Differential input			40KHz
	*	JG1-14	—	—	—			

	*	JG1-15	—	—	—			
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3.4.10 SPG-V33 interface card circuit

SPG—V33 interface card circuit is shown in Fig 3.16

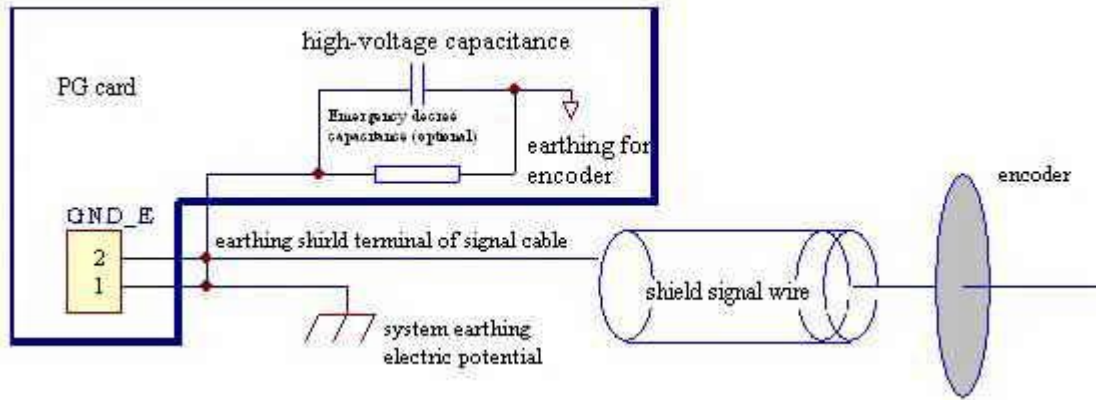


Fig 3.16 SPG-V33 card



The resistance of earthing terminal GND_E is not necessary component. There is difference between different versions.

3.4.11 Precautions for SPG-V33 interface card

1. Sine and cosine PG card can provide the power supply output of $5V \pm 2.5\%$ /500mA for encoder. If the current that the circuit need exceeds the current output capacity that PG card appointed, it may damage the PG card; even it may damage the encoder.
2. The cut-off frequency of the analog input signal is not more than 40KHz. it is decided by flitter circuit. The input signal bandwidth exceed the range, PG card can't analyze the signal correctly. If so, the VVVF system will not work.
3. if the frequency output function that PG card provide is used, make sure that the current of frequency interface output is in the appointed range. Otherwise, circuit protection will be caused. At the same time, the Max. Speed rate is 250Kb/S, beyond the limit of the speed rate, frequency output will be abnormal.
4. PG card can set two different kinds of gain for A ,B ,C and D signal, A and B signal by setting the status of the jumper "SGA" and "SGB". Generally, "SGA" and "SGB" are not shorted. At this time, the gain is the smaller value. If the signal cable of the encoder at site is longer than 7 meters, it is better short the "SGA" and "SGB", at the moment, the gain value is the larger one, which can compensate the signal attenuation caused by the increased line impedance. Being affected by different Conductor characteristic, please short "SGA" and "SGB" to get the signal with larger value. At the moment, compared with the status of the load compensation, if the effect of shorting and un-shortening "SGA" and "SGB" is same, it is better to short "SGA" and "SGB". Short or open "SGA" and "SGB" at the same time, otherwise, PG card will work abnormally; C and D signal by setting the status of jumper "SGC" "SGD", The specific situation as A,B.
5. The two earthing terminal of GND_E is shorted. At actual site, one terminal is required to connect to the system earthing wires, and the earth resistance should be minimized and earthing in one point independently. The other one terminal is connected to the shielding layer of the signal cable for the sine and cosine encoder. For the shielding layer of the signal cable, it is better that the two terminals are earthing connected or the end terminal at GND_E side is earthing connected. Because the output signal of the sine and cosine encoder is small analog signal, so it is east to be interfered by electromagnetic. It is better to use the signal cable manufacturer recommended cable, the cable should

have complete shield layer and have lower and twin twisted transfer impedance. Inappropriate earthing connection or unsuitable signal transfer cable using will result in the system work abnormally.

6. Older version SPG_V33 gain the function of determine the position encoder through C,D signal (UPC of PG card on the back is 20106_**_01050000_*****_**_***_*****), if you want to use this function, need to make the jumper S_CD short-circuited before power on; New version SPG_V33 gain the function of determine the position encoder through C,D signal (UPC of PG card on the back is 20106_**_01060000_*****_**_***_*****), the function of encoder break testing and starting no load of the new version, if you want to use this function, need to make the jumper S_CD disconnect before power on.

7. Jumper S_XF is retention function, keep disconnected by state.

3.5 Precaution when wiring

1. Before connection, pay much attention to the precaution in chapter 1.2, especially the “warning” and “caution”.
2. Before connection, make sure that the power supply is off.
3. Please ask the professional person who is trained and be authorized to wire.
4. The wire size and clapping torque should be as per the regulation of Table 3.2 and the Table 3.4
5. To wire conveniently and reliably, it is better to use round crimp type terminal (for main circuit) and club-shaped terminal (for control circuit).
6. Wiring the control circuit and main circuit and power supply separately.
7. CAN communication wire、encoder wire、encoder frequency output wire and analog input wire should use TWISTED-PAIR SHIELD CABLE.
8. Please earthing the cable shield wire reliably. And maximize the contact area.
9. Make sre the signal wire of PG (encoder) in 30m, and minimize the length.
- 10 Make sure the connection between the controller and the motor in 100m, and minimize the length.
11. When connect the DC reactor option, please cut-off the short between the P1 and (+), and connect the DC reactor.
12. Make sure the brake resistance is between B1 and B2.
13. Make sure Connect earthing terminal PE well, please does not share the earthing wire with other devices, such as welding machines or power tools. The minimize length and dimension of earthing wire. And Earthing at one point.
14. After connection, make sure check the following:
 - 1) Correctness and reliability of connection.
 - 2) Whether there is leftover, such as wire, screw and metal filing
 - 3) Whether the connection of the screw, the terminals and the connection parts is loose.
 - 4) Whether the bare conductor of terminals is connected with other terminals.

4 Digital operators

4.1 Digital Operator Display

BL3—U series elevator in-built controller is equipped with LCD digital operator OP-1. It is a kind of special tool for commissioning and maintenance of control system. It provides kingly man-machine conversation passage with Chinese and English.

4.1.1 The key names and functions of the Digital Operator are described below

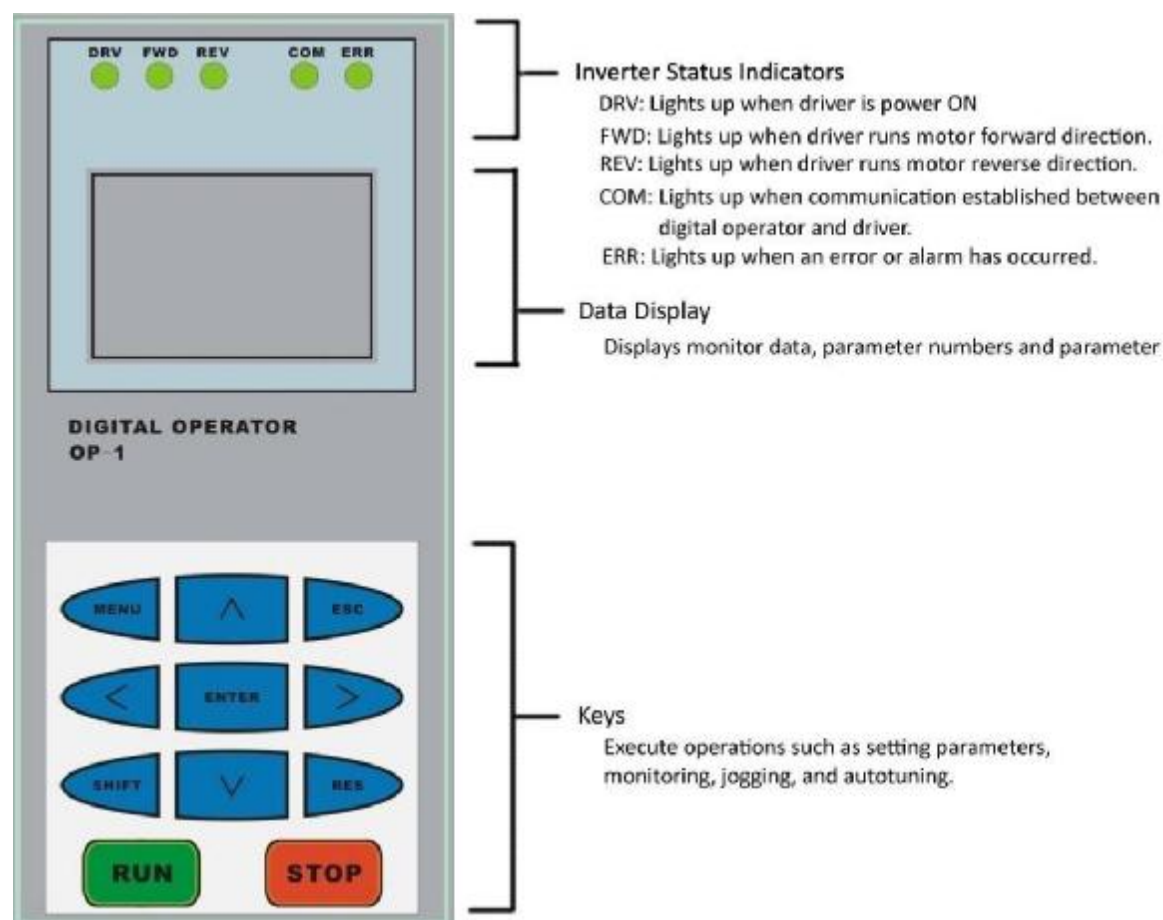


Fig 4.1 digital operator

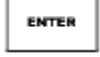
4.1.2 Digital Operator Keys

The names and functions of the Digital Operator Keys are described in Table 4.1.

Table 4.1 Key Functions

Key	Name	Function
	[RUN] key	Start the drive in keypad control mode. Start the auto-tuning
	[STOP] key	Stop the drive in keypad control mode. Stop tuning. Reset the fault alarm state (in any control mode)

Table 4.1 Key Functions (continual)

Key	Name	Function
	[MENU] key	Shift between Stop/Running state and Menu state. In menu operation, return to the last menu
	[SHIFT] key	Sets the active digit when programming parameters.
	[RES] key	Enter the interface of bit parameter setting for some parameter.
	[UP] key	Increase the function code, function group or parameter
	[DOWN] key	Decrease the function code, function group or parameter
	[LEFT] key	Toward the left to select the bit of the setting parameter.
	[RIGHT] key	Toward the right to select the bit of the setting parameter.
	[ENTER] key	Save the change of parameters. In menu operation, enter submenu
	[ESC] key	Return to the Previous Menu of the current menu.

4.1.3 LCD Display

After the controller is power-on, enter the main interface. The main interface is shown in Fig 4.2

The main interface display the main information of the elevator in present status, including the floor No., the elevator speed, running direction, door lock status, running model and fault code, and so on. Under the main interface, the above information is updated timely.

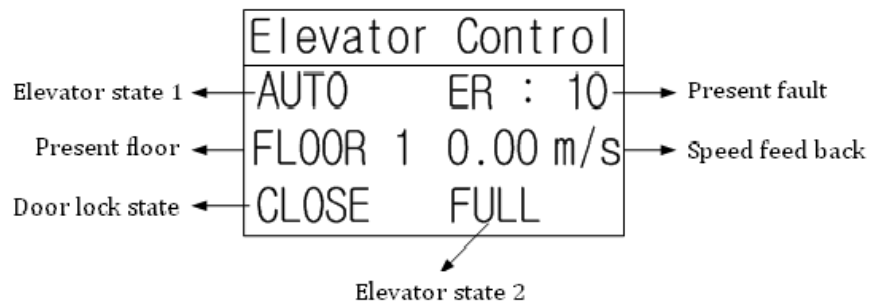


Fig 4.2 Digital operator LCD display interface

The instructions for the information shown in the Digital operator LCD display interface are as following:

1. The status of the elevator display1:

INSP	MANU	AUTO
FIRE	STOP	USER
2. Present floor display:

Display the floor number that the elevator is at.
3. Door lock status:

CLOSE:	the door lock is closed.	OPEN:	the door lock is opened.
--------	--------------------------	-------	--------------------------
4. The status of the elevator display 2:

FULL:	full load
OVER:	overload
GROUP:	group control
TWINS:	Duplex control
5. Present error:

ERR: XX	display the present Error.
---------	----------------------------

4.1.4 Function of digital operator

The function of digital operator is as follows:

- 1.LCD display with English and Chinese
- 2.Setting visit grade and password.
- 3.Adjusting the setting of the shortcut menu
- 4.Monitor the status of the elevator and the controller.
- 5.Prameter check, setting and save.
- 6.Hoistway learning
- 7.Motor prameter learning
- 8.Weighing data learning
- 9.System time setting
- 10.Error history record and check.
- 11.Prameter copy, upload and unload.
- 12.Restore the factory value.

4.1.5 Installation and connection of digital operator

The installation and unload of the Enclosed digital operator please refer to page 2 to 9 in the chapter 2.7.5

The connection for the digital operator of the opening controller is as follows: first unload the front cover of the controller, then connect the special communication to one terminal of the digital operator, connect to the other one terminal of the J232_T connection part of the main board. After confirming connection reliably, install the front cover of the controller well.

1. The installation, unload, insert and pulling out of the digital operator can be carried out when the controller is power-on. That is to say, hot plug is accepted by the digital controller.
2. Please install, unload, insert and pull out the digital operator carefully to avoid the damage of the digital operator.
3. Please save the unloaded digital operator and communication cable well, never pres or damage or put in bad environment.
4. Please never use the over 3m communication cable.



IMPORTANT

4.2 Structure and switch over of the digital operator menu

The structure and switchover of the digital operator menu is shown in Fig 4.3

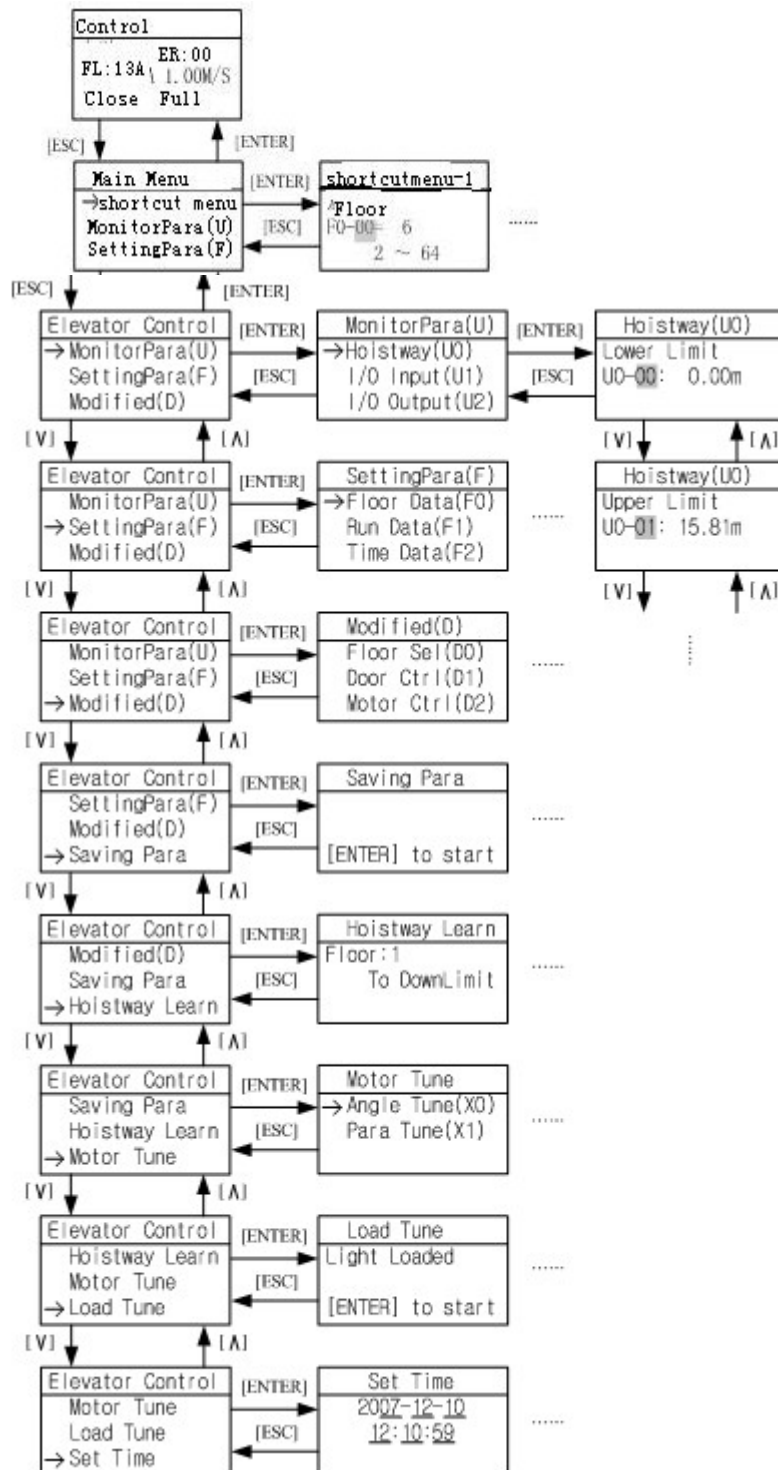


Fig 4.3 structure and switchover of the digital operator menu

4.3 Parameter setting

4.4 Commissioning parameter

In this commissioning interface, move the arrow by pressing [UP] key or [DOWN] key to carry out the operation, press [ENTER] to enter the selected interface.

In floor selection interface, press [UP] key or [DOWN] key to check the car call status in present floor. If some floor should set car call, press [ENTER] key in the interface to save the car call information. In the interface, the information bellow F is the present floor the elevator car is in.

Floor selection flowchart is shown in Fig 4.6

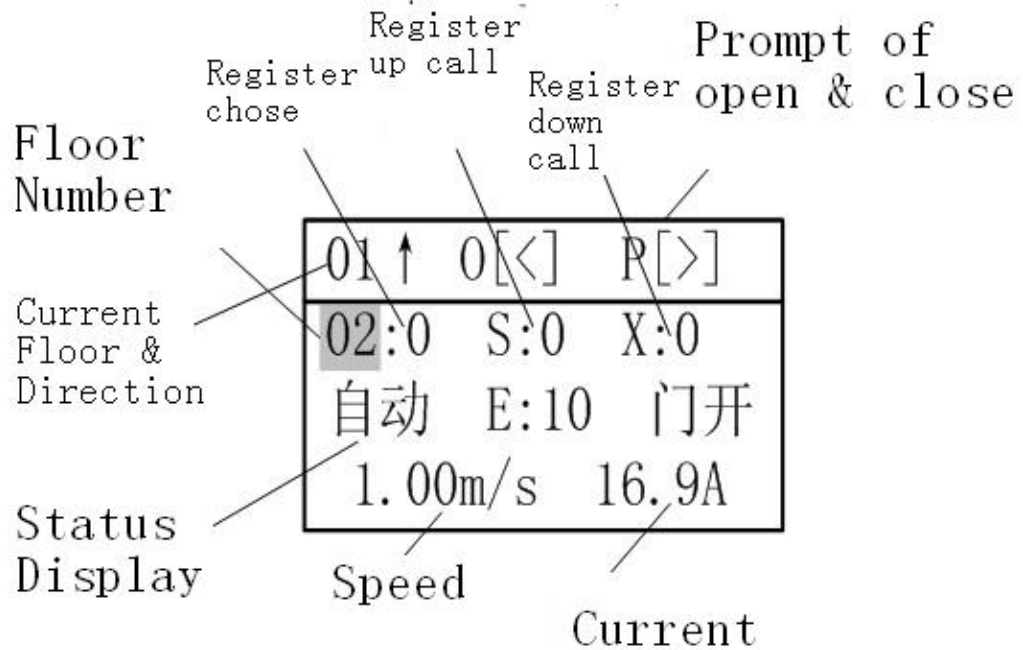


Fig 4.6 Floor selection flowchart

In the interface of door open and close, press the [LEFT] key as per the guide in the interface to carry out the command of “opening the front door”. Press [RIGHT] key to carry out the command of “close the front door”. Press [UP] key to carry out the command of “open the rear door”. Press [DOWN] key to carry out the command of “close the rear door”, Press [RES] key to cancel all the commands.

Door open and close flowchart is shown in Fig 4.7

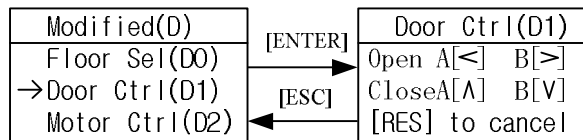


Fig 4.7 Door open and close flowchart

In the motor running interface, press [UP]/ [DOWN] key to select the running mode is continuous or inching. In the mode of continuous running, first it can press [RUN] key to start the motor, press [UP]/ [DOWN] key to make the motor datum speed up or down. The datum speed can be negative (negative speed means the motor reversely), press [ENTER] key to save the datum speed. Press the [STOP] key to stop the motor. In the mode of inching, press [UP]/ [DOWN] key to increase or decrease the inching operation frequency, after press the [ENTER] to save the setting, and then press the [RUN] key to start, and press the [STOP] to stop the operation.

The flowchart that the controller controls the motor running is shown in the Fig 4.8.

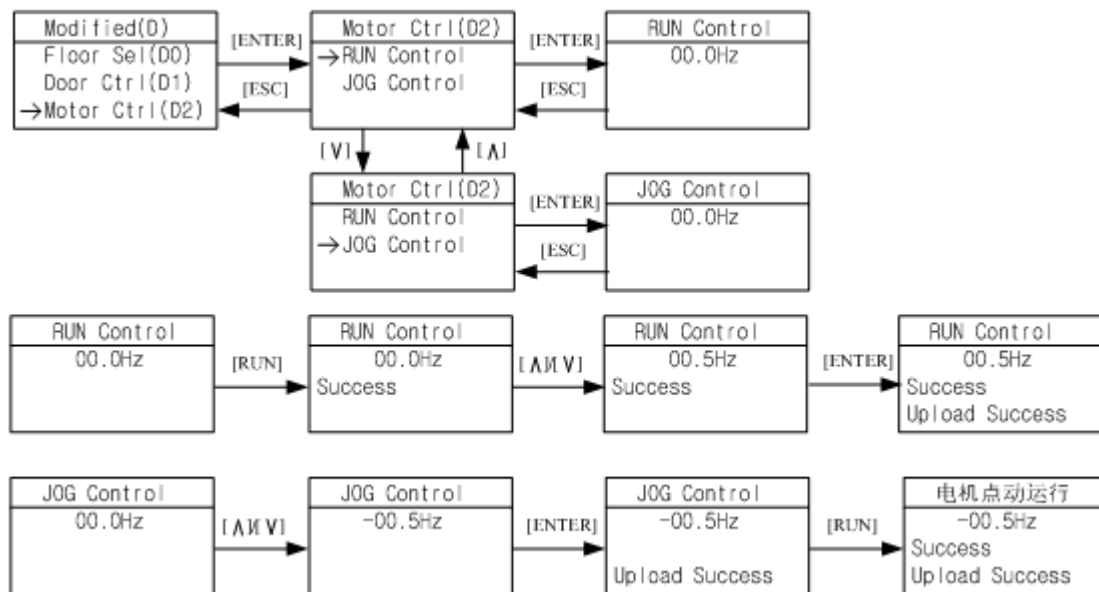


Fig 4.8 the procedure of using digital operator run the moto

In the interface of car call testing, it can press [UP]/ [DOWN] key to check the connection of all floors. Normally, it displays running successfully. Or it fails. Press [ESC] key to return the Previous Menu.

The flowchart of car call testing is shown in Fig 4.9

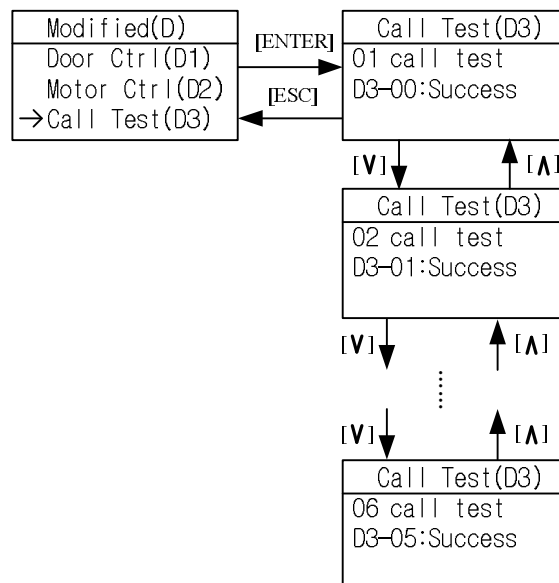


Fig 4.9 car call testing interface

The interface of communication testing is shown in the Fig 4.10

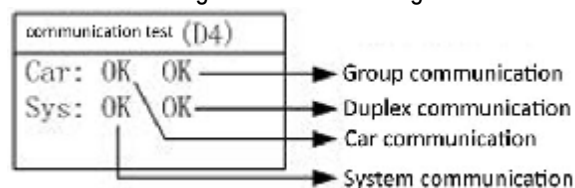


Fig 4.10 communication testing interface

The display of the elevator control communication is as follows:

OK	communication normally.
ER	the controller error of transmit (please check communication connection and the main board) , when there appears a number, that is the times of the error of the system communication.

System communication display

OK	communication normally
ET	the controller error of transmit (please check car call communication connection) , when there appears a number, that is the times of the error of the system communication.

Group control communication

OK	Group control system communication normally
ER	Group control system communication abnormally (option)

Duplex control communication

OK	Duplex control system communication normally
ET/ER	Duplex control system communication abnormally (option)

4.5 Save parameter

In the interface of saving parameter, press the [ENTER] to save all the parameters in the F menu. During the course of saving, it will display “running”, after saving, it will display “running successfully” or “running fail”.

The flowchart of saving parameter is shown in the Fig 4.11

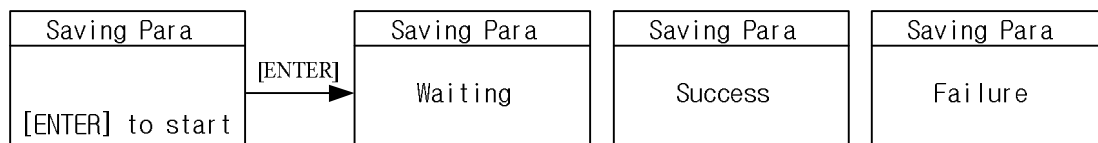


Fig. 4.11 saves parameter.

4.6 Hoistway learning

After enter the hoistway learning interface, the status display “to the down limit”, the hoistway learning can’t start directly. The elevator should run to the down limit position, at this time the “to the down limit” status disappear, the screen will remind “press the enter key to start”, press the [ENTER] key to start hoistway learning. After starting hoistway learning, the interface will display the learning floor timely and display the status of “running”, when the hoistway learning is completed, the status will display “success”, when there are errors during the hoistway learning, the status will display “error” and display error code. Press [ESC] key to return the Previous Menu.

The flowchart of the hoistway learning is shown in the Fig 4.12.

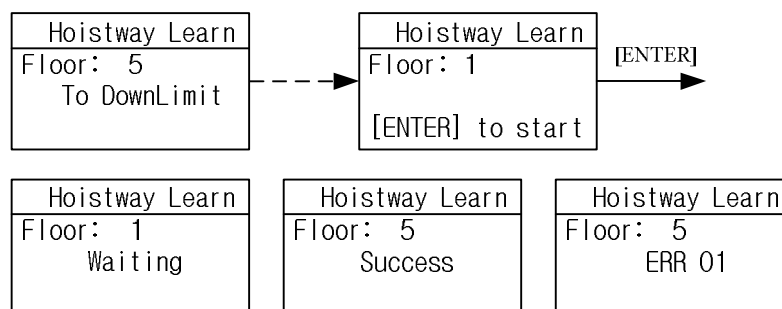


Fig 4.12Hoistway learning

4.7 Motor auto tuning

The motor learning includes two parts: the motor angle learning and motor parameter learning. In the interface of motor learning, press [UP] or [DOWN] key to move the arrow to select related learning information. It can Press [ENTER] key to enter the interface.

In the interface of motor angle learning and motor parameter learning, press [ENTER] to start learning. After learning, the operator display “running” of learning. After learning, it will display “success”. During learning, if there is error, the learning will stop, and it will display “learning finish”, and the status is “error”, and the error code will be displayed.

Note: : Before learning, make sure set the datum in the F5 menu.

In the status of motor no-loading, first carry out motor parameter learning, then carry out the motor angle learning. Press [ESC] key to return the Previous Menu.

The flowchart of the motor learning is shown in Fig 4.13

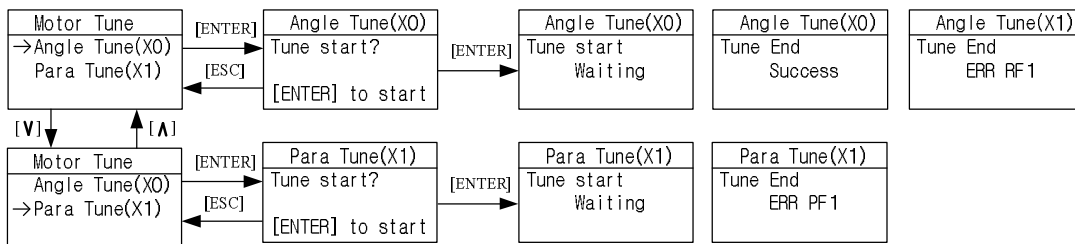


Fig 4.13 motor learning

4.8 Weighing learning

Weighing learning also includes two parts: non-load learning and full-load learning.

In the interface of weighing learning, it can press [UP] or [DOWN] move the arrow to select related learning information. Please press [ENTER] key to enter the interface.

After enter the interface of non-load learning or full-load learning, press [ENTER] to start learning, the operator display the present status. Press [ESC] key to return the Previous Menu.

The flowing chart of weighing learning is shown in 4.14.

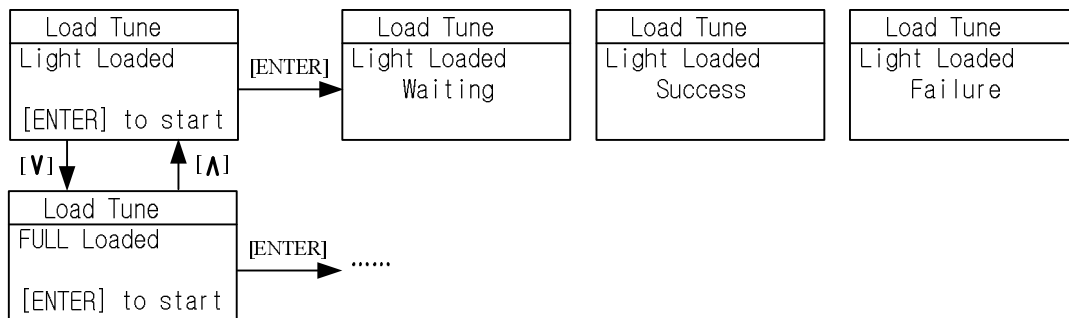


Fig 4.14 weighing learning

4.9 Time setting

In the interface of time setting, the last two, “month”, “day”, “hour”, “Minute” and “second” all can be set, press [LEFT] or [RIGHT] to move the arrow to the right place. Press [UP] or [DOWN] key to change the value of the arrow. After setting, it can press the [ENTER] key to save. Press [ESC] key to return the Previous Menu.

The flowchart of time setting is shown in the Fig 4.15

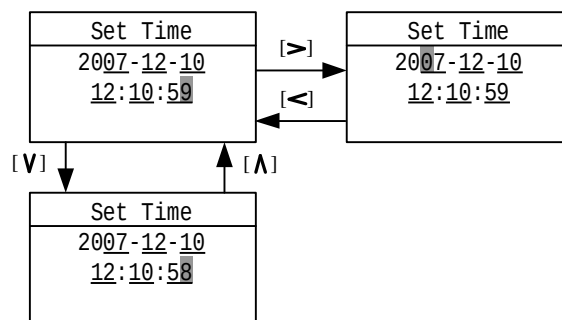


Fig 4.15 Time setting

4.10 Error record check

In the interface of error record, it can press [UP] or [DOWN] key to check the elevator error record and the controller error record. Press [ENTER] to enter the interface of error record, press [UP] or [DOWN] key to check 30 error history record.

Each elevator error record include the error No. time, the status when error happen(such as the floor , running direction, running speed, input and output state, and so on), information like that. It is easy to analysis the reason of elevator error to guide the elevator maintenance.

Each controller error record includes the error No. time, the status when error happen and some important data (such as the speed, the voltage and current, and so on), information like that. It is easy to analysis the reason of controller error to guide the elevator maintenance.

The flowchart of the error record check is shown in the Fig 4.16

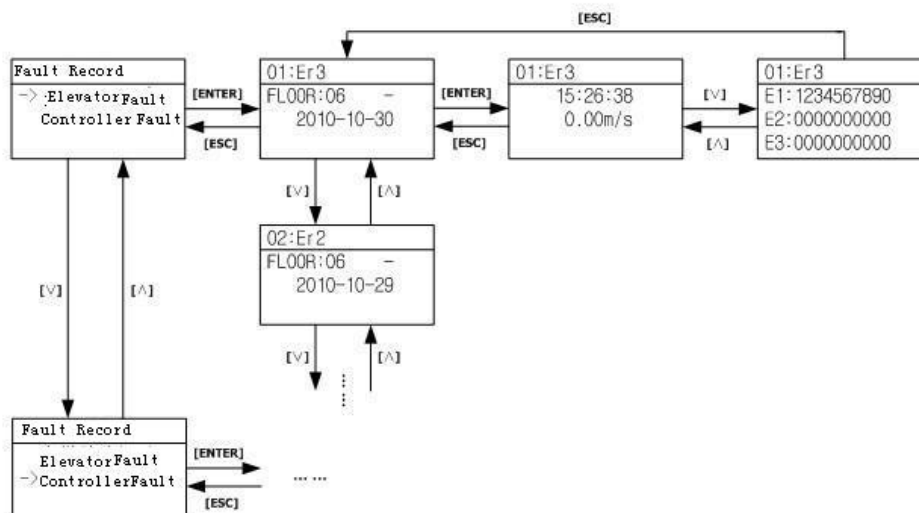


Fig 4.16 error record checking

Press [ESC] key to return the Previous Menu.

4.11 Environment setting

After entering the interface of environment setting, the display language, the parameter visit grade, input password and the shortcut menu of the operator can be set.

The flowchart of environment setting is shown in the Fig 4.17.

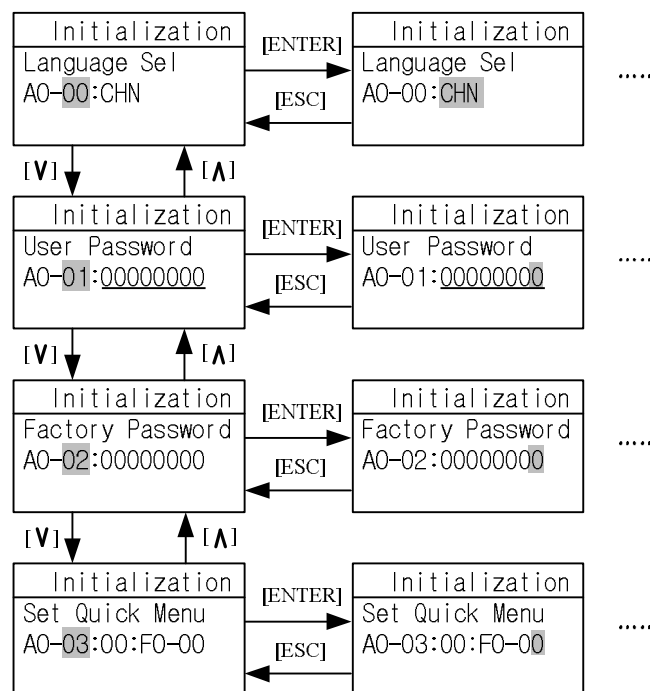


Fig 4.17 environment setting

Display language

The flowchart of the display language is shown in the Fig 4.18.

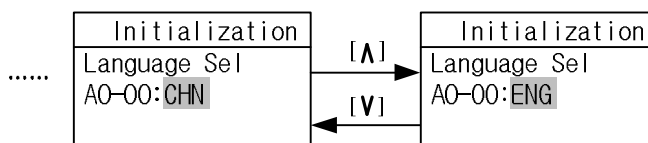


Fig. 4.18 Language selection

Press the [UP] or [DOWN] to select the language, and press [ENTER] key to save the selection.

User password input and setting

In the interface of password, press the [LEFT] or [RIGHT] key to move the arrow to the necessary position, press [UP] or [DOWN] key to increase or decrease the value that arrow is pointing, press the [ENTER] key to input the password. When the password is input wrongly, password error will be displayed, at this time, the password can't be changed. When the password is input correctly, password right will be displayed, then press the [RES] key to enter the password setting interface and reset the password. Resetting the password is similar with the password input. it can Press the [ENTER] key to save the new password.

The flowchart of the password input and change is shown in the Fig 4.19.

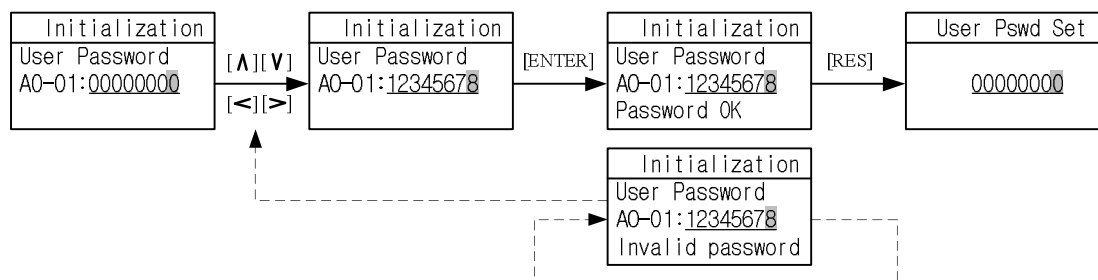


Fig 4.19 user password input and change setting

Factory password input and setting

The flow chart of the factory password input and setting refers to the user password input and setting.

Shortcut menu setting

Setting shortcut menu is to simplify the parameter setting and commissioning of the elevator. Forward the necessary and basic elevator parameter and normal commissioning parameters to the shortcut menu, if so, only setting and modifying the parameter for the menu can finish the elevator running and commissioning, which avoid to carry out the parameter setting and modifying through several menus and modified the course of the parameter setting and commissioning.

The controller is with shortcut menu before it goes out. Generally, the user needn't modify.

Set the view interface in the shortcut menu, press [UP][DOWN] key to view the setting shortcut menu number, press the [ENTER] key to the setting and modifying interface of the shortcut menu, press [LEFT]/[RIGHT] / [UP]/[DOWN] key to move the arrow and modify the value in the position to the setting of signal shortcut menu. After the modifying, it can press [ENTER] key to save the setting. And the system will jump to the next interface.

The flowchart for setting of the shortcut menu is shown in the Fig 4.20

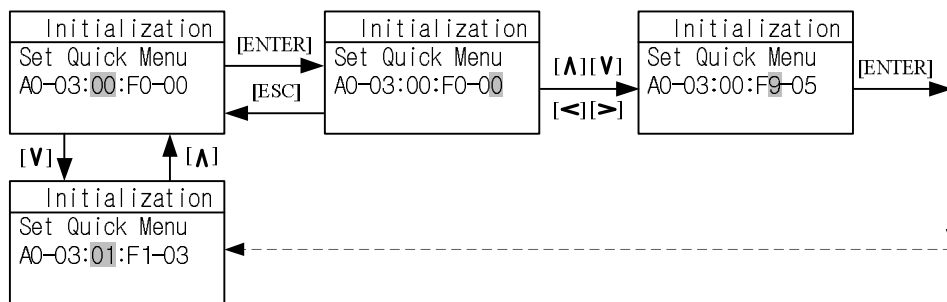


Fig 4.20 setting of the shortcut menu

Press [ESC] key to return the Previous Menu.

4.12 Parameter copy

The function parameter is to simplify the parameter setting and commissioning, especially for several elevators with the same configuration. After the parameter setting and commissioning of one elevator, this function can copy all the parameters (to the digital operator memory) , take the digital operator and connect to other elevators, make use of this function to copy the parameter to the controller, after checking, the elevator will run normally.

In the interface of parameter copy, it can press [UP] or [DOWN] key to move the arrow to carry on the necessary operation, press the [ENTER] key to start operation.

The flowchart of parameter copy is shown in the Fig 4.21.

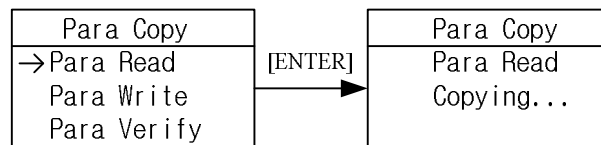


Fig 4.21 parameter copy

After operation, it will display as Fig 4.22.

Para Copy	Para Copy
Para Read	Para Read
Copy ok!	Copy Error

Fig 4.22 parameter copy

Press [ESC] key to return the Previous Menu.

4.13 Restore the factory setting

If necessary, the controller can be restored the factory setting. Press the [ENTER] key to restore the factory setting, the interface will display the status and the result.

The flow chart of restore the factory setting is shown in the Fig 4.23.

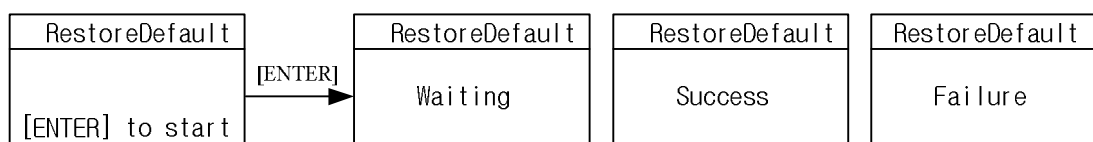


Fig 4.23 restoring the factory setting

Press [ESC] key to return the Previous Menu.

4.14 Blue-ray machine input

Using blue-ray all-in-one machine, if your tractor is also made by blue-ray, you need to input the model number and the information of the encoder of the machine name board to finish the parameter set of the machine.

The interface of the blue-ray machine input is showing in fig4.24(a). The content of inputting consist of three part, separated by ".". The first part is the model number(separated in 4 entering), the middle part is the information of the encoder, the last part is the mold of PG. The detail information is showing in fig4.24(b)

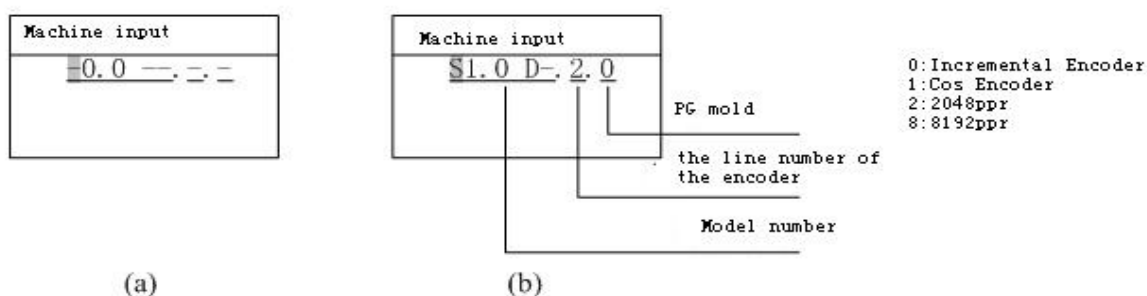


Fig 4.24 the operator interface of the blue-ray machine input

Use [UP] or [DOWN] key to set the content of the cursor position, then press the [ENTER] key to confirm. The cursor is moved to next bit by itself. If the cursor position isn't set, the cursor doesn't move even you press the [ENTER] key (Excluding the 4th number of the machine model, eg. S1.0D-the last number is empty, you can't press the [ENTER] key directly to set the next bit).

The flow diagram of the blue-ray machine inputting is showing in fig.4.25(S0.75D as an example)

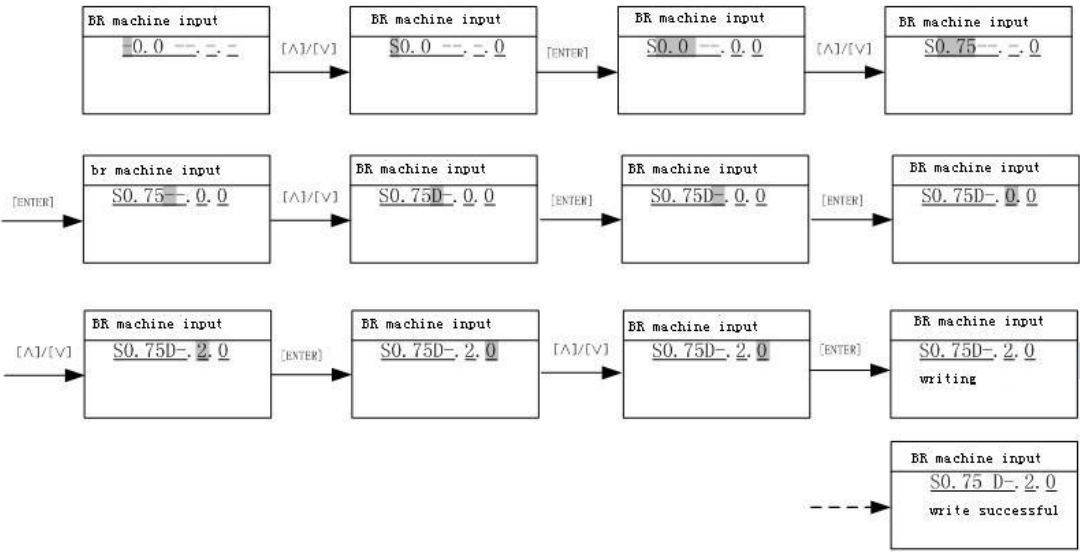


Fig.4.24 the flow diagram of the blue-ray machine inputting

5 parameters

This section describes the contents of the parameter tables.

5.1 Parameters of function classification

Parameters as per different function, classify in group as table 5.1

Table 5.1

Functions symbol	U	F	A	D
Function Name	monitoring parameter	Setup parameter	environment setup	commissioning parameter

5.2 Parameters of Hierarchical structure

Controller's all parameters of hierarchical structure as fig. 5.1

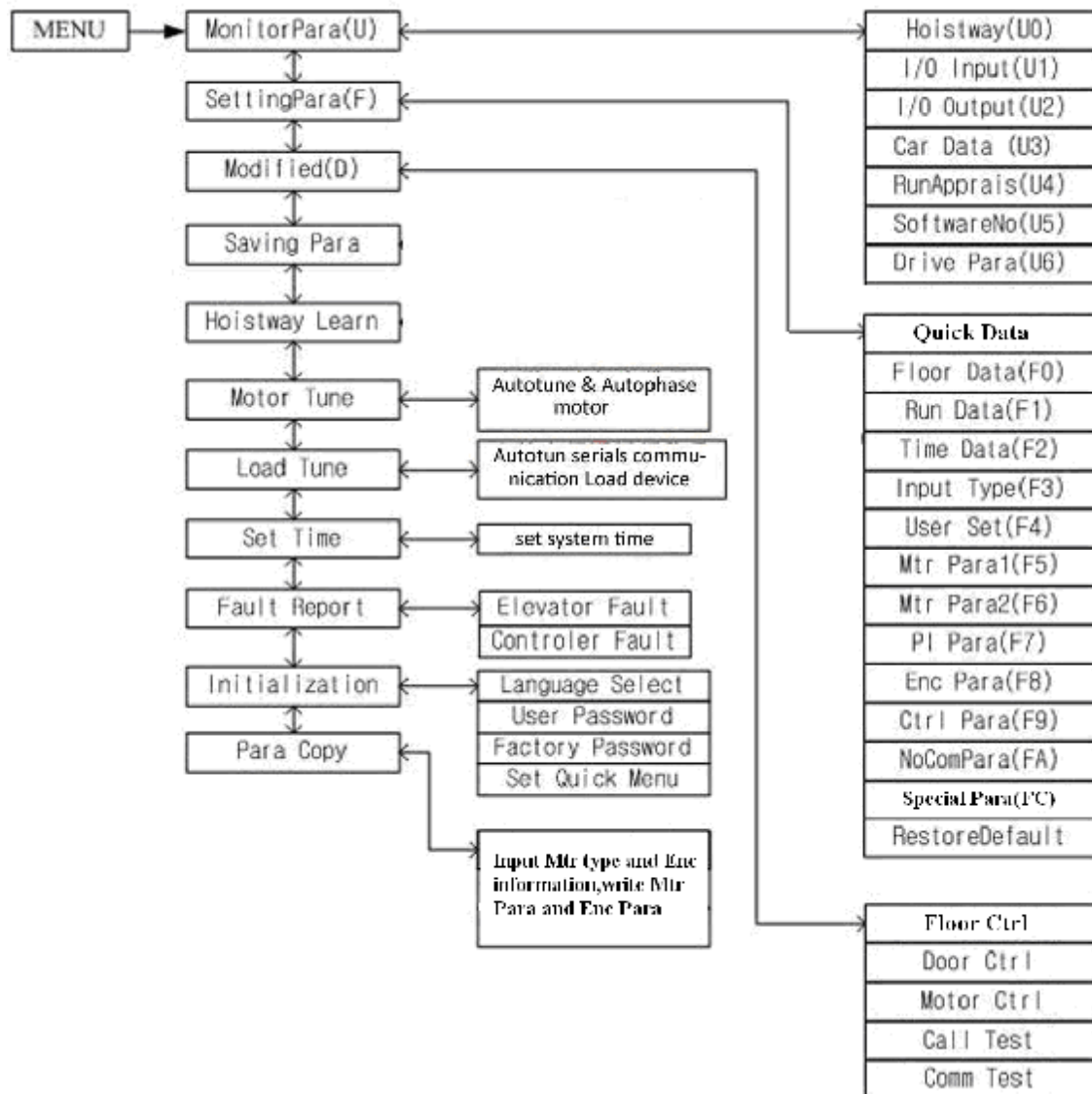


Fig.5.1 Controller's all parameters of hierarchical structure indication

5.3 Monitoring Parameters

5.3.1 Shaft Location Parameters (U0)

Shaft location parameters means the shaft switch location after shaft finished its self learning and the calibration data of each floor location.

After shaft finished self-learning, please check the floor calibration is correct or not, if the calibration shows too high, it is possible the elevator's rated speed setup is higher than the actual value or elevator encoder pulse setup is smaller than the actual pulse. If the calibration shows too low, it is possible that the elevator rated speed setup is lower than the actual value or its encoder pulse setup is larger than the actual pulse.

Shaft location parameter's parameter number, description and content as table 5.2

Table 5.2 U0 monitoring parameter table

Parameter No.	Display	Content	range	unit	Reference page
U0-00	Lower Limit	The location of lower limit under the elevator shaft. the data will auto access after it finishes self learning	0.00 ~ 655.35	m	--
U0-01	Upper Limit	The location of upper limit above the shaft .the data will auto access after it finishes self learning	0.00 ~ 655.35	m	--
U0-02	Lower Slowdown 1	The location of lower slowdown 1 in the shaft . the data will auto access after it finishes self learning	0.00 ~ 655.35	m	--
U0-03	Lower Slowdown 2	The location of lower slowdown 2 in the shaft . the data will auto access after it finishes self learning	0.00 ~ 655.35	m	--
U0-04	Upper Slowdown 1	The location of upper slowdown 1 in the shaft . the data will auto access after it finishes self learning	0.00 ~ 655.35	m	--
U0-05	Upper Slowdown 2	The location of upper slowdown 2 in the shaft . the data will auto access after it finishes self learning	0.00 ~ 655.35	m	--
U0-06 ... U0-69	Floor Data 1...64	The location of Floor Data 1...64 in the shaft . the data will auto access after it finishes self learning	0.00 ~ 655.35	m	--

5.3.2 Monitoring parameter of I/O Status, cabin signal and interference appraise (U1~U5)

U1-U5 monitoring parameter Ref. NO., description and content as table 5.3.

Table 5.3 U1~U5 monitoring parameter table

Parameter No.	Display	Content	range	unit	Reference page
U1-00	Input Data	Controller input data show as the decimal system. it will be turned into binary system so that it can show the logical status of the input port.	--	--	3-6
U1-01	Input Bin	Input port data show as the binary system .This parameter each data correspond to each logical status of the input port.	--	--	3-6
U1-02	Input App	"ON/OFF" Each line correspond to one input port, "On/Off" states the current status of the port, and N states the signal app to input level."10" states the good working environment, input port no interference, App data is near "0",it states the larger interference.	--	--	3-6
U2-00	Output Data	Show the output port Y0-Y15 current status. if it has the output port, then it has the indication of corresponding port. Port without output will be hided.	--	--	3-7
U3-00	Car Input Data	It wil show input signal status of the cabin. The cabin signal monitoring limiting surface show the status of the current cabin input signal. if it has the input port, then it will has the show of the corresponding port. and the port without input will be hided.	--	--	5-5
U4-00	Run Times	It will show the accumulated running times of the elevator. it adopts 10 digital of decimal system as indication	--	times	--
U4-01	Run Hours	It will show the accumulated running time. it adopts the 10 digital of decimal system as indication.	--	hours	--
U4-04	SendApp1	SendApp1,if this data is larger, then it will send more mistakes.	--	--	--
U4-05	ReceiveApp2	ReceiveApp2,if this data is larger, then it will receive more mistakes.	--	--	--
U4-06	Interfer Apprais	Appraise the value of interference strength is big or small. The big value states strong interference, otherwise it states weak."0" states no interference and good earthing condition.	--	--	--
U4-07	Encoder Apprais	The interference degree of encoder appraise. When the elevator runs steadily, the data larger states the encoder signal weakly.	--	--	--

Table 5.3 U1-U5 Monitoring parameter table (To be continued)

Parameter No.	Display	Content	range	unit	Reference page
U4-09	Lock Timer	The current stop timer	--	--	--
U5-00	CtrlSoftWare NO	It will show the elevator control software No. information. It will supply the current software version No. when factory maintaining and uprising.	--	--	--
U5-01	Drive CodeVer	It will show the drive code version information. it will supply the current software version No. when factory maintaining and uprising.	--	--	--
U5-02	Cpld Edition	It will show the Cpld Edition information. it will supply the current software version No. when factory maintaining and uprising.	--	--	--



As per the meaning of input status U1-01 AND OUTPUT STATUS U2-00, please refer to the control terminal definition table3.3 (3-6page)
Cabin signal symbol definition and content is as table5.4 (5-5page).

Table5.4 U3-00cabin signal's definition and content tale

Cabin signal	Symbol signal	Cabin terminal No.	content
C00	IGM1	J3-4	door close 1 input
C01	IKM1	J2-4	door open 1 input
C02	IGM2	J5-4	door close 2 input
C03	IKM2	J4-4	door open 2 input
C04	GMV2	J10-6	door close 2 input
C05	KMV2	J10-5	door close 2 input
C06	GMV1	J9-3	door close limiter 1 input
C07	KMV1	J9-2	door open limiter 1 input
C08	SZY	J10-1	professional input
C09	IGMYS	J6-4	door open prolong input
C10	SZH	J9-10	drive input
C11	--	--	(standby)
C12	SZS	J10-2	Bypass input
C13	MZ	J9-6	loading input
C14	QZ	J9-8	unloading input
C15	CZ	J9-5	overloading input
C16	KZ (50%)	J9-9	50%overloading(unloading) input
C17	KAB2	J9-7	Door safety edge 2
C18	KAB1	J9-4	Door safety edge 1

5.3.3 Drive monitoring parameter

Drive monitoring parameter No., name and content as table 5.5

Table5.5 U6Drive monitoring parameter table

Parameter No.	Display	Content	range	unit	Reference page
U6-00	Power	Power class	--	kW	--
U6-01	Ref Speed	Ref Speed	--	RPM	--
U6-02	Feedback Speed	Feedback Speed	--	RPM	--
U6-03	Load	the percentage of overloading	--	%	--
U6-04	DC Voltage	DC LINK VOLTAGE	--	V	--
U6-05	Output Current	Output Current	--	A	--
U6-06	Temperature	temperature in VVVF	--	°C	--
U6-07	Output Torque	Output Torque	--	N·M	--

5.4 Parameters of Setup functions instruction

5.4.1 Parameter of Building setup table

The Parameters No. of Building setup (F0) , description and content as table 5.8

Table5.8 parameter of Building setup table

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F0-00	Total Floor	Total Floor	2~64	6	--	N	--
F0-01	Homing Floor	Without HOP and inner selection the elevator will return this floor.	1~Total floor	1	--	N	6-8
F0-02	Fire Floor	When the fireman switch closes, the elevator will return to the floor automatically in the extinguishing status.	1 ~ Total floor	1	--	N	6-9
F0-03	Parking Floor	When elevator closes the electric lock in the process of running, the elevator hauls at the lock stair level.	1 ~ Total floor	1	--	N	6-8
F0-04	VIP Floor	VIP floor setup	1 ~ Total floor	1	--	N	--
F0-05 ... F0-68	set Indication 1~64	Set indication 1-64.	---	1 ... 64	--	N	6-13



“N” states the parameter cannot be changed in the process of running.”Y” states the parameter can be changed in the process of running. It states the same in the following table.

5.4.2 Parameter of running setup (F1)

Parameter of running setup No.F1, Description and content as table 5.9.

Table5.9 Parameter of running setup table

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F1-00	Car Speed	Elevator speed of motor rated speed. To calculate as per motor rated revolution, traction ratio ,decelerating ratio and traction sheave diameter	0~4.0	1.6	m/s	N	6-1
F1-01	Motor Speed	Motor rated rev. and set the parameter as per motor name brand	1~9999	1450	RPM	N	6-1
F1-02	Pulses	Encoder received by main board, pulse per revolution	0~9999	1024	PPR	N	6-1
F1-03	Insp Speed	the running speed of inspection cannot be larger than 0.6m/s as per relative standards and regulations	0~0.6	0.3	m/s	Y	6-1
F1-04	Start Speed	While the resistance of the traction is large, the starting speed can be arises properly. The speed setting "0", it is not workable.	0~0.2	0.00	m/s	Y	--
F1-05	Leveling Speed	When the elevator has problems and cannot stayed at the door area, if it conform to this running condition, the elevator can be leveling by itself to the door area	0.01 ~ 0.6	0.3	m/s	Y	6-1
F1-06	Least Speed	the steadiest value of curled at lowest speed	0~1.0	1.0	m/s	N	6-2
F1-07	Open Door Speed	the allowance speed at the time of the elevator open the door in advance	0~0.3	0.15	m/s	N	Appendix 2 -III
F1-08	Relevelst Speed	The speed protection value of re-leveling to set the elevator's protection speed of re-leveling , upon the re-leveling speed surpass this protection value, re-leveling stop will produce 03# protection	0~0.3	0.3	m/s	N	Appendix 2 -III
F1-09	Relevelrun Speed	the running speed of elevator re-leveling	0~0.10	0.05	m/s	N	Appendix 2 -III
F1-10	Acceleration B1	Acceleration B1,it means the index of speed verification of curve is smaller, then this acceleration rang of starting can be slower and you can feel more stable within this range	0.1~1.0	0.6	m/s ²	N	6-2

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F1-11	Deceleration B2	Deceleration B2,It means the index of speed verification of curve is smaller, then this deceleration range of braking can be slower and you can feel more stable within this range	0.1~1.0	0.6	m/s ²	N	6-2
F1-12	S Curve P1	S Curve P1,It means the increasing changed speed index of curve is smaller, hen this beginning range of starting can be slower, and you can feel stable within this range	0.1~1.0	0.7	m/s ³	N	6-2
F1-13	S Curve P2	S Curve P2,It means the decreasing changed speed index of curve is smaller, then this end range of starting is slower and you can feel stable within this range	0.1~1.0	0.7	m/s ³	N	6-2
F1-14	S Curve P3	S Curve P3,It means the decreasing changed speed index of curve is smaller, then this start range of braking is slower and you can feel stable within this range	0.1~1.0	0.7	m/s ³	N	6-2
F1-15	S Curve P4	S Curve P4,It means the decreasing changed speed index of curve is smaller, then this end range of braking is slower and you can feel stable within this range	0.1~1.0	0.7	m/s ³	N	6-2
F1-16	Zero Speed	When revolution of motor is less than this value, the system will regard this speed of elevator as zero and it will output the signal of no power	0~10	5	RPM	Y	6-4
F1-17	Leveling Adj	Adjust difference of up/down leveling	0~100	50	mm	N	6-13
F1-18	Load Adj	Generally this parameters are used in synchronous controlling system ,the weight of changeable weight of steel ropes of each floor can be adjusted within this range	0~20	0	--	Y	6-12



When elevator rated speed and monolayer running speed is lower than 0.5m/sec, please properly decrease the acceleration B1, deceleration B2; otherwise it will affect the system calculation of rated speed. When the user modifies the two speed value out of limit, the system will return to the data before modification.



Acceleration B1, deceleration B2, S curve P1, P2, P3, P4 co-work to decide the verification tend of rated curve. They will restraint each other and not allow to setup freely. When the modification value is out of limit, the value will recover to the former data as before.

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F1-21	Drive Mode	The selection of driving mode ,when setting “1”,the driver close the door manually; when setting “3”,the elevator automatically tests and works ,other setup is nonsense.	0~9	0	--	N	--
F1-22	Two Door Mode	Setup two door mode, you can go outside from front or back door at the same floor, as per customer's requirement, set from mode“0” to mode“5”.	0~5	0	--	N	6-9 6-10
F1-23	Fire Mode	There are three fire modes to be chosen from: 1.Mode“0” :after returning to fire protection floor, then the elevator run the fire fighting function 2.Mode “1”: after returning to fire protection floor, then the elevator stop running this function 3. Mode “2”: after returning to fire protection floor, then the fireman transfer switch to check running or not.	0~2	0	--	N	6-9
F1-24	Parallel No.	While Parallel enable setting YES, Set parallel no. as zero or one, and when group controlling, set zero or seven.	0~7	0	--	N	6-12
F1-25	Twins Control	Elevator's twins control of on/off 1: On 0:OFF	0/1	0	--	Y	6-12
F1-26	Group Control	Group control of on/off 1:ON 0:OFF	0/1	0	--	Y	6-12
F1-27	Far Monitor	Far monitor Mode of On/Off 1: On 0: Off	0/1	0	--	Y	--
Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F1-28	Auto Parking	Auto parking of on/off 1:ON 0:OFF	0/1	0	--	Y	6-8
F1-29	Load Enable	Load enable of ON/OFF 1:ON 0: OFF	0/1	0	--	Y	6-12
F1-30	Open Delay Able	Open delay able of ON/OFF,1:ON 0:OFF	0/1	0	--	Y	6-8
F1-31	Brake Feedback	Function enable 1: open 2: close	0/1	0	--	Y	6-4
F1-32	Rerun Password	Remove the current rerun password	0~9999	0	--	N	--

5.4.3 Parameter of time setup (F2)

Parameter of time setup F2's parameter number, description and content as table 5.10

Table 5.10 Parameter table of time setup

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F2-00	Brake ON Time	Gate opening and made time run curve, gate opening times in advance to improve the starting point of comfort, allowing the system to adapt to different brake traction machine of the open time.	0.00~9.99	0.50	s	Y	6-4
F2-01	Brake OFF Time	Brake with the withdrawal of driving time so that it will enable the system to wait for the next gate after the tractor wheel brake to hold fast to remove the drive after the output, so as not to open the door when the stall torque converter to withdraw prematurely, slide car, the impact of parking comfort.	0.00~9.99	0.50	s	Y	6-4
F2-02	Insp Brake Time	The delaying time of inspection mode before braking off	0.00~9.99	0.05	s	Y	6-4
F2-03	Run Input Time	while main board receives drive running signal, the brake is on after delaying	0.00~9.99	0	s	Y	6-4
F2-04	Zero Time	When system inspected the post phoned time after zero speed, adjust the parameter properly, after the zero speed is stable, then brake off to get zero speed off	0~9.99	0.30	s	Y	6-4
F2-05	Open Door Time	The elevator automatically open the door when stopping at one floor, after the door is opened safely the elevator will automatically close the door within this instructed time	0~999	3	s	Y	6-7

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F2-06	Open Delay Time	After open delay function is enable, Press the open delay button that can prolong the time of door opening	0~999	30	s	Y	6-7
F2-07	Homing Time	The waiting time before the elevator return to homing without hall call and car hall, while setting "0", there is no homing function.	0~999	60	s	Y	6-8
F2-08	Door Run Time	1.After sending the opening or closing order, it will stay the time 2, when there are not opening or closing door limit of door system, it will stay sometime of ON/OFF relay. 3. When there are opening/closing door limit of door system, the value should be longer 1S or so than the actual time of opening or closing.	0~999	5	s	Y	6-7
F2-09	Beep Delay Time	After the operated elevator changed speed to instructed floor, it can delay this instructed output time of signal to instructed floor, and it will make system of destination calling machine or bell broadcast later.	0.00~9.99	0.15	s	Y	--
F2-10	Enable Delay	It means the delaying time before sending enable signal, after cancel the inverter signal, then the enable signal cancelled. Within this time, the elevator will decrease the output current in order to avoid the current noise.	0.00~9.99	0	s	Y	6-4
F2-11	Lamp Off Time	In the condition of auto running, the elevator will power off the illumination through the control panel without car call and hall car within this time.	0~999	15	min	Y	--

F2-12	Over Time	In order to prevent wire rope skidding or elevator car stuck harm on the system should each express elevator to stop running from the start-up time is limited. This parameter is set a time limit for this value, if the elevator is running for more than a single value, the system will immediately stop access to the protection of the state, and only re-power the system in order to withdraw from the protection of state	0~999	45	s	Y	--
F2-13	Smooth Start Time	It means the sustaining time of smooth start speed	0.00~9.99	0	s	Y	6-4
F2-14 F2-15	Start Time	Setting the starting time. System will automatically make the elevator start(electrical lock :ON)	00:00 ... 23:59	00:00	h:m	Y	6-8
F2-16 F2-17	Stop Time	Setting the stopping time. System will automatically make the elevator stop (electrical OFF).The time of starting time and stopping time is same, this function is unavailable.	00:00 ... 23:59	00:00	h:m	Y	6-8
F2-18 F2-19	Start Time1	Setting the starting time. System will automatically make the elevator start(electrical lock :ON)	00:00 ... 23:59	00:00	h:m	Y	
Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F2-20 F2-21	Stop Time1	Setting the stopping time. System will automatically make the elevator stop (electrical OFF).The time of starting time and stopping time is same, this function is unavailable.	00:00 ... 23:59	00:00	h:m	Y	



The automatic switch elevator of F2-14, F2-15 F2-16, F2-17 was settled separately as per hours and minutes. Please make the operation just according to the indication of the manipulate exerciser

5.4.4 Parameter of Input type setup (F3)

Parameter of input type setup F3's parameter number description and content as table 5.11

Table 5.11 Parameter table of input type setup

Parameter No.	Display	Content	Setting Range	Factory setting	Change during operation	Reference page
F3-00	Input Type	Setting the input type of main control panel. Each bit corresponds to one terminal. Setting level status of main board input port, Available when it is ON, available when it's OFF.	0~4294967295	3974102631	N	6-10 6-11
F3-01	Car Input Type	Setting the input type of cabin. Each bit corresponds to one terminal. Available when it is On, available when it is Off.	0~4294967295	4294573839	N	6-10 6-11
F3-02	Input select 1	X19 Input selection	0~32	19	N	
F3-03	Input select 2	X22 Input selection	0~32	22	N	
F3-04	Input select 3	X23 Input selection	0~32	23	N	
F3-05	Input select 4	X24 Input selection	0~32	24	N	
F3-06	Input select 5	X25 Input selection	0~32	25	N	
F3-07	Output select 1	Y0 Output selection	0~32	0	N	
Parameter No.	Display	Content	Setting Range	Factory setting	Change during operation	Reference page
F3-08	Output select 2	Y11 Output selection	0~32	11	N	
F3-09	Output select 3	Spare Output selection	0~32	11	N	



When using X22 and X23 as multifunctional input, please make sure the re-leveling device is not used.

5.4.5 Parameters of Service setup (F4)

Parameters of service setup F4's parameter number, description and content as table 5.12

Table 5.12 Parameter table of service setup

Parameter No.	Display	Content	Setting Range	Factory setting	Change during operation	Reference page
F4-00	Set Stop Floor1	Setting the elevator will stop corresponding floor or not as per each bit (1-32 floors)	0~ 429496 7295	429496 7295	Y	6-12
F4-01	Set Stop Floor2	Setting the elevator will stop the corresponding floor or not as per each bit (33-64floors)	0~ 429496 7295	429496 7295	Y	6-12
F4-02	TIM Stop Floor1	Setting the elevator will stop corresponding floor or not as per each bit in the customer appointed period time. (1-32floors)	0~ 429496 7295	0	Y	6-12
F4-03	TIM Stop Floor2	Setting the elevator will stop corresponding floor or not as per each bit in the customer appointed period time (33~64 floor)	0~ 429496 7295	0	Y	6-12
F4-04	Door Select A1	Setting the elevator will be open or not at front door of corresponding floor as per each bit .(ON means the front door of this floor open, Off means the door of the floor close)	0~ 429496 7295	429496 7295(1~32floor)	Y	6-10
Parameter No.	Display	Content	Setting Range	Factory setting	Change during operation	Reference page
F4-05	Door Select B1	Setting the elevator will be open or not at front door of corresponding floor as per each bit.(ON means the front door of this floor open, Off means the door of the floor close)	0~ 429496 7295	429496 7295(1~32floor)	Y	6-10
F4-06	Function Select	Setting corresponding function as per each bit is open or not.(ON means open, OFF means close)	0~ 429496 7295	4	Y	6-13
F4-07	Function Select2	Setting corresponding function as per each bit is open or not.(ON means open, OFF means close)	0~ 429496 7295	0	Y	6-13

5.4.5 Parameter of motor setup (F5~F6)

Parameter of motor setups parameter number, description and content as table 5.13.

Table5.13 Parameter table of motor setup

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
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F5-00	Motor Type	Setting motor type(0:synchronization outer rotor,1:asynchronism,2:synchronization inner rotor)	0~2	0	--	N	6-14
F5-01	Poles	Moto poles set as per motor nameplate	1~99	20	--	N	6-14
F5-02	Sync Freq	Motor synchronization set as per motor nameplate	0.001~50	16	Hz	N	6-14
F5-03	Rated Power	Motor rated power set as per motor nameplate	1~50	6.7	kW	N	6-14
F5-04	Rated Speed	Motor rated speed set as per motor nameplate	1~1999	96	RPM	N	6-14
F5-05	V IN	Motor VIN set as per motor nameplate	1~380	280	V	N	6-14
F5-06	L_phase	L-phase(self learning got or filling in by hand)	self learning /set as per brand		mH	N	6-14
F5-07	R_phase	R-phase(self learning got or filling in by hand)	Self learning /set as per brand		Ω	N	6-14
F5-08	Rated FLA	Rated FLA set as per motor nameplate	0~99.999		A	N	6-14
F5-09	NO-Load Current	No-load excitation current of a synchronism motor	0.1~50	0	A	N	6-14
F5-10	Rated Slip	asynchronies' motor rated slip as per nameplate	0.1~10	1.3	HZ	N	6-14

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F6-00	Carrier Freq	set carrier Freq of controller	5~15	8	kHz	N	--
F6-02	SpeedZoom	Speed Zoom(lower elevator's actual running speed)	0~100	100	%	Y	6-14
F6-03	DirSel	Select motor running direction (0: while motor rotated by clockwise, the cabin goes down,1: while motor rotated by counter clockwise, the cabin goes up)	0/1	0	--	--	6-14
F6-04	Kp	Speed loop perpetration enhanced (No use subsection PI, then it will take effect all the way.	0~65535	700	--	--	6-15
F6-05	KI	Speed loop integration (No use subsection PI, Then it will take effect all the way)	0~65535	260	--	--	6-15

5.4.7 Parameters of Segmentation PI (F7)5.4.7.

Setting segmentation PI's parameter number, description and content as table 5.14

Table5.14 Segmentation PI setting parameter table

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F7-00	PI Mul Enable	Multiple PI parameters enable,1 a vailable,0 unavailable	0/1	0	--	N	6-15 6-16
F7-01	PI1 Range	PI 1working scale (start-middle speed running PI ALT frequency)	0~ Rated frequency	0	Hz	Y	6-15 6-16
F7-02	PI2 Range	PI 2working scale(middle speed running-high speed running PI ALT frequency)	0~ Rated frequency	0	Hz	Y	6-15 6-16
F7-04	PI4 Range	PI 4 working scale(standby)	0~ Rated frequency	0	Hz	Y	6-15 6-16
F7-05	Kp1	PI working scale ,Kp1	0~2000	700	--	Y	6-15 6-16
F7-06	Kx1	PI working scale,KX1	0~2000	260	--	Y	6-15 6-16
F7-07	Kp2	PI working scale,Kp2	0~2000	300	--	Y	6-15 6-16
F7-08	Kx2	PI working scale,Kx2	0~2000	0	--	Y	6-15 6-16
F7-09	Kp3	PI working scale,Kp3	0~2000	0	--	Y	6-15 6-16
F7-10	Kx3	PI working scale,Kx3	0~2000	0	--	Y	6-15 6-16
F7-11	Kp4	PI working scale,Kp4	0~2000	700	--	Y	6-15 6-16
F7-12	Kx4	PI working scale,Kx4	0~2000	260	--	Y	6-15 6-16

5.4.8 Parameter of encoder setup (F8)

Parameter of encoder setup F8's Parameter number, description and content as table 5.15.

Table 5.15 Parameter table of encoder setup

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F8-00	Encoder PPR	the pulse count per revolution of encoder	100~8192	8192	--	N	6-18
F8-02	PGType	PG Type (0:incremental encoder, 1: sin&cos encoder)	0/1	0	--	N	6-18

5.4.9 Parameter of control setup (F9)

Parameter of control setup F9's parameter number, description and content as table 5.16.

Table 5.16 Parameter table of control setup

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
F9-00	Max Torq Comp	Max Torque Comp	0~100%	0	%	N	6-17
F9-01	SPDSourceSel	Speed given source selection 0: simulation 1: Multi-segment 2: interior 3: manipulator	0~3	2		N	6-14
F9-03	Spderr Set	Spderr Set(100% correspond to motor rated revolution)	0~100	5	%	Y	--
F9-11	Load Comp Enable	Load Comp Enabl,1 enable,0 unavailable	0/1	1	--	N	6-17
F9-13	Load Source Sel	Load source(0:SJT Weighing,1:-10-10V weighing,2:0-10V weighing)	0/1/2	0	--	N	6-17
F9-19	UP Comp Bias	UP Comp Bias	-100~100	0	--	Y	--
F9-20	DOWN Comp Bias	DOWN Comp Bias	-100~100	0	--	Y	--
F9-21	FULL Comp Pro	FULL Comp PRO	0~200	100	--	Y	--

5.4.10 Parameter of Unloaded compensation (FA)

Parameter of unload compensation FA's parameter number, description and content as table 5.17.

Table 5.17 Parameter table of control setup of unload compensation

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
FA-00	Start KP	Start KP without compensation	0~5000 0	30	--	N	6-19
FA-01	Start KI	Start KI without compensation	0~5000 0	750	--	N	6-19
FA-08	PLKP	PLKP Without compensation	1~6500	3600	--	N	6-19

FA-09	PLTime	PL Time without compensation	1~500	900	ms	N	6-19
FA-11	IntegralGain	Integral Gain	0~65535	800	--	N	6-19
FA-12	FluxGain	Flux Gain	0~65535	125	--	N	6-19

5.4.11 Special Parameter (FC)

Special parameter(FC) is mapped a part of factory parameter(FX),users can access by password.During these parameters,FC-00~FC-06 only can be checked,but they can't be set.Special parameter (FC)'s parameter number, description and content as table 5.18.

Table 5.18 Special Parameter table

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
FC-00	Zpulse_Init	The result of angle self-learning.FX-00	0~3277	--	--	N	--
FC-07	Kplreg	Kplreg. Be cautions to modify.FX-07	0~65535	2500	--	N	--
FC-08	Kxlreg	Kxlreg.Be cautions to modify.FX-08	0~65535	550	--	N	--
FC-13	Auto TuneModeSel	The selection of sine and cosine PG angle self-learning.0:Rotary self-learning.1:Stationary self-learning.FX-20	0/1	0	--	N	--
FC-14	N Temp Alarm Ena	N temp alarm ena.1:When the temperature below negative 15 degree, the bell alarms.0:When the temperature below negative 15 degree, the bell doesn't alarm.FX-21	0/1	1	--	N	--
FC-15	InitTuneEnable	When using sine and cosine PG card, whether we need CD signal to posit electricity.0:Yes.1:No(Just above SPG-V33 can be set to 1)	0/1	1		N	FX-24
FC-16	CD DirSel	When FX-24 is set to 1,it can work, When AB signal and CD signal are same-phase, it is set to 0.When AB signal and CD signal are opposite-phase, it is set to 1.	0/1	0		N	FX-25

5.4.12 Parameter of Environmental setup (A)

Parameter of environmental setup (A)'s parameter number, name and content as table 5.19.

Parameter Table of environmental setup

Parameter No.	Display	Content	Setting Range	Factory setting	unit	Change during operation	Reference page
A0-00	Language Sel	Language selection of operator panel	--	Chinese	--	Y	--
A0-01	User Password	Input/setting user password	000000 ~ 999999	000000	--	Y	--
A0-02	Factory Password	Input/setting user password	000000 ~ 999999	111111	--	Y	--
A0-04	Contrast	Setting the LCD contrast	0~10	5	--	N	--

6 parameters description

6.1 Elevator running speed

6.1.1 Elevator rated speed, motor rated RPM and pulses of encoder setting

1. Elevator rated speed (F1-00) can be calculated out by motor RPM, traction ratio, reduction ratio and traction sheave diameter.



Elevator rated speed is for calculating the value of ratio between motor RPM and elevator speed, changing this parameter can't change the actual running speed of elevator; if you want to modify the running speed of elevator, please modify the parameter F6-02 (Speed Zoom)

2. Motor RPM (F1-01), it is the RPM of traction motor under rated speed of elevator.
3. Pulse of encoder (F1-02), it is the rated pulse of encoder's one circle rotating for hoistway counting while the elevator is running.
4. Motor rated rev and elevator rated speed much follow the condition below:

$$\text{Elevator rated speed} = \frac{\text{Motor RPM} \times \frac{\text{traction sheave diameter}}{60 \times 1000} \times 3.14 \times \text{speed reduction ratio}}{\text{traction ratio}}$$

For example: Motor RPM 1370 rev/min., traction sheave diameter 590mm, speed reduction ratio: 2/53, traction ratio 1/1, then:

$$\text{elevator rated speed} = \frac{1370 \times 590 \times 3.14 \times 2}{60 \times 1000 \times 1 \times 53} = 1.6\text{m/s}$$

After setting for elevator rated speed, motor rated rev, encoder pulse number, system record the relationship between traction ratio, speed reduction ratio, traction sheave diameter and relationship between shaft counting pulse and running distance, if the setting is incorrect, the following phenomenon's will happen:

1. The speed indication on elevator operating panel or commissioning software is different with the elevator actual running speed.
- For example, if encoder pulse or motor rated rev. is set by a smaller value, speed indicated on the display will be higher than the actual speed, after shaft-learning, the floor height learned by the system will be bigger than actual floor height.
2. Because the floor height learned by the system is different with the actual floor height, the elevator deceleration curve will be steeper than the setting, maybe elevator can't landing sometimes (running beyond the leveling zone).

6.1.2 Inspection Running Speed

Under inspection mode, elevator runs in slow inspection speed (F1-03), as per international standard, it shouldn't be more than 0.6m/s. After elevator reach the top or bottom terminal, the running speed will reduced to half.

If the bottom terminal (X8) or top terminal (X7) is valid, elevator speed will get down to 1/2 inspection speed to prevent it exceeding the limit.

When up jog or down dog is over, speed given will be instantly 0, no deceleration curves.

6.1.3 Self-Rescue Speed

Under the two situations below, elevator will run in Self-rescue speed (F1-05):

1. Because of fault or other reasons, elevator stops at the no-leveling zone, after elevator restores, it runs to the nearest floor and open to evacuate the passengers;

2. When there is a hoistway counting error (ER14, ER18), elevator will run at self-rescue speed to the bottom floor to do adjustment; After elevator reach the bottom terminal, elevator will run at 1/2 of this speed to the bottom leveling position.

6.1.4 Least speed

Least speed curve (F1-06), limit the least speed curve in the system. System will choose the speed as per the running distance, but least speed is the min. speed elevator can reach. If at the jobsite, some floors' distance is very small, please reduce its value to realize the normal running for a single floor. If its value is too high, the elevator can't finish the running within a small traveling, it will stops at the next floor.

6.2 Normal Running Curve

Elevator can give out several optimal running curves as per rated speed setting to fit different speed and different distance running. But the least speed running curve is limited by F1-06.

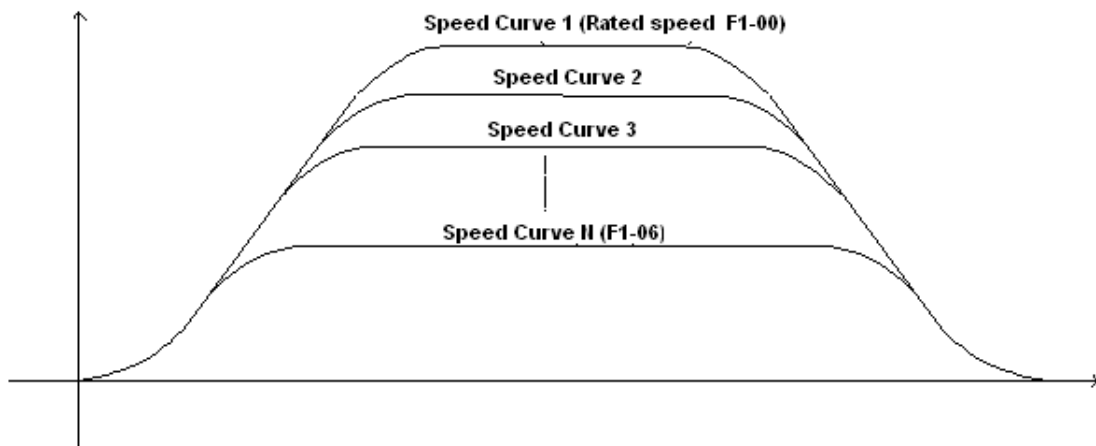


Fig. 6.1 Running Curve

Acceleration, deceleration and Acce/Dece, S curve for normal running curve are set by the following parameters:

- F1-10 (Acceleration slope B1)
- F1-11 (Deceleration slope B2)
- F1-12 (S curve P1)
- F1-13 (S curve P2)
- F1-14 (S curve P3)
- F1-15 (S curve P4)

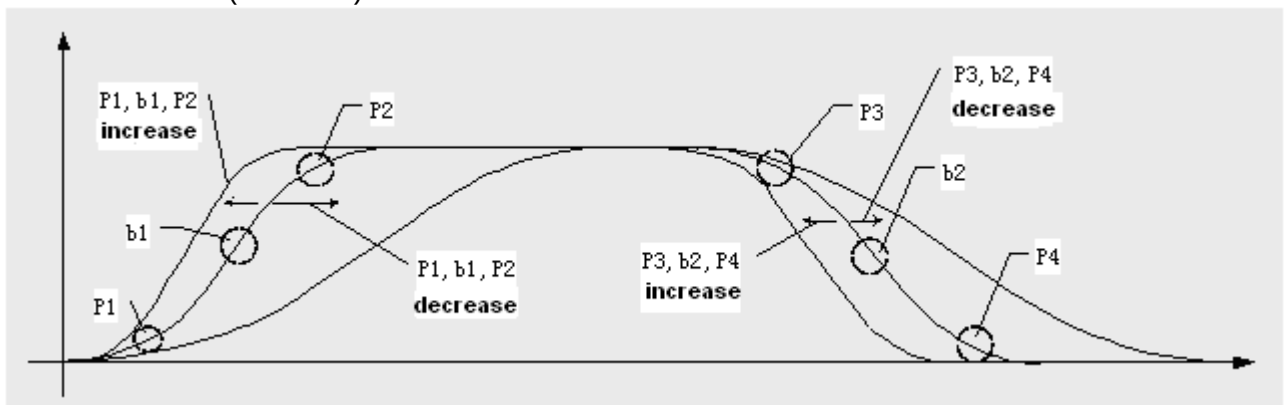


Fig. 6.2 Parameter Adjustments for Speed Curve

- ◇ Elevator running comfort can be got by adjustment of the above 6 parameters. (Comfort is only related to the running curve also related to driving control parameters.)
The relationship between b1, b2, P1, P2, P3, P4 and running curve refer to the Fig. 6.2.
- ◇ When increasing them, the corresponding curve will be steeper. When decreasing them, the corresponding curve will be smoother. Adjusting the above six parameters properly can reach good comfort of elevator running, and meet the requirement of standard.

6.3 Timing diagram of elevator running

6.3.1 Timing diagram of normal running

Timing diagram of normal running Fig. 6.3

If smooth running speed for start (F1-04) is "0", smooth running for start function will be invalid, smooth running time for start (F2-13) will be invalid too.

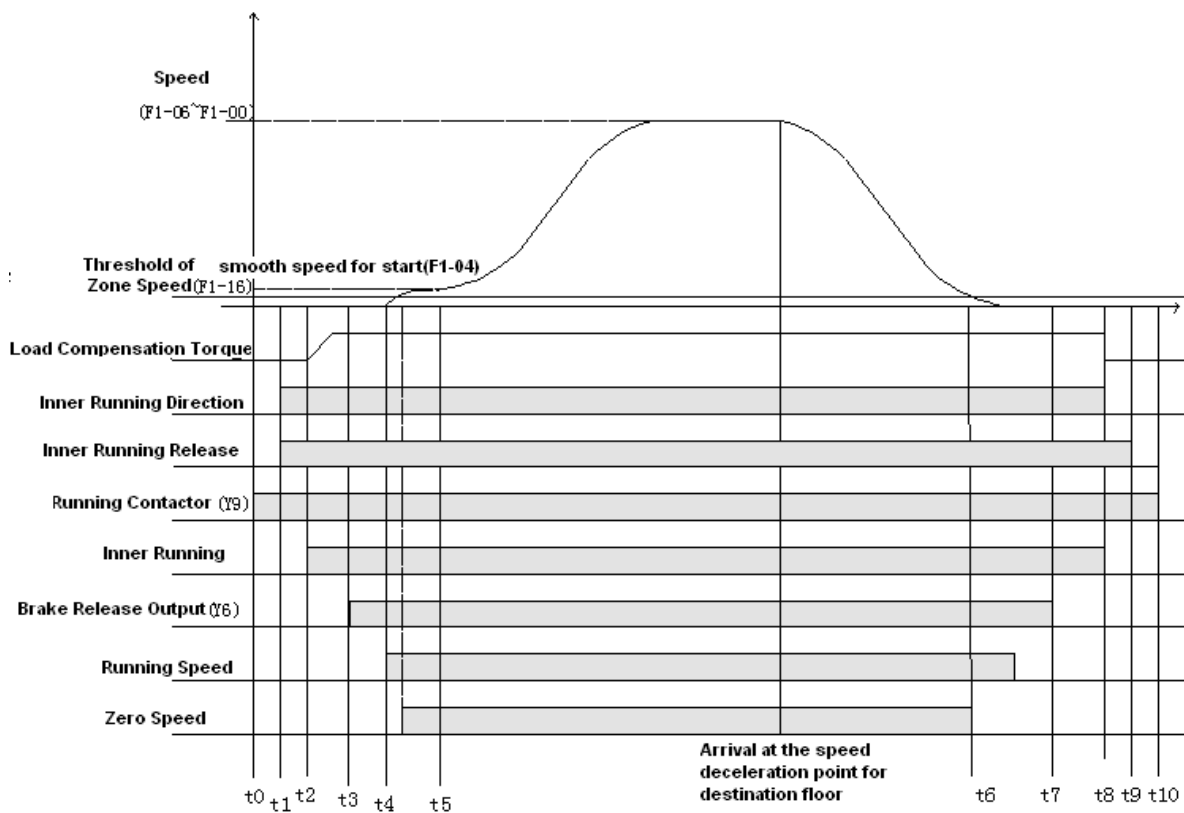


Fig. 6.3 Timing diagram of normal running

The instruction on above is Figure 6.3 referring to Table 6.1:

Table 6.1 Instruction Table for Timing diagram of normal running and parameter setting

Time	Definition and Setting Instruction
t0~t1	Control system output running contactor (Y9) control, the feedback time of running contactor, after feedback is valid, system gives running direction and release driving to control the running according to the orientation.
t1~t2	Driving module motivates the output, and start to output torque.
t2~t3	Waiting time for inner running signal (F2-03): In this period, driving module complete the excitation or load compensation torque output.
t3~t4	Advance time for brake release (F2-00):

	System output brake release (Y6), after brake contactor and brake arm give feedback (for example, F1-31=1), waiting for the advance time for brake release (F2-00), then running speed is given. There are two functions of advance time for brake release (F2-00): 1. Brake has enough time to completely release, it can avoid elevator startup before brake realest. 2. After brake release, traction sheave may rotate due to the load, with enough time traction sheave can be steady at zero speed then start in order to achieve comfortable start. According to the brake situation, for synchronous traction machine, it should be 0.8~1.5s, for asynchronous traction machine, it should be 0.3~0.5 s.
t4~t5	Smooth start time (F2-13): the running time under start smooth speed (F1-04), overcome part of static friction. Set it as "0.2~0.4 s", setting it so long will lower the elevator start efficiency.
T6~t7	Zero speed time (F2-0): When elevator runs to the destination floor and speed is lower than threshold of zero speed (F1-16), and then zero speed time (F2-04) pass, brake release output is cancelled. If zero speed time is too small, it will cause braking under running, set it as "0.2~0.4 s".
t7~t8	Braking time (F2-01): After braking, due to the subsequent flow and demagnetization, brake can lock on the traction sheave firmly, in this period, torque output remains, after this period, inner direction release is cancelled, torque output is eliminated. This time can prevent the elevator slip when braking lag. According to the braking situation, for synchronous traction machine, it should be 0.8~1.5 s, for asynchronous traction machine, it should be 0.3~0.5 s.
t8~t9	After inner direction is cancelled, if output current is eliminated at once, there will be big current noise, after time delay (F2-10 time delay for release), inner release is cancelled, in the delay time, current output decrease slowly, until the release is invalid, it stops.
T9~t10	Time delay for running contactor open 0.4s, it prevents electric arc when contactor open with current. But braking, emergency stop, door lock protection don't have time delay, they act at once.

6.3.2 Timing diagram of inspection running

Timing diagram of inspection running Fig. 6.4

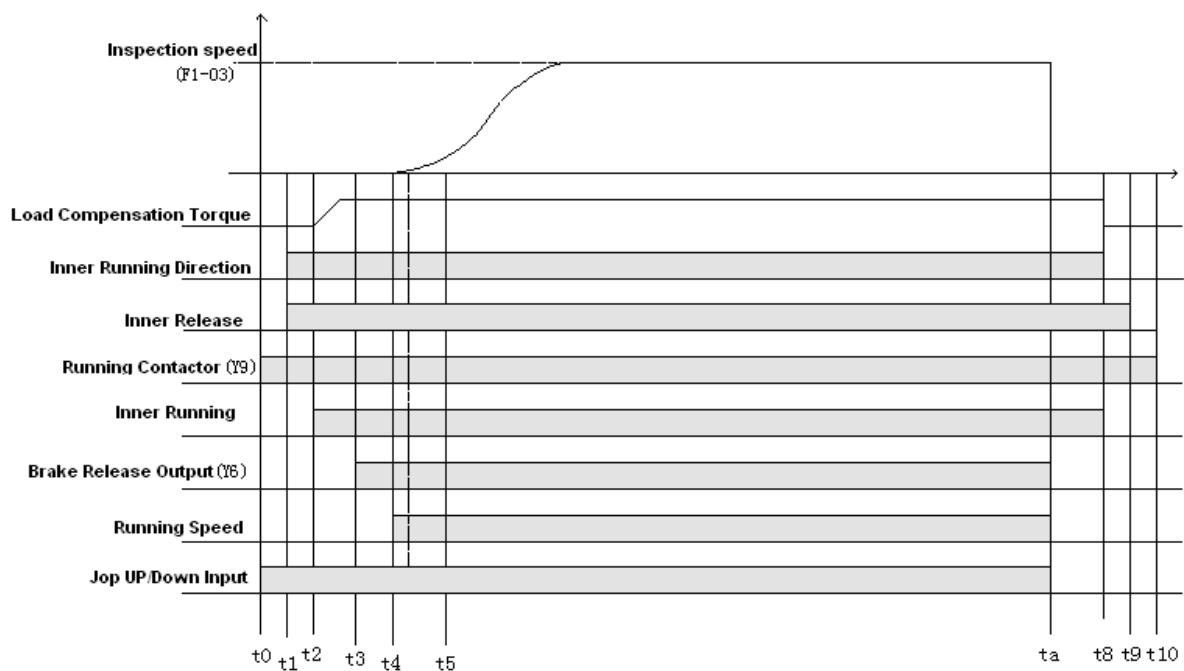


Fig. 6.4 Timing diagram of inspection running

The instruction on above is Figure 6.4 referring to Table 6.2:

Table 6.2 Instruction Table for Timing diagram of inspection running and parameter setting

Time	Definition and Setting Instruction
t0~t9	t0~t9 have the same definition with them under normal elevator running.
ta~t8	Inspection braking time (F2-02): When inspection running stops, elevator braking is not activated under zero speed, when jog up or down instruction is eliminated, brake will act with speed at once, for some asynchronous traction machines, because the holding time for torque output is too long, it will cause over-current protection of driving output, so we can decrease the braking time. For synchronous control, it should be same with the value under normal running 0.8~1.5 s, for asynchronous control, it should be 0.1~0.3 s.

6.3.2 Timing diagram of self-rescue

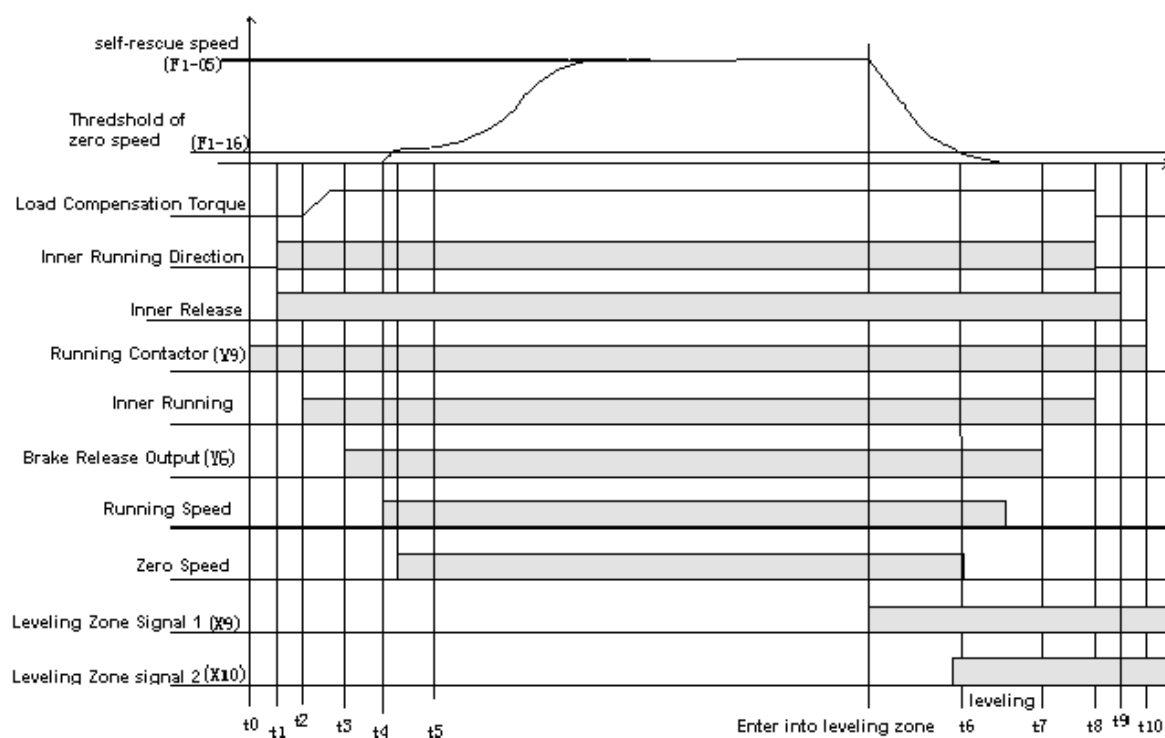


Figure 6.5 Time diagram of self-rescue

(Every time section has the same definition and same setting with them under normal running state.)

6.3.4 Timing diagram of verify the counting

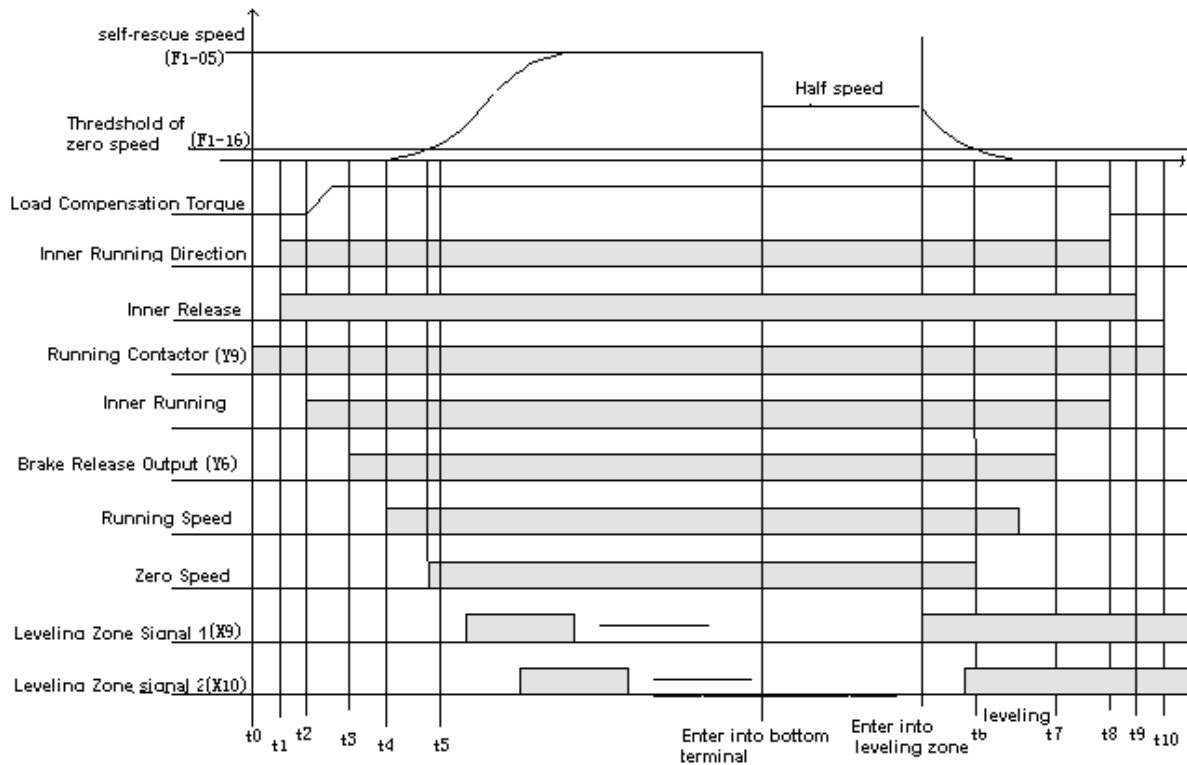


Figure 6.6 Timing diagram of verify counting

(Every time section has the same definition and same setting with them under normal running state.)

6.4Door control

6.4.1 Door control

The parameter rated to door open and close is door open and close time, beside that, the parameter related to door open and close control is door limit.

Timing diagram of door open and close is Fig. 6.7 and Fig. 6.8.

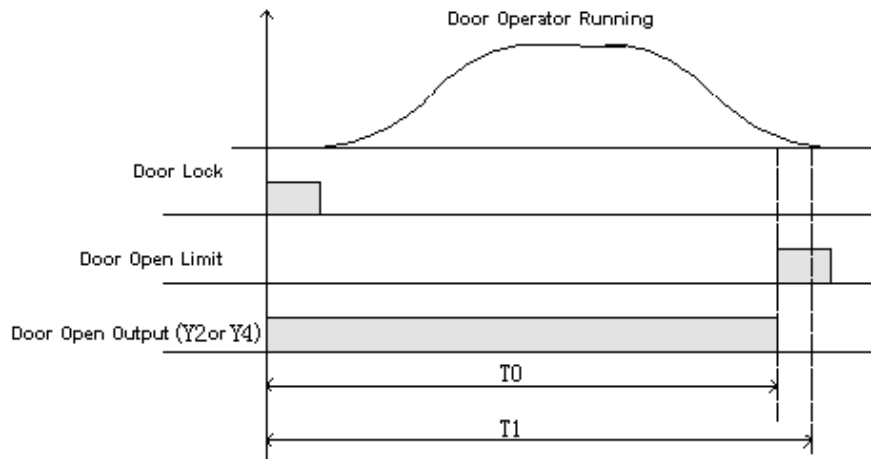


Figure 6.7 Timing diagram of door open

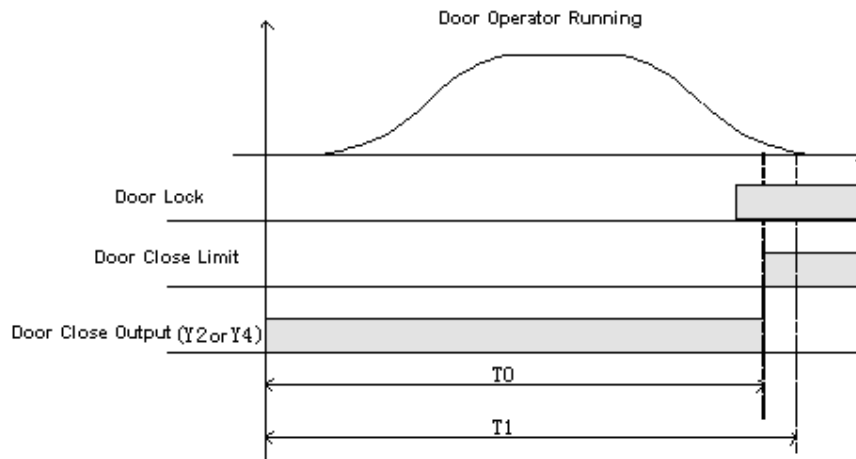


Figure 6.8 Timing diagram of door close

T0: The actual running time of door open or close completely; T1: Door open or close time (F2-08).

T1 should be 1 s more than T0 to ensure the normal door open or close, otherwise,

1. After door close signal is output, T1 door close time (F2-08) pass, but the door still doesn't close completely, elevator will open the door again, door can't close.
2. After door open signal is output, T1 door open time (F2-08) pass, but door still doesn't open completely, elevator stops door opening, door can't reach the self-lock position.

Door close limit should be fixed on the position where it is valid only door is completely closed, otherwise there will be the following phenomenon:

1. Door can't close in inspection.
2. Door close limit is valid, system stops output of the door close signal, door can't close, and system open the door again.
3. Door close limit is valid, system won't open door again at once, after the time F2-08, it will open door again. Under the holding force of door operator, door can close properly, system will show a door close error, but elevator can run normally.

If the door open and close limit switches can't be valid after fixing, elevator will control the output of door open and close according to F2-08 time. Door open and close limit signals are normal; the control of door open and close output is as per the door open and close limit signals.

6.4.2 Door open holding time

When elevator runs normally, after it lands to the floor and open its door, it will waiting for the following time then close its door automatically:

1. After landing, there is no landing call of the same direction with elevator running on current floor; the waiting time is door open holding time (F2-05).
2. After landing, there is only landing call of the same direction with elevator running but no car call, waiting time is door open holding time (F2-05) minus 2 s.
3. After landing, there are both car call and landing call of the same direction with elevator running, waiting time is the twice of door open holding time (F2-05) minus 2 s.
4. If door open delay release (F1-30) is valid (value: 1), meanwhile after door is open, press the door open delay button, waiting time will be door open delay time (F2-06). After the above period or press door close button to close the door in advance, door will close and open again, door open delay function is cancelled, unless you press the door open delay button again after door is open.
5. When elevator is under function for disabled persons, if there is car call or landing call of disabled persons on current floor, waiting time will be door delay time (F2-06).

6.4.3 Door open under by-attendant state and special use state

Under by-attendant and special use state, door can't close automatically after it is open, door must be closed manually. If door driving mode (F1-21) is 1, door close will be jog mode, if it is 0, door will close completely after an unbinding button push, no need to push the button continuously.

6.4.4 Door close under inspection and fireman state

Under inspection state, door open and close are both of jog mode.

Door open and close under fireman state:

1. When system enters into fireman state, if elevator is running, elevator will land to the nearest floor but doesn't open door, then elevator return to the fireman floor.
2. When system enters into fireman state, if elevator stops at other floor, door can't open after it is closed, then elevator return to the fireman floor. If the door is open, door will close at once, but before the door is completely closed, the door can be opened again by pushing the door open button.
3. If elevator is at the fireman floor, it will open its door and wait.
4. In fireman running, except on fireman floor, on all other floors door can't open and close automatically, but door can open and close by jog control.

6.5 Homing function

Under automatic running state (not duplex or group control), in the set time, there is no car call or landing call, elevator will return to the main floor.

1. Main floor setting (F0-01)
2. Homing time setting (F2-07), when homing time (F2-07) is 0, there will be no homing function.

6.6 Parking function

Normal elevator lock input:

1. When lock input (X20) is invalid, elevator return to the elevator lock floor (F0-03) after finishing all the car calls' service under normal running state.
2. When lock input (X20) is invalid, because under by-attendant and special use state, door can't close automatically, so after finishing all the car calls' service, the door must be closed manually, then elevator return to the elevator lock floor.
3. When lock input (X20) is invalid, elevator will be locked at once in inspection state.

Automatic start or off:

When lock input (X20) is valid, set automatic start or off release (F1-28) as 1, automatic start time (F2-14), automatic off time (F2-15), setting them to motivate this function.

When lock input (X20) is invalid, elevator will enter into lock state normally; it can't affect auto-start or auto-off.

If the elevator is in auto-off time and elevator need to run in that period, turn the lock key to invalid position then restore it to be valid, elevator will run, after the running, operate the lock key same again, elevator enters into lock state, until the auto-start time begins, then it starts running automatically.

6.7 Fireman operation function

Controller has three kinds of fireman operation mode; fireman mode (F1-23) can be set as “0, 1, and 2”:

0: Normal fireman mode: When fireman input is valid, elevator lands to the nearest floor but doesn't open door, then it returns to the fireman floor(F0-02), it opens door and stops, meanwhile it enters into the fireman mode, elevator can run in fireman operator standard.

1: Fire emergency stop mode: When fireman input is valid, elevator lands to the nearest floor but doesn't open door, then it return to the fireman floor (F0-02), it opens door and stops. Meanwhile elevator stops running, until fireman input is eliminated.

2: Fireman control mode:

It includes two steps, step 1 is fire emergency return, and step 2 is fireman operation. Only after elevator returns to the fireman floor in step 1, then step 2 can be operated.

1. Fireman input switch is of two states ON/OFF; it is connected to the X12 of main control board. Fireman operation switch is of three states ON/START/OFF, ON is connected to the fireman input (by-pass input) on C.O.P. board, START is connected to the door close input in parallel (it is for fireman close door, when fireman turns the key to the START position, point ON is still valid, after removing the hand, switch will automatically return to ON.)

1. Step1: Fireman input X12 is valid, if elevator is running, it will cancel all car call and landing call and stop at the nearest floor, but doesn't open door and then return to the fireman floor; if elevator is under stop and door open state, elevator will close door at once and return to the fireman floor. After elevator returns to the fireman floor, it opens door and stop running.

2. Step 2: Elevator enters into fireman state, and return to the fireman floor then stop running; At that time, fireman operation running can be motivated by fireman operator switch, switch it to ON/STAR, elevator will run under fireman operation(it only serve one car call, door open and close follows fireman operation function). After it is switched to ON/STAR, if elevator already leave the fireman floor, then switch is turned to OFF, elevator will keep running, only when elevator returns to the fireman floor again then it stops running.

2. After elevator enters into fireman state, light curtain will be invalid.

When fireman input switch, fireman operation switch are both OFF, elevator will escape from fire emergency state.

6.8 Two door mode

6.8.1 Mode selection

On the same floor, there are both front/rear door opening, mode 0 to mode n can be set by actual requirement.

Two door mode F1-22

n=0: single door

n=1: two door mode 1, only one door acts at every floor.

n=2: two door mode 2, two door can open on some floors, but they can't open synchronously, if you open one door, the other door must be closed.(There are two set C.O.P., each one for one door)

n=3: two door mode 3, two doors can act at some floors, when elevator runs to such floors, two doors will open synchronously.(There is only one set C.O.P., H.O.P. address setting of both side doors are same.)

n=4: two door mode 4, two doors can act at some floors, when elevator runs to such floors, two doors will open synchronously. (There are two set C.O.P., each one for one door)

n=5: two door mode 5, two doors can act at some floors, when elevator runs to such floors, door will open as per requirement, for example: when there is a car call or landing call at front

door, front door opens, when there is a car call or landing call at rear door, rear door opens, when there is a car call or landing call at other front and rear doors, two doors open synchronously. (There are two set C.O.P., each one for one door).

According to the two door mode, H.O.P address setting and C.O.P. wiring are different to:

H.O.P address setting

- 1) When two door modes are 0, 1, 3, H.O.P address setting is same as normal method.
- 2) When two door mode is 2、 4、 5, H.O.P address setting should follows the rules as below:
 - a. 1~32 is the absolute floor no. at the front door side:
1 is the bottom floor, 2 is the floor above the bottom floor, and it can be max. 32, total 32 floors.
 - b. 33~64 is absolute floor no. at the rear door side:
33 is the bottom floor, 34 is the floor above the bottom floor, it can be max. 64, total 32 floors.
 - c. If there is only one single door on some floors, then the H.O.P address for the non-existing door is vacant.

For example 1 One elevator has one basement floor, and on this floor, both front and rear doors act, so on this floor, H.O.P address at front door is 1, H.O.P address at rear door is 33.

For example 2 One elevator has one basement floor, on this floor, there is only front door, but on upper floor, there are both front and rear doors, then on the basement floor, H.O.P address at front door is 1, H.O.P address at rear door is vacant. On the upper floor, H.O.P address at front door is 2, H.O.P address at rear door is 34.

- 3) C.O.P button wiring:
 - a. When two door mode is 0, 1: 1~N floor buttons are connected to the C.O.P. ports normally.
 - b. When two door mode is 2, 4, 5: set the total floor no. N, then the 1~N car call ports correspond to 1~N floor car call buttons at front door, 1 is the bottom floor, N is the top floor. N+1~2N floor car call ports correspond to the 1~N floor car call buttons at rear door, N+1 is the bottom floor, 2N is the top floor.
For example, one elevator has 6 floors (1~6), no basement, 3rd floor has double entrance; other floors have only front entrance. Then car call buttons 1~6 at front door should be connected to 1~6 car call ports, car call button 3 at rear door should be connected to N+3=6+3=9 car call port.



IMPORTANT

When two door mode is 1, 2, 3, 4, 5, please install two sets of door open and close buttons, in that way, front door and rear door can be opened optionally in inspection running.

If mode is 1, and there is one set of door open and close buttons, please set parameter FU18 of function selection "ON". In other mode, if there is only one set of door open and close buttons, in inspection running, only front door can be opened by door open button, rear door can't be opened by the button.

6.8.2 Setting of two doors

Front door setting parameter (F4-04), rear door setting parameter (F4-05), set if the front or rear door open on certain floors. Under single door mode, the two parameters are invalid. Under two door mode, if it is set "ON" for some floors, on these floors, front door can open, if it is set "OFF", on these floors, front door can't open, rear door setting is same. Please set it as per the actual situation to avoid mistake.

6.9 Input type setting

Input type of main control board (F3-00), input type of C.O.P board (F3-01), their setting is as per the actual switch on/off state. When switch is at validity position or at validity state, switch is closed, this input type is ON. When switch is at validity position or at validity state, switch is open, this input type is OFF. Input typesetting of main control board and C.O.P. board is as the following table 6.3

Table 6.3 Input type setting

Name	port	Position	Definition	parameter	Input level default	Outer Contactor State	Indicator Normal state
Main control board F3-00	X0	J3-8	Inspection state	F3-00-00	ON	NO	OFF
	X1	J3-7	Up running input	F3-00-01	ON	NO	OFF
	X2	J3-6	Down running input	F3-00-02	ON	NO	OFF
	X3	J3-5	Top terminal 2 input	F3-00-03	OFF	NC	ON
	X4	J3-4	Bottom terminal 2 input	F3-00-04	OFF	NC	ON
	X5	J3-3	▲ Top limit input	F3-00-05	ON	NC	ON
	X6	J3-2	▲ Top limit input	F3-00-06	ON	NC	ON
	X7	J3-1	Top terminal 1input	F3-00-07	OFF	NC	ON
	X8	J2-8	Bottom terminal 1input	F3-00-08	OFF	NC	ON
	X9	J2-7	Upper leveing input	F3-00-09	ON	NO	OFF
	X10	J2-6	Lower leveling input	F3-00-10	ON	NO	OFF
	X11	J2-5	Spare	F3-00-11	ON	NO	OFF
	X12	J2-4	Fireman input	F3-00-12	ON	NO	OFF
	X13	J2-3	Emergency stop input	F3-00-13	ON	NO	OFF
	X14	J2-2	Door interlock input	F3-00-14	ON	NO	OFF
	X15	J2-1	*Brake feedback input/Main power contactor input	F3-00-15	ON	NO	OFF
	X16	J1-10	Subsidiary contactor input	F3-00-16	ON	NO	OFF
	X17	J1-9	Brake contactor feedback input	F3-00-17	ON	NO	OFF
	X18	J1-8	Emergency landing input	F3-00-18	ON	NO	OFF
	X19	J1-7	Spare	F3-00-19	ON	NO	OFF
	X20	J1-6	Electric lock input	F3-00-20	ON	NO	OFF
	X21	J1-5	Thermal switch input	F3-00-21	OFF	NO	OFF
	X22	J1-4	Re-leveling condition input	F3-00-22	ON	NO	OFF
	X23	J1-3	Re-leveling inductor input	F3-00-23	ON	NO	OFF
	X24	J1-2	Spare	F3-00-24	ON	NO	OFF
	X25	J1-1	Spare	F3-00-25	ON	NO	OFF

Name	port	Position	Definition	parameter	Input level default	Outer Contactor State	Indicator Normal state
Main control board F3-00	X29+	J6-3	Emergency stop input +	F3-00-29	ON	NO	OFF
	X29-	J6-4	Emergency stop input -				
	X30+	J6-5	Door interlock input +	F3-00-30	ON	NO	OFF
	X30-	J6-6	Door interlock input -				
	X31+	J6-7	Spare +	F3-00-31	ON	NO	OFF
	X31-	J6-8	Spare -				
C.O.P. board F3-01	KMV1	J9-2	Door open limit 1input	F3-01-07	OFF	NC	
	GMV1	J9-3	Door close limit 1input	F3-01-06	OFF	NC	
	KMV2	J10-5	Door open limit 2 input	F3-01-05	OFF	NC	
	GMV2	J10-6	Door close limit 2input	F3-01-04	OFF	NC	
	KAB1	J9-4	Safety edge 1input	F3-01-18	OFF	NC	
	KAB2	J9-7	Safety edge 2 input	F3-01-17	OFF	NC	
	CZ	J9-5	Overload input	F3-01-15	ON	NO	
	MZ	J9-6	Full load input	F3-01-13	ON	NO	
	QZ	J9-8	Low load input	F3-01-14	ON	NO	
	KZ(50%)	J9-9	50% load input			NO	
	SZH	J9-10	By-attendant input			NO	
	SZY	J10-1	Special use input			NO	
	SZS	J10-2	By-pass input			NO	
	ZHS	J10-3	By-attendant, up			NO	
	ZHX	J10-4	By-attendant, down			NO	



If the top/bottom limit is normal, their circuit should be closed, it shows "ON".
If the top/bottom limit is abnormal, their circuit should be open, it shows "OFF".

6.10 service floor setting

Non-stop floors can be set by F4-00, F4-01, and some floors which elevator can land, set "ON", some floors which elevator can't land, set "OFF". After setting F4-00, F4-01, elevator can't land to the floors in any case. On this base, by function selection (refer to the function selection table), timing for non-stop function is available too. Between the beginning time (F2-16) and ending time (F2-16), if elevator can't land to some floors, please set the non-stop floors by setting F4-02 and F4-03 "OFF".

6.11 Weighing device

If there is SJT-150 or SJT-300 weighing device, they can communicate with the main control board through CAN to realize the load detection. Setting method is as follows:

1. Set weighing device release F1-29=1, F9-11=1;
2. Perform low load and full load self-learning;

3. According to the compensation situation, adjust the compensation coefficient F9-00.
4. If there is no compensation chain with elevator, adjust F1-18 according to the compensation situation on top floor and bottom floor:
 - a) Adjust the simulative load compensation gain in inverter, it should prevent elevator from slipping when it runs down from the top floor, then the value is proper;
 - b) Moving the no-load elevator to the bottom floor, increase the load compensation adjustment parameter, it should prevents elevator from slipping when it runs up from the bottom floor, then the value is proper.
 - c) The adjustment range of this parameter is 0~12.

6.12 Duplex control

Controller has inner integrated duplex module, two elevators are connected by CAN1 in parallel, and after setting corresponding parameters, duplex control can be realized.

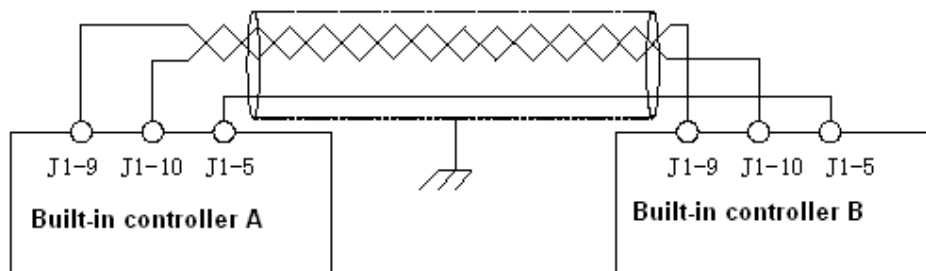


Figure 6.9 Connection for duplex control

Setting process:

1. Set duplex control release, F1-25=1
2. Set duplex elevator no. F1-24, one elevator is set A, the other elevator is set B.
3. Fireman floor should be set same for both elevators. (otherwise, landing call will mixed to other floor)
4. H.O.P address should be set same with single elevator.

After completing the setting and saving the parameters, please reset the power and ensure the updating of the communication address.

6.13 Group control

Main control board and group control board are connected through CAN1 port. (refer to the manual of group control board), and after setting some parameters, group control can be realized:

Setting process:

1. Set group control release , F1-26=1;
2. Set elevator no. F1-24, set elevator no. by A, B, C, ~H
3. The fireman floor should be set same for the elevators (otherwise, landing call will mixed to other floor)
4. H.O.P address should be set same with single elevator.

After completing the setting and saving the parameters, please reset the power and ensure the updating of the communication address.

6.14 Leveling adjustment

After elevator lands to the floor normally, if the running curve has no problem (for example there is no emergency stop or overrun), if elevator overruns the leveling position (it stops higher in up-running, lower in down-running), decrease leveling adjustment parameter F1-17 (default 50); if elevator can't reach the leveling position(it stops lower in up running, higher in down running),

increase leveling adjustment parameter F1-17, generally the range of this parameter is 40~60, if the adjustment is big, please adjust driving parameter PI, or curve (F1-10~F1-15).

6.15 Floor indication setting

In parameter F0-05~F0-6, we can set third bit display, first two bit display can be digit, word or “-”, the third bit only can be the following words: ABCDEFGHIJKLMNO. If only need to indicate two bits, set the first two bits only, third bit is of no display. (Third bit indication should be supported by the program of H.O.P. board, otherwise it can't be indicated normally.)

6.16 Special function selection

To meet special requirements of customers, our control system has some non-standard functions (F4-06) for option.

Function No	Instruction
F4-06-00	After elevator stops, the current floor will be the reference floor. If there is no landing call or car call ahead of the current floor in the previous running direction, then system will cancel all the car calls.
F4-06-03	ON: shielding ER29 fault; OFF: When communication excessive interference report ER29 fault.
F4-06-04	ON: Two elevators in spare time in parallel, when the same floor on both call and call registration two go elevators; OFF: one elevators go.
F4-06-05	Set to ON: elevator shielding the car cabin overload signal, when used in elevator acceptance 125% load operation. OFF: Call the overload signal.
F4-06-06	Set to ON: When the elevator in the current floor can't open the door, automatic enrollment, realize open and close the door in the recently floor.
F4-06-07	ON: The elevator door in area then change the floor show ; OFF: The elevator change speed then change the floor show
F4-06-08	Set to ON: When overhaul run stop and waiting for zero speed signal gate under, reduce the impact.
F4-06-09	Set to ON: In the operation of the elevator can cancel the registration within selected (If all the registered call canceled, the elevator stop in the recently floor)
F4-06-14	ON: Function for disabled people are available. OFF: Function for disabled people is not available
F4-06-15	ON: When in Fire state elevator leave fire floor the fire linkage output layer revoked, when the elevator return to fire floor the fire linkage restore output layer
F4-06-16	ON: When system ensures the lock is closed, door close limit must be valid too. OFF: Door lock state is unrelated to door close limit.
F4-06-17	ON: When elevator stops in inspection state, release direction and brake release are eliminated together. OFF: When elevator stops in inspection state, there will 0.5s interval ,then release direction is eliminated.
F4-06-18	ON: In two door mode, there is only one set of door open& close buttons. OFF: In two door mode, there are two sets of door open & close buttons.
F4-06-19	ON: Door open and elevator re-leveling function OFF: No re-leveling function
F4-06-20	ON: Release for door open in advance

	OFF: No door open in advance function
F4-06-21	ON: In inspection state, door can't open at no-levelling zone. OFF: In inspection state, door can open at any position.
F4-06-22	
F4-06-23	ON: the weighing device matching to the main control board adopts SJT-300 of CAN main cable.
F4-06-24	ON: sub-terminal switch is adopted for elevator speed below 2m/s (it is generally adopted in 1.75m/s elevator for two steps deceleration.) OFF: Sub-terminal switch is not adopted for elevator speed below 2m/s.
F4-06-25	ON: Repair the car doors when in key is invalid OFF: Repair the car doors when in key effective
F4-06-28	ON: Use light curtains and touch boards, the original driver in the form of separate directional, under the front door and the back door respectively as input of the input touch boards; OFF: Light curtains and touch boards series input (blue light default Settings)
F4-06-29	ON: Use operation contactor and motor seal star contactor separate configuration, X11 as sealing star contactor feedback, Y8 as motor seal star contact device output control (see appendix hookup timing and detailed) OFF: Operation contactor bring their own motor seal star function (blue light default Settings)
F4-06-30	ON: All-in-one PC motherboard digital pipe display content upside down, used for ultra-thin equipment (motherboard reverse when placed use) OFF: All-in-one PC motherboard digital tube display properly (Blue light default Settings)
F4-07-00	ON: Increase the emergency smooth layer actually slip before judge overloaded direction and start the car, when the rescue function setting to ON, when the elevator into a state of emergency rescue, the system will have control of the gate 1 seconds (when actually slip car speed is greater than 0.1 m/s, immediately after the brake), through the judge the direction of the elevator car free slipped to find the direction of the heavy, and then system through the battery running along the direction to save heavy crawling, save energy, save to prevent battery depletion.
F4-07-01	ON: Enable the elevator data recorder function with equipment PC debugging software for debugging and after-sales personnel can provide fault diagnosis and fault capture tools
F4-07-02	ON: Cancel top and bottom limit signal, with top station and under door are effectively at the same time invalid combined from cap out of a signal. Below the station and the door area with effective invalid under the lower combined from out of a signal
F4-07-03	ON: Enable electric lock serial function
F4-07-04	ON: When serial electric lock take the negate
F4-07-05	ON: Enable fire serial function
F4-07-06	ON: When fire serial take the negate

Table 6.4 Special function selection

6.17 Motor parameter setting

First, please set the motor parameters as per the motor:

F5-00: motor type, 0: synchronous external rotor motor, 1: asynchronous motor, 2: synchronous internal rotor motor.

Please set the parameter as per the actual type of the motor, system will perform vector control as per the setting, meanwhile when system perform motor parameter self-learning, system will learn according to the different motor type setting. Therefore, please make sure its setting is correct.

Secondly, please input the follow motor parameters according to the motor nameplate or manual to ensure the accuracy of the motor digital module and the precise coupling of the motor by vector control.

1. F5-01: Motor pole
2. F5-02: Motor rated frequency
3. F5-03: Motor rated power
4. F5-04: Motor rated RPM
5. F5-05: Motor back-EMF (this parameter is only valid for asynchronous machine)
6. F5-06: Motor phase inductance (this parameter can be acquired from motor self-learning)
7. F5-07: Motor phase resistance (this parameter can be acquired from motor self-learning)
8. F5-08: Motor rated current
9. F5-09: No-load current (this parameter is only valid for asynchronous machine, it can be acquired from motor self-learning)
10. F5-10: Slip difference (this parameter is only valid for asynchronous machine)

Beside the above, when some parameters can't be acquired, please perform motor parameter self-learning.

For the driver versions at 0007 or more, can avoid the parameters self-learning

6.18 Elevator running speed given

Elevator rated speed (F1-00), motor RPM (F1-01), their setting is adopted to calculate the ratio of elevator rated running speed and motor RPM. Mean while, by the pulse of one circle F1-03, we can calculated out the relationship between pulse and traveling distance, to realize the distance detecting of elevator running. Therefore changing F1-00, F1-01 will change their proportion; elevator speed can't be changed by changing these two parameters.

When F1-03 for hoistway counting is inner pulse, pulse no. in one circle (F1-03) = pulse no. of motor encoder (F8-00) / PG frequency division ratio (F8-01). Rated speed of traction machine (F1-01)= Motor rated speed (F5-04).

If F1-03 for hoistway counting is outer pulse (for example, governor encoder, then we must adopt special PG card), F1-03 should be set by pulse no. in one circle of the outter encoder, Rated speed of traction machine (F1-01) = Outer rotor speed.

Speed given diagram Figure 6.10

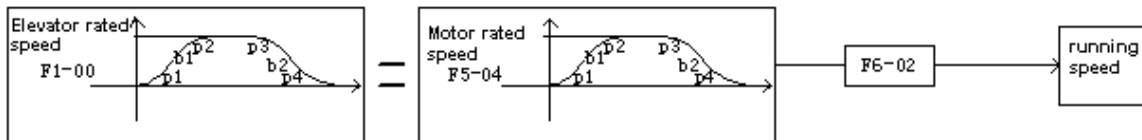


Figure 6.10 Speed given diagrams

If it is necessary to lower the elevator actual speed, please change the speed ratio F6-02 value, when F6-02=100%, elevator runs in rated speed, if decrease F6-02, elevator speed will decrease in corresponding ratio.

6.19 Speed control adjustment (PI adjustment)

6.19.1 Speed control adjustment by changing PI

Speed control under vector control, its speed adjustment has two ways, F7-00: when multi-section PI release is 0, PI won't be changed as per the speed given change, in the full range of the elevator speed, PI parameter is an unchangeable value, it makes adjustment simple. Generally, this speed adjustment method is adopted.

Speed adjustment diagram Figure 6.11

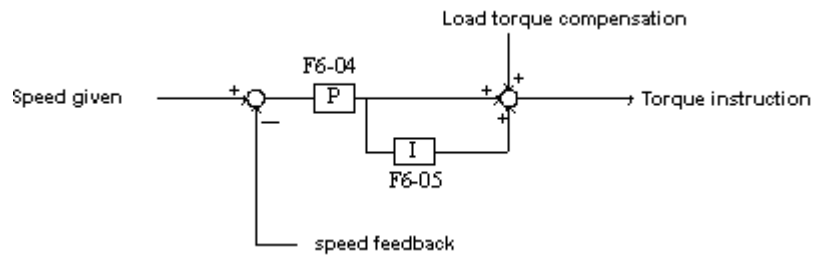


Figure 6.11 Speed adjustment diagrams 1

6.19.2 Speed control adjustment by changing PI in multi-section

Speed control under vector control, F7-00: when multi-section PI release is 1, speed control function can be performed by changing PI in multi-section, after multi-section control, F6-04 and F6-05 setting won't take effect in full trip, please adopt 4 groups of PI value F7-05~F7-12 to perform speed control.

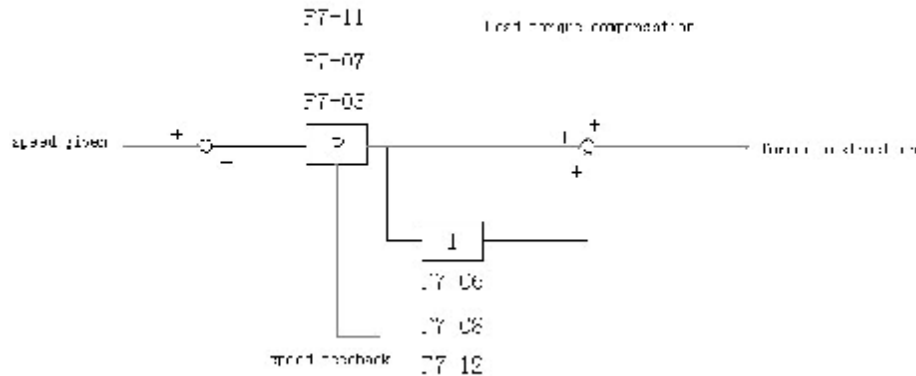


Figure 6.12 Speed adjustment diagrams 2

F7-05, F7-06: Group 1 of PI parameter, it is generally adopted in low speed section in starting.

F7-07, F7-08: Group 2 of PI parameter, it is generally adopted in middle speed section of acceleration or steady-speed section of low speed running.

F7-11, F7-12: Group 4 of PI parameter, it is generally adopted in deceleration section.

Switching frequency of the above groups of PI parameter adopts F7-01~F7-04: frequency setting of PI effect range.

Group 1 Effect range of PI parameter:

When motor starts from zero speed or accelerates, if the current frequency given is smaller than F7-01, system will adopt F7-05, F7-06 of group 1 to adjust the speed. To make elevator steady in zero speed when brake release in starting and follow the running curve quickly in initial acceleration, F7-05 and F7-06 can be set bigger properly.

Group 2 Effect range of PI parameter:

After motor starts to run (acceleration), when speed is bigger than F7-01 and smaller than F7-02, system adopts F7-07, F7-08 of group 2 to adjust the speed. When motor running speed is bigger than F7-02, system adopts F7-09, F7-10 of group 3 to adjust the speed. If F7-02 is zero, then system adopts F6-04 and F6-05.

Group 3 Effect range of PI parameter:

When motor speed is smaller than F7-03 (deceleration), system adopts F7-11, F7-12 of group 4 to adjust the speed. If F7-04 is 0, system will adopt the last group of PI parameter before deceleration.

Effect diagram of different groups of PI parameter Figure 6.13

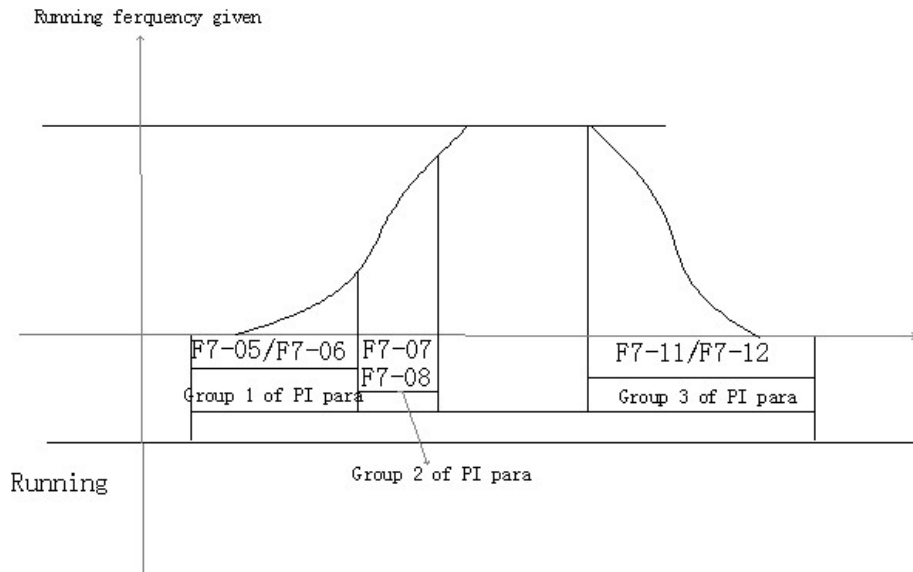


Figure 6.13 Effect diagrams of different groups of PI parameter

6.20 Output control of load compensation torque

Parameters which are related to output control of load compensation torque:

1. The related parameters of the weighing device which specially match to our system:
 - 1) F1-29: Weighing device release (1: release, 0: no release)
 - 2) U6-03: weighing value, check the current load situation
 - 3) F1-18: weighing adjustment, adjust the compensation as per the floor no., it is suitable for elevator without compensation chain.
2. Load simulative input, input range +10V~-10V or 0V~+10V, it is unchangeable setting.
3. Load compensation source selection F9-13: 0 inner serial signal, adopt specially matching weighing device of our company, 1 outer simulative input +10V~-10V, 2 outer simulative input 0V~+10V.
4. Max. Compensation torque F9-00, it limits the max. Compensation torque, if it is set 60%, in full load, system outputs 60% of the max. Rated torque output as compensation torque.
5. Release torque control output F9-11, when setting it "1", system will output torque with corresponding direction by F9-13 multiplying F9-00; when this parameter is , load compensation is not released.

Output control diagram of load compensation torque Figure 6.14

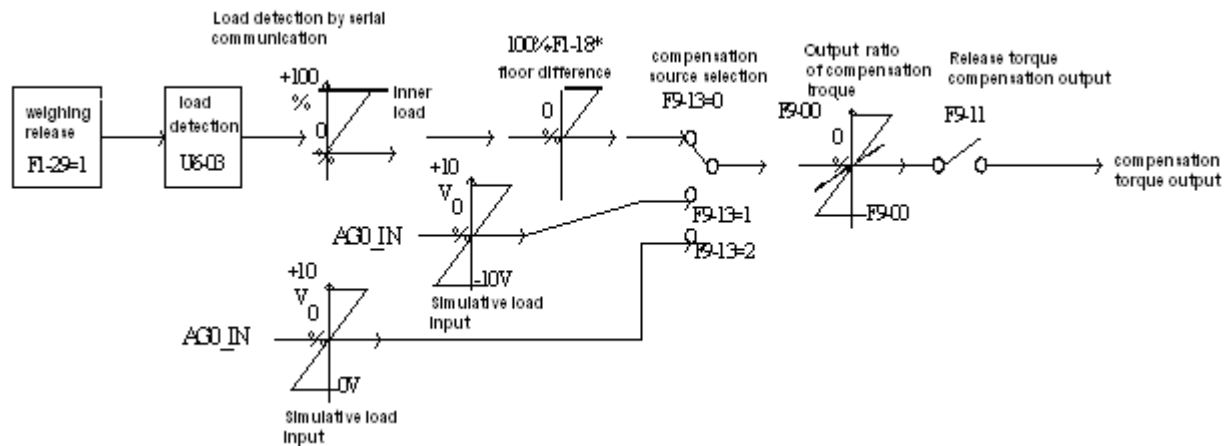


Figure 6.14 Output control diagram of load compensation torque

6. In synchronous control system, because there is no compensation chain when building is low, our weighing device only can weigh the load in the cabin, it can't detect the rope weight change on different floor, system perform adjustment by load compensation parameter F1-18 in running parameters.

Adjustment method:

- 1) Perform no-load, full-load learning.
- 2) Make the no-load elevator run to the top floor.
- 3) Adjust simulative load compensation gain in inverter to prevent elevator from slip when it runs down from top floor, then its value is proper.
- 4) Make the no-load elevator run to the bottom floor, increase the load compensation adjustment parameter (F1-18) to prevent elevator from slip when it runs up from bottom floor, then its value is proper.

6.21 Encoder parameter

Set F8-00 as per encoder pulse no., confirm the frequency division ratio F8-01 as per the precision requirement of hoistway counting.

Encoder pulse distributing Figure 6.15

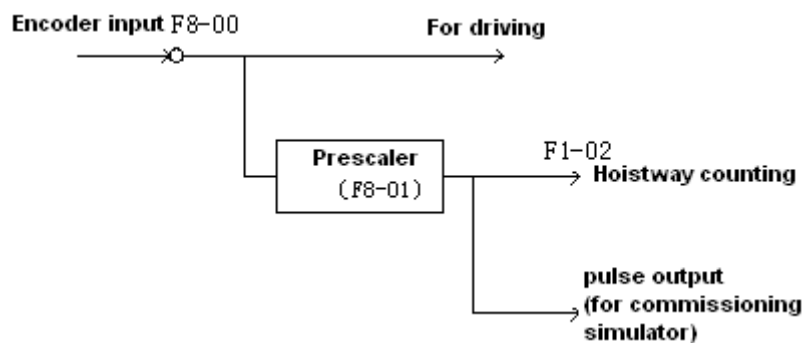


Figure 6.15 Encoder pulse distributing diagram

6.22 Adjustment of starting without load compensation

When this product adopts Sine/Cosine PG card, it can reach comfortable starting without load compensation by proper setting of FA group parameters. (Namely, in condition of no weighing device, it can reach same effect of load compensation.)

1. Notice for starting without load compensation:
 - 1) PG card type confirmation, confirm F8-02 is "1" (namely, Sine/Cosine PG card)
 - 2) Weighing compensation release confirmation, confirm F9-11 is 0, namely release is invalid. In that time, FA group parameters can take effect.
 - 3) Driving software version confirmation, confirm version is 13 or advanced version.
2. Adjustment method for starting without load compensation:
 - 1) Working Principle: refer to Figure 6.16. When brake is released, through the position feedback from Sine/Cosine PG card, it calculates the necessary torque of the motor to remain the invariableness of traction machine position under the current load, and it gives corresponding torque at once and make the traction sheave have less moving to achieve comfortable starting.

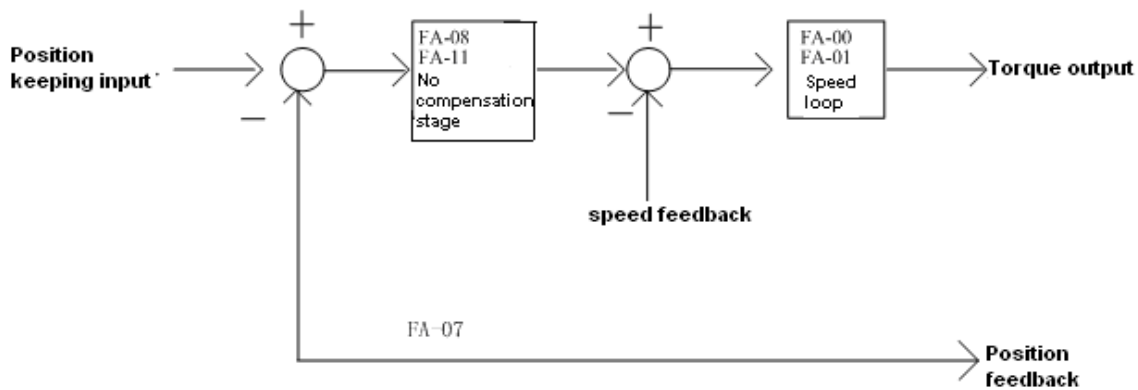


Fig. 6.16: Working principle diagram for starting without load compensation

- 2) Related parameters: Parameter table for adjustment of starting without load compensation refer to table 6.5

Table 6.5: Parameter table for adjustment of starting without load compensation

Parameter NO	Name	Product Value	Faster Brake Recommend	Lower Brake Recommend
FA-00	StratKP	30	Keep	Keep
FA-01	StratKI	750	Keep	Keep
FA-08	PLKP1	3600	4800	3600
FA-09	PL Time	900	700	Keep
FA-11	PLKP2	800	Keep	Keep

FA-12	PLKPMOD	125	Keep	Keep
F2-00	Brake ON Time	0.5	0.9	1
F9-00	Max Trop Comp	0	Keep	Keep
F9-11	Load Comp Enable	1	0	0

3) Adjustment method: Mainly adjust FA group of three parameters : FA-08、FA-09、FA-11.

FA-09

This parameter is the working time of the starting without load compensation after brake opening, it must be set according to real time of brake opening time, if the value of FA-09 is too short, it will slip because the starting without load compensation will be over before the brake fully opened; at the same time the value of F2-00 (brake opening time before running) must be 100ms longer than the value of FA-09, so that the action of the starting without load compensation can be over before speed curve.

FA-08 and FA-11

Two gain parameters of the starting without load compensation, these two parameters can be adjusted according to the slipping situation of traction sheave and the comfortable feeling of taking the elevator in job side, if the slipping is more please increase the value of FA-08, if the traction machine gets vibration, please reduce its value; during the period of torque keeping, if there is light slipping or swing on traction sheave, please increase the value of FA-11, if there is vibration, please reduce its value.



1. During commissioning, FA group parameters can be kept with factory setting except above 3 parameters
2. For different versions of programming, the name of FA group parameters might be different, but their position has no change, so, no matter the version of hand operator is, please just adjust FA-08, FA-09, FA-11 only.
3. The setting value of above parameters is just for reference, because the PG card is not same in different job side, please adjust above parameters according to the real situation in job side.
4. F9-00 is the pre-set torque when the starting without load compensation function is acting, in general, not need to change its value, please keep it with factory setting (0)

7 Commissioning

7.1 important reminding

1. Thanks for purchasing our product, please read this manual and related instruction thoroughly before processing with installation, connection (wiring), operation or maintenance and inspection, Ensure you have sound knowledge of the device and familiarize you with all safety information and precautions before processing to operate the control system to avoid any damage or loss.
2. Before commissioning and running, read the manual carefully and refer to this manual during commissioning and running.
3. Make sure that all the mechanical devices are installed reliably specially the devices in hoistway (the devices which should be set in the machine room depends on the situation of the machine room).
4. Make sure that the installation and commissioning for the devices, which should be finished before commissioning of the control system, have been completed.
5. Before commissioning, it is necessary to get signature confirmation from the persons who are responsible for mechanical installation and commissioning.
6. Make sure that all the mechanical devices and other derives which are related to the system commissioning are installed and tested properly.
7. Make sure that there are no unsafe factors which could cause injuries and damages to body and devices.
8. Qualified electricians should carry out commissioning.
9. Circumstance should meet the conditions for commissioning and running.
10. When doing commissioning for both electric and mechanical parts, technician should collaborate.
11. If the manual couldn't meet your requirement, please contact with our company in time to acquire help and to avoid accident and loss.
12. Before system commissioning, make sure all the conditions are sufficiently prepared.

7.2 inspections before powering on

After electric system is installed, please check the electric system carefully and pay attention to the following items:

1. Comparing with the manual and electric diagram, check if the connection is all correct.
2. Check if there is interference between the high voltage part and low voltage part. Use a multi-meter to measure the resistor in different circuit, resistor to earth is ∞ .
3. Please check if wiring of power supply to the control cabinet and traction machine is correct, that is to avoid damaging the controller after powering on.
4. Check wiring between encoder and controller, coaxial degree of encoder and traction machine shaft, wiring between encoder and traction machine.
5. Check if the enclosure of the controller and motor, cabin, landing door are reliably earthed to ensure the body safety.



Controller enclosure and motor enclosure should be earthed to the one point.

6. Make sure correct wiring of the terminal block J1 on control board to avoid damage to the main control board.



Controller has special operation panel, after operation panel is connected to the socket J232 on main control board by a special cable, then start commissioning, or adopt USB serial communication cable to connect the port USB0 on main control board to computer, then do commissioning by special software. (please refer to commissioning software manual)

7. Make elevator stop at the middle floor.
8. Turn electric lock to position "ON".

7.3 Power on and inspection

7.3.1 Confirmation before power on

1. Confirm all the breakers in control cabinet are open.
2. Confirm switch on the controller is at "inspection" position, emergency stop button is pressed.
3. Confirm inspection switch on car top and cabin are both "normal" positions.
4. Confirm terminal resistor of the bottom H.O.P is already connected
5. Check the voltage of the main power supply: 3 phase voltage: $380 \pm 15\% \text{VAC}$, deflection should be smaller than 15VAC.
The voltage between each phase and N wire is $220 \pm 7\% \text{VAC}$.
6. Confirm the wire specification and main switch capacity match the design requirement.

7.3.2 Inspection after power on

1. Switch on power switch Q1, if phase relay KXX green light is on, it means the phase sequence is correct, otherwise red light is on, it means phase sequence is incorrect, in that way, please switch off the main power switch, exchange any two phase. Repeat the above inspection.
2. Inspect the terminals' voltage of transformer TC1, the values should be in the range of $\pm 7\%$ of the indicated value, if it exceeds the range, find out the problem and correct it.
3. if the above inspection are normal, then do the following process:
 - 1) Switch on F4 the voltage between terminal 100~101 $110 \pm 7\% \text{VAC}$
the voltage between terminal 103~102 $110 \pm 7\% \text{VDC}$
 - 2) Switch on F5 the voltage between terminal 200~201 $220 \pm 7\% \text{VAC}$
 - 3) (Remark: connect the key pad operation panel before power on) after power on, first see if the main menu display is correct or not on the LCD indicator, for example: elevator state, fault state, door lock state, current floor, running speed, etc, by that, we can judge if the controller work and indication are normal, 24V power supply is normal or not, etc.

Interface of operation panel Figure 7.1

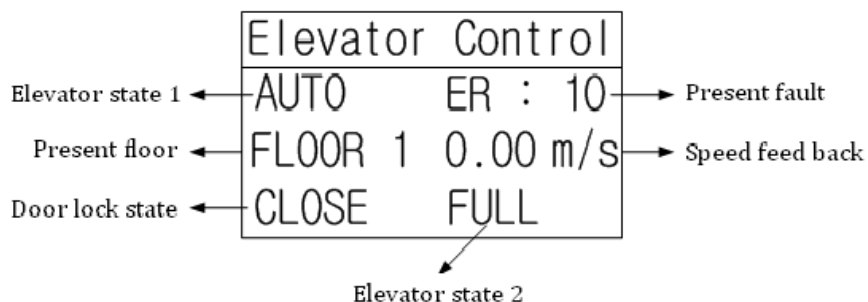


Figure 7.1 Interface of LCD indicators on operation panel

- 4) Voltage of the terminals on switch power unit:

Table 7.1 Voltage of the terminals on switch power unit

Terminal	L~ N	24V~ G
Voltage	220±7%VAC	24.0±0.3VDC

- 5) After the above inspection, do the next inspection as follows:
- u Inspect if the door lock circuit is normal or not.
 - u Inspect leveling zone signal, top/bottom limit signal are normal or not.
 - u Inspect electric lock is normal or not, set the elevator auto-start/off time "0", set electric lock switch "ON", then LCD indicator will show elevator state "INSP", if set electric lock switch "OFF", LCD indicator will show "STOP".
 - u Inspect if the door open/close system is normal or not, if it is abnormal, please do corresponding adjustment.

7.4 Parameter setting

According to the actual situation at the jobsite to set the parameters is very important, it is the base of controller or control system adequately exerting its characters. Parameters' setting refer to chapter 4~6. When setting parameters, please pay attention to the following points:

1. Set the motor basic parameters as per the motor nameplate namely motor parameter 1 content (F5 group). For example, motor type, pole no., rated frequency, rated power, rated RPM, rated current, Back-EMF (only for synchronous machine), no-load current (only for asynchronous machine), phase inductance, phase resistance, etc. there into, phase inductance, phase resistance, etc which are unknown can be acquired from motor self-learning; If moment of inertia is unknown, set it default.
2. Set encoder parameter correctly according to the encoder (F8 group), for example encoder pulse, PG frequency division coefficient.
3. Set elevator running parameters correctly. For example, speed, motor rated RPM, encoder pulse (after frequency division.)
4. Set input type correctly, they should be uniform with the actual contactors, relays and hoistway contacts and switches.

7.5 Motor parameter tuning

Controller has motor parameter tuning function. By the basic motor parameters' input, the inner program in controller automatically performs control, detecting and calculation; recognize the unknown motor parameters which are necessary. For example, phase inductance, phase resistance, no-load current (only for asynchronous machine), etc.

Unless all the necessary motor parameters are known (F5 group), for example, phase inductance (F5-06) is known, phase resistance (F5-07) and all other parameters are known too, please set the parameters directly, otherwise tuning must be performed.

7.5.1 Rotation tuning

Motor rotation tuning process is as follows.

1. Make motor (traction machine) without load (don't hang on the steel rope);
2. Jump out running contactor output Y 9(J4-7) and COM3 (J4-6), make running contactor closed;
3. Jump out brake output on controller Y6(J4-10) and COM3 (J4-6), make braking contactor closed, release the brake;

By operation panel, perform the motor tuning according to Figure 7.2

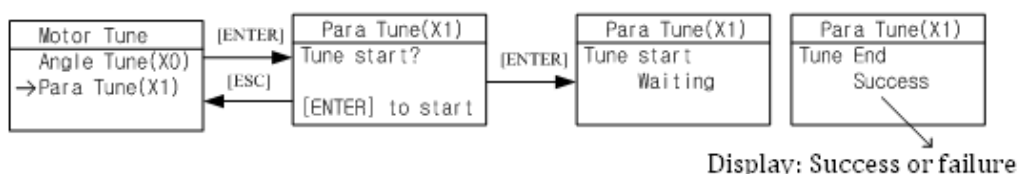


Figure 7.2 Operation diagram of motor tuning

After pressing “Enter” button, it performs motor parameters tuning, motor become static after a short vibration, continue to supply current, motor will give current noise, it takes about 5 s for the self-learning.

If self-learning is successful, return to parameter setting menu, check motor phase inductance (F5-06), motor phase resistance (F5-07) etc. should be all corrected to the acquired value. Generally, one time learning success is enough. If you’d like to acquire more accurate results, you can record the results of this learning, then repeat the above process, then take the average value of the results from several times’ learning.

If self-learning fails, operation panel indicates the error code, please refer to the troubleshooting table in chapter 8, find out the reason and solve it correspondingly, then start motor self-learning again.

7.5.2 Static tuning

Motor static tuning process is as follows.

1. Make contactor closed.
2. Select tuning method parameter FX-20 “1” by operation panel.
3. Jump out running contactor output Y 9 (J4-7) and COM3 (J4-6), make it closed; (If it has single star short circuited contactor, also make it closed).

By operation panel, perform the motor tuning according to Figure 7.2

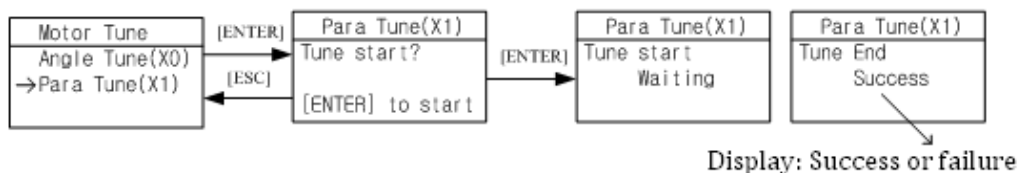


Figure 7.2 Operation diagram of motor tuning

If self-learning is successful, return to parameter setting menu, check motor phase inductance (F5-06), motor phase resistance (F5-07) etc. should be all corrected to the acquired value. Generally, one time learning success is enough. If you’d like to acquire more accurate results, you can record the results of this learning, then repeat the above process, then take the average value of the results from several times’ learning.

If self-learning fails, operation panel indicates the error code, please refer to the troubleshooting table in chapter 8, find out the reason and solve it correspondingly, then start motor self-learning again.

7.6 Motor initial degree tuning

Refer to synchronous motor, besides the above motor tuning, it is necessary to tune offset angel of magnet pole. Otherwise synchronous motor can run normally, and even slip. Therefore, tuning offset angel is very important for synchronous motor. Before proceed to running a synchronous motor, tuning must be finished successfully. After motor can run normally without load, then motor can run with load. For the tuning procedure, please refer to figure 7.3.

7.6.1 Tuning procedure for Incremental encoder

1. Make synchronous motor(traction machine) without load (don’t hang on steel ropes);
 2. Jump out running contactor output Y 9 (J4-7) and COM3 (J4-6), make it closed;
 3. Jump out brake contactor output Y6 (J4-10) and COM3 (J4-6), make it closed, brake releases;
- By operation panel, tuning offset angel of magnet pole as follows:

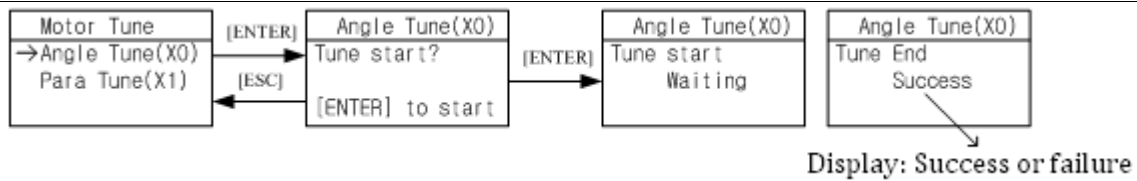


Figure 7.3 Tuning offset angel (position)

After pressing “Enter”, tuning starts. Motor vibrates at beginning or runs forward / reverse a little, then it accelerates forward to a certain speed (facing to traction sheave, anticlockwise rotation is forward direction), after 20s running in constant speed, it stops; Then it accelerates forward to a certain speed, and after 20s running in constant speed, it stops again; Then it accelerates forward to a certain speed in third time, and after 20s running in constant speed, it stops and indicates “Success”. The whole tuning procedure lasts 80 s.

After tuning is successful, perform a test run according to chapter 4 “4.4 Operation panel control of commissioning parameter”. Make traction machine accelerate forward from zero speed to rated speed and run it in constant speed for a while, observe if it runs normally or not; make traction machine accelerate reverse from zero speed to rated speed and run it in constant speed for a while, observe if it runs normally or not.

After test running by operation panel, please eliminate short circuit of “running contactor output” and “brake control output”. Press jog up button on control cabinet to perform inspection running, observe running is normal or not; Then press jog down button on control cabinet to perform inspection running, observe running is normal or not.

When jog up (down), the real running direction of cabin is down (up), please set F6-03 to correct it, 0: anticlockwise rotation is down, 1: anticlockwise rotation is up, as per the actual situation.

After the above test running is finished, hang on the steel rope and run motor with load.

Refer to the scene already hanged on the steel rope, also can use the static initial degree tuning, the processing and notes, please check chapter 7.6.2.2.

7.6.2 Tuning procedure for Sine/Cosine Encoder

Refer to synchronous motor which has been fixed with Sine/Cosine Encoder, there are two methods for tuning offset angle.

7.6.2.1 Rotation tuning

For this tuning method, make sure the motor without load and brake is released.

Procedure before rotation tuning:

1. Make synchronous motor(traction machine) without load (don't hang on steel ropes);
2. Jump out running contactor output Y 9 (J4-7) and COM3 (J4-6), make it closed;
3. Jump out brake contactor output Y6 (J4-10) and COM3 (J4-6), make it closed, brake releases;

Select tuning method parameter FX-20 “0” by operation panel (0: rotation tuning, 1: static tuning), after setting motor parameter (F5) and encoder parameter (F8) correctly, turn offset angle as the following procedure diagram Figure 7.3:

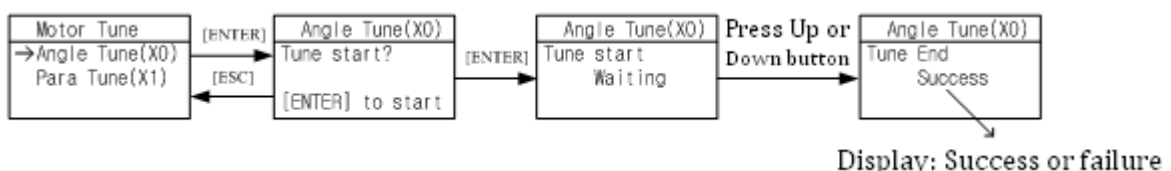


Figure 7.3 Operation procedure of rotation tuning

After pressing “Enter”, tuning starts. First, motor rotates to a firm position, then it rotates forward (facing to driving shaft, anticlockwise rotation is forward direction) in a constant speed, rotation speed and time depends on the pole no. And initial position, it stops after maximum one round

rotation, then it rotates to one position and remains for 2 s again, motor stops and indicates success. The whole tuning procedure lasts less than 20 s.

After tuning is successful, perform a test run according to chapter 4 “4.4 Operation panel control of commissioning parameter”. Make traction machine accelerate forward from zero speed to rated speed and run it in constant speed for a while, observe if it runs normally or not; Make traction machine accelerate reverse from zero speed to rated speed and run it in constant speed for a while, observe if it runs normally or not.

After test running by operation panel, please eliminate short circuit of “running contactor output” and “brake control output”. Press jog up button on control cabinet to perform inspection running, observe running is normal or not; Then press jog down button on control cabinet to perform inspection running, observe running is normal or not.

When jog up (down), the real running direction of cabin is down (up), please set F6-03 to correct it, 0: anticlockwise rotation is down, 1: anticlockwise rotation is down, as per the actual situation.

After the above test running is finished, hang on the steel rope and run motor with load.

7.6.2.2 Static tuning

For this tuning method, tuning can be carried out with steel rope hanged on the motor, but please make sure the following procedures are finished correctly at very beginning:

1. Wiring in control cabinet is completely correct, and it is under inspection state;
2. Running parameter (F1), motor parameter (F5) and encoder parameter (F8) are set correctly ;
3. All mechanical faults in hoistway have been eliminated; cabin and counterweight locate at the center of the hoistway.

Select tuning method parameter FX-20 “1” by operation panel (0: rotation tuning, 1: static tuning), turn offset angel as the following procedure diagram Figure 7.4

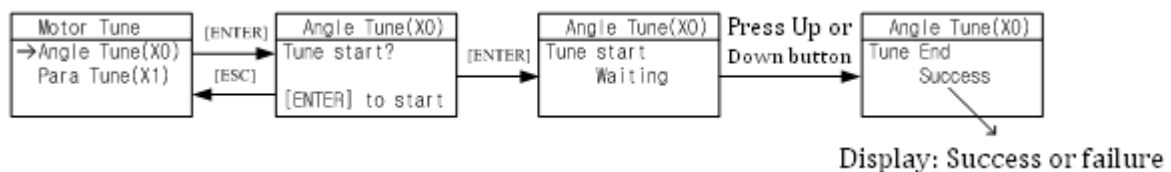


Fig. 7.4 Operation procedure of static tuning

After pressing “Enter”, tuning starts. When operation panel indicates “ running” , press jog up or down button , contactor KDY closes, motor will vibrate a little and give a noise, the lasting time depends on motor rated power and rated current, but maximum is 5 s, this is static tuning period; Make sure jog up or down button being pressed continuously, the button can’t be released during the period, motor will start and run in inspection speed , jog up or down, until operation panel indicates success, this is test running period. Finally, release the jog up or down button, finish the tuning procedure.

Please note the following items when static tuning:

1. To ensure safety, during tuning process, people is strictly forbidden to stay in cabin or hoistway;
2. Up or down running can be confirmed as per the current cabin position;
3. The whole tuning procedure can be divided to two steps, namely static tuning step and motor test running step, make sure there is no interval between two steps, if there is no default, before operation panel indicates success, press the jog up or down button continuously;
4. To achieve optimal control effect, do the above operation 5 times, if the deflection of the results is small, take the average value.

If fault occurs in tuning, please refer to chapter 8 troubleshooting table, find out the fault reason and solve it correspondingly, then repeat tuning of offset angel again.

7.7 Inspection running

7.7.1 Inspection running in machine room

1. Confirmation items before inspection running in machine room.
 - 1) Inspection switch should be at “inspection” position, inspection switch on car top and cabin should be in “normal” position.
 - 2) Safety circuit, door interlock circuit are normal, please don't jump out door interlock.
 - 3) After power on, KJT emergency stop contactor in controller, KMB door interlock contactor, KMC power contactor are closed, check if the controller is normal and parameters' setting is correct, in LCD indicator, elevator state is “INSP”.
 - 4) Connect the brake to control cabinet properly.
2. Inspection running in machine room

When the conditions for inspection running in machine room are full, press the Jog Up/Down button on the control cabinet, elevator will run in set inspection speed.

7.7.2 Inspection running on car top and cabin

If inspection running in machine room is normal, then you can perform inspection running on car top and cabin. If the up or down direction of the buttons are reverse with the actual running direction, please inspect the jog buttons' wiring, don't change the wiring in control cabinet.

7.8 Hoistway self-learning

Hoistway learning means elevator runs at a self-learning speed and measures every floor height and record the position of every switch in the hoistway. Because the floor position is the base of elevator normal starting, braking and floor display. Therefore, before normal running, hoistway self-learning must be performed at first. Before hoistway self-learning, inspection running in full trip must be performed too; it must run normally from bottom limit to top limit.

Hoistway learning procedure is as follows:

1. Make sure elevator meet the conditions of safety running;
2. Make sure all the switches in hoistway are installed and connected correctly, traveling cable and hoistway cable are connected corrected, and set the H.O.P address;
3. Make elevator enter into inspection state, jog elevator down to the bottom limit (bottom limit is valid);
4. Enter into self-learning menu by LCD indicator, operate according to clew, and refer to Figure 7.4.

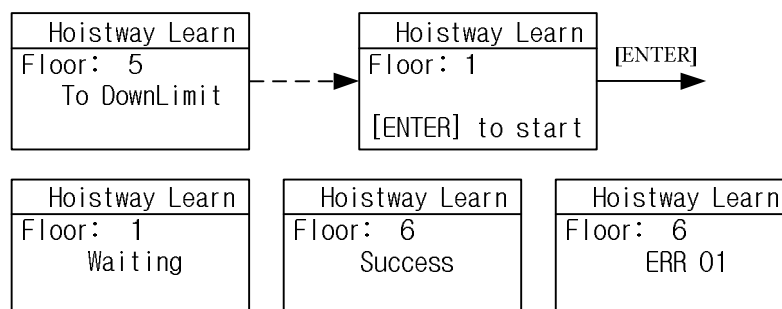


Fig. 7.4 Hoistway self-learning diagram

5. The results of self-learning can be observed in hoistway position parameter U00~U69 under monitor menu, its unit is meter, please observe the switches' position and check them correct or not after self-learning.
6. In self-learning process, if control system detects any abnormal phenomenon, self-learning will stop and give fault code, please refer to troubleshooting table in chapter 8, find out the reason and solve it correspondingly, then start hoistway self-learning again.



When self-learning process stops, only indicator shows “success”, self-learning is completed successfully.

IMPORTANT

7.9 High speed running

After hoistway self-learning is completed successfully, high speed running can be carried out.

Procedure as follows:

1. Make the elevator under attendant mode (Manual)
2. By floor selection parameter D0 on operation panel, destination floor can be set (details refer to chapter 4.5 commissioning parameter), so single floor traveling, double floor traveling, multi-floor traveling and full trip traveling can be performed. Through D1 parameter interface, input door open and close instruction to control the door.
3. Make sure elevator can start, accelerate, decelerate and stops normally in the above sections.
4. If running is abnormal, please check if there is an error of the parameters' setting.

7.10 Comfort adjustment

If comfort and leveling accuracy of elevator running are not perfect, first, please check the mechanical situation: for example, clearance of guide shoes, lubricating, steel rope, position of the rope hitch plate, etc. which influence the comfort of elevator running. After checking all the mechanical parts, then do adjustment in controller.

Because the controller control the motor running according to the given starting and braking curve, therefore, the shape of starting and braking curve, the tracking degree of motor feedback speed to running curve and the timing logic of controller signals directly influence the comfort of elevator running.

7.10.1 Adjustment of starting and braking curve

Running curve diagram Figure 7.5

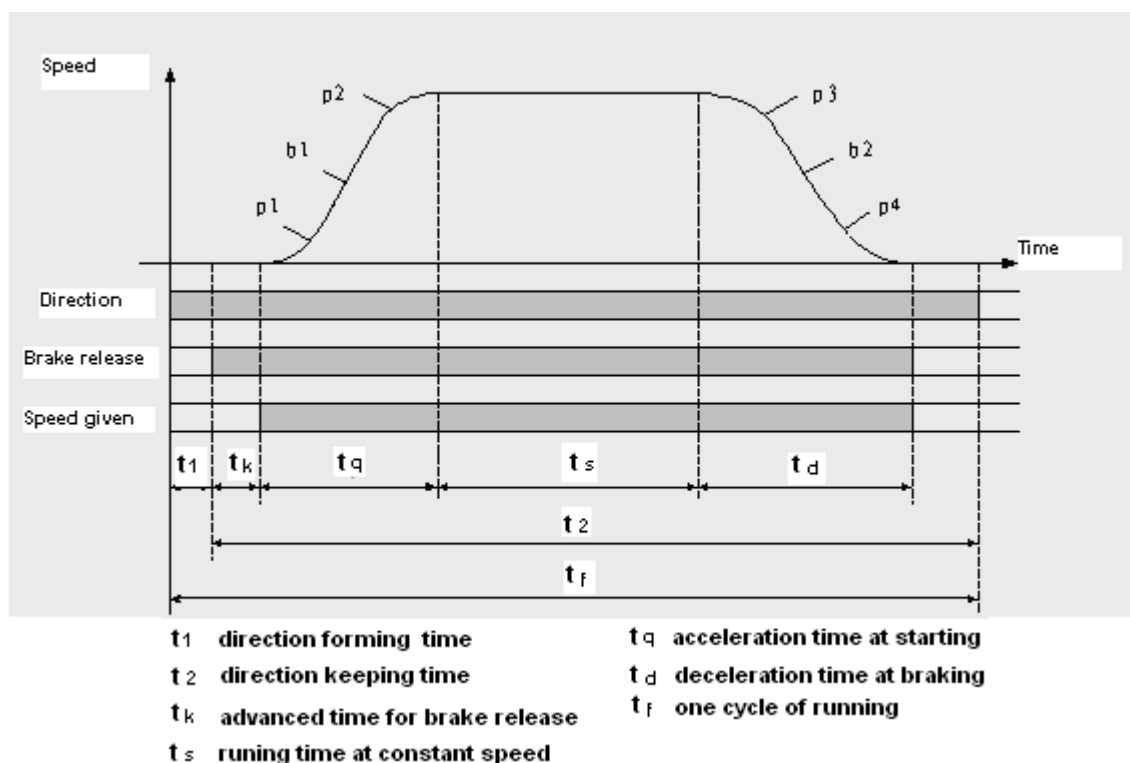


Fig. 7.5 running curve diagram

1. Three parameters adjustment for starting section of S curve:

- 1) P1: Acceleration increase in starting section, it means the rate of acceleration change, the smaller the value is, the slower the starting is, the smoother the running is, the lower the efficiency is. Conversely, the faster the starting is, the higher the efficiency is.
- 2) b1: Acceleration in starting section, it means the rate of speed change, the smaller the value is, the slower the starting is, the smoother the running is, the lower the efficiency is. Conversely, the faster the starting is, the higher the efficiency is.
- 3) P2: Acceleration decrease in ending of starting section, it means the rate of acceleration change, the smaller the value is, the slower the ending of starting section is, the smoother the running is, the lower the efficiency is. Conversely, the faster the ending of starting section is, the higher the efficiency is.

2. Three parameters adjustment for braking section of S curve:

- 1) P3: Deceleration increase in starting of braking section, it means the rate of deceleration change, the smaller the value is, the slower the starting of braking section is, the smoother the running is, the lower the efficiency is. Conversely, the faster the starting of braking section is, the higher the efficiency is.
- 2) b2: Deceleration in braking section, it means the rate of speed change, the smaller the value is, the slower the braking is, the smoother the running is, the lower the efficiency is. Conversely, the faster the braking is, the higher the efficiency is.
- 3) P4: Deceleration decrease in ending of braking section, it means the rate of deceleration change, the smaller the value is, the slower the ending of braking section is, the smoother the running is, the lower the efficiency is. Conversely, the faster the ending of braking section is, the higher the efficiency is.



In commissioning at jobsite, on the premise of running efficiency, adjust the above 6 parameters to achieve optimal elevator running curve.

7.10.2 Tracking adjustment of running curve

In-built controller must control the motor and make feedback speed strictly following the change of running curve setting to achieve expected comfort.

Because the controller establish the mathematic module according to the motor parameters input by the customers, and controller perform decoupling control by this module on motor and control its starting and braking. Therefore, customer should input the motor parameters correctly. (When motor parameter is not precise or not confirmed, we suggest customer perform motor self-learning.)

Proportion gain of the speed circle F6-04 and integral gain F6-05 or F7-05~F7-12 of PI section parameters, they influence the tracking to curve. Generally, increasing the proportion gain will improve the reaction of the system, promote the tracking speed. But big proportion gain will cause system vibration with high frequency, noise become bigger. If increasing integral gain can improve the anti-interference and tracking ability, improve the leveling precision, but big integral gain will make system vibration, it is like over adjustment of speed or vibrates and wave.

Generally, first adjust the proportion gain, on the premise of no vibration; please increase this value as more as possible. Then adjust the integral gain, make the system both with quick reaction and without big over adjustment.

In starting or stop, if running in low speed section is not perfect, they can be controlled by multi-section PI.

7.10.3 Control timing adjustment

The control timing of this system is same with the chapter 6.3, the program in the controller control the elevator performance according to timing diagram, customer can adjust any timing parameters and zero speed setting.

1. Timing adjustment

Refer to 6.3 Timing diagram under different state

2. Zero speed setting

This parameter is the threshold of zero speed. Main control board judges the braking time by this value. If this value is too big, elevator will stop with speed, if it is too small, it will cause the delay of door open.

Generally, for asynchronous motor, it is 5 RPM, for synchronous motor, it is 1RPM.

7.11 Leveling precision adjustment

Leveling precision adjustment should be done after comfort adjustment.

7.11.1 Basic conditions for elevator leveling

1. First important to ensure the correct position of leveling inductor and shelter board.
2. Shelter board length on every floor is completely same.
3. Shelter board must be installed vertically.
4. Shelter board should be installed precisely. When elevator is at the leveling position, the center of the shelter board and center of the two inductors should match together (refer to appendix), otherwise elevator leveling will have deflection, namely, in up or down running, elevator stops higher or lower than leveling position.
5. If magnetic inductor is adopted, please make sure the shelter board inserting to the inductor sufficiently, otherwise it will influence the reaction time of inductor, in that way elevator will overruns the leveling position.
6. To ensure precise leveling, there must be a crawl distance before stop.
7. In practice, first make adjustment for a middle floor, until leveling is precise. Then, adjust the other floors on the base of these parameters.

After adjustment of curve selection, ratio and integral gain in the above context, please make sure whatever elevator run up or down, stop positions are repeated.(namely every time, the deflection $\leq \pm 2\sim 3\text{mm}$).

7.11.2 Leveling parameter adjustment

If after adjustment according to 7.9.1 instruction, elevator still can't achieve perfect situation, leveling adjustment can be done by parameters, after elevator stops in normal running, if running curve has no problem (for example, no sudden stop or overrun beyond leveling), if elevator overruns the leveling position (it stops higher in up-running, lower in down-running), decrease leveling adjustment parameter F1-17 (default: 50). if elevator can't reach the leveling position(it stops lower in up running, higher in down running), increase leveling adjustment parameter F1-17, generally the range of this parameter is 40~60, if the adjustment is big, please adjust driving parameter PI, or curve (F1-10~F1-15).

7.12 Terminal switch position

Top and bottom terminal switch signal is for elevator forced deceleration and floor position adjustment, it should be installed in the position where it is motivated when elevator is 2.5m ahead of top (bottom) leveling position. (1.6m/s) Testing method is as follows:

1. Make the elevator in inspection state.
 2. Set the inspection speed at 0.3m/s, run up (down).
 3. Until top (bottom) switch is mounted, stop elevator at once.
 4. Distance between car sill and landing door sill should be $2.5\pm 0.1\text{m}$.
- Under different speed, the position of the terminal switch refer to the append

8 Troubleshooting

The contents include faults display of MU, and their possible reasons and solution.

Due to elevator system errors, Hoistway learning errors, driver errors, motor parameters setting and Encoder phasing errors.

If those errors should occur, use this section as a reference and apply the appropriate measures.

8.1 Elevator system faults detection

Elevator system error code and their possible reasons and solution are flowing table 8.1.

Table 8.1

Display	Meaning	Deal method
Er2	Door inter-lock faults: Door inter-lock circuit open when car is running	Check the work condition of car door retractable cam and landing door lock. When the cam running pass the roller, the roller should have enough clearance at both side to cam
Er3	Driver faults	The details is in table 8.3
Er4	Elevator running direction is contrary to command	1. Exchange phase "V" and "W" of motor 2. Exchange phase "A" and "B" of encoder on terminal block.
Er5	Traction machine brake is not working properly 1. After Y6 output 0.5s, no input to X17, or after Y6 output 2s, no input to X15, or 2. Y6 is not ready for output, but there is input to X15 or X17.	1. Check the traction machine brake detection switch and wiring of it; 2. Check the mechanical parts of brake, if no feedback, set OFF
Er6	During elevator running, there always has input to X9 or X10	Check the signal circuit and induction switch
Er7	During car is running, the pulse from inverter is too less	Check the wiring come from inverter for pulse
Er9	Contactor KDY is not working properly 1. After Y9 output, X16 is no feedback within 0.4s 2. Y9 is not ready for output, X16 is invalid.	Check the contactor KDY coil and its wiring.
Er10	Safety circuit open X13 and X29 is not lighting ON	Check phase relay and all safety switch
Er11	Leveling switch signal is missing: Elevator is running pass the floor, but there is not input at X9 or X10 or both of them	Check the leveling switches and its wiring.
Er12	1. Stop running by top limit switch (X5 is invalid)	Check the encoder and parameter in inverter, the up limit switch and its position and wiring.
Er13	1. Stop running by down limit switch (X6 is invalid)	Check the encoder and parameter in inverter, the down limit switch and its position and wiring.
Er14	Hoistway count error if this case happen, the car will goes to bottom floor correct the counter	1. Check the wiring arrangement as in chapter 10; 2. Check the leveling switch and working condition with flag; 3. Check the traction rope and main sheave.
Er17	The running command is being output from main board, but driver module doesn't work	Check the parameters inside of driver, or call supplier;
Er18	Hoistway count error (if this case happen, the car will goes to bottom floor correct the counter	Check the encoder and its wiring.
Er19	The deceleration distance to destination floor is not enough	1. Decrease the value of "Least Speed" in user menu; 2. Do hoistway learning again.
Er20	When elevator runs to the top floor or bottom floor and	1. Check the braking resistor specification,

	it get deceleration instruction, but it doesn't slow down; After adjust the position of limit switch, forget do hoistway learning.	change it if need; check the wiring of braking resistor; 2. Check the parameter of PI and deceleration, set the proper value; 3. Do hoistway learning again.
E21	Running time is over setting time	1. Check the traction rope, dose it slid on sheave or not. 2. Check parameter in drive; 3. Check the value "Over Time"
Er22	There is inspection instruction input at X0 during high speed running	Check inspection switch and circuit
Er23	One of two leveling switch is invalid	Check leveling switch and the wiring
Er25	Temperature switch for braking resistor, the resistor or the motor is over heat (X21 is invalid)	Check the wiring; when the fault occurred 90s, still cannot restore, main board Y10 relay output, then closed contactor KMC
Er26	Door inter-Lock contactor working state does not confirm to its coil (X14 and X30 are input different)	1. Check the wring of this contactor 2. Check the contactor for X14, is there any dust inside of it; 3. Check the surge absorb for this contactor; 4. If door inter-lock is coming and X14 is light ON, you already have done all above, but X30 is not ON, call supplier.
Er27	Safety contactor KMB working state does not confirm to its coil (X13 and X29 are input different)	1. Check the wring of this contactor; 2. Check the contactor for X13, is there any dust inside of it; 3. Check the surge absorb for this contactor; 4. If door inter-lock is coming and X13 is light ON, you already have done all above, but X29 is not ON, call supplier.
Er28	Slow down or second slow down switch on top or bottom floor always in close state X7 or X8 always are invalid in installing floor	1. Check the switch and its working state with cam; 2. Check the switch in shaft installation position.
Er29	There electrical noise on communication cable level is too high to system	1. Correct actions detail in chapter 8.3 2. Check the hall station and COP communication board.
Er30	Open the door fault (car is not open the door)	1. Check the door open order if Y4 output or not 2. If Y4 has not output, check the set for installation of the limit switch and valid type of the signal. 3. For the double through opening, check the front/back door if it sets opposite or not.
Er31	Close the door fault	Generally, the car door is not installed, then blocked it forced the door inter-lock, please check if the door close output is consistent with door inter-lock contactor running or not.
Er32	Protect the fault in calculate the floor level	Peripheral station and limit are power failure instantly, can lead to building date error, after this fault, the elevator will return to the main floor to count.
Er33	Motor star short circuited contactor fault	KDY fault: KDY order is different with feedback: 1. Y8 output, X11 is no feedback within 0.4s 2. Y8 has not output, X11 is valid 3. Y8 cancelled, X11 is no feedback within 0.4s

8.2 Hoistway learning faults detection

Hoistway learning error code and their possible reasons and solution are flowing table 8.2.

Table 8.2

Display	Meaning	Probable Causes	Corrective Actions
LER=0	System running protection		Press 'esc', check fault record as in table 8.1
LER=1	Pulse input phase reverse		Exchange A phase and B phase
LER=2	Bottom slow down switch signal shaking		Check the switch and its work condition with cam
LER=3	Bottom slow down switch fault (only for rated speed >2m/s)	1. The bottom slow down switch is always valid or always invalid 2. The bottom second slow down switch is valid before slow down switch	Check the switch and its installation position
LER=4	Bottom second slow down switch signal shaking		Check the switch and its work condition with cam
LER=5	Bottom second slow down switch fault (only for rated speed >2m/s)	The up second slow down switch is valid before bottom second slow down switch	Check the switch and its installation position
LER=6	Up second slow down switch signal shaking		Check the switch and its work condition with cam
LER=8	Up second slow down switch fault (only for rated speed >2m/s)	1. The up second slow down switch is always valid or always invalid 2. The up slow down switch is valid before up second slow down switch	Check the switch and its installation position
LER=9	Bottom slow down switch fault	1. The bottom slow down switch is always valid or always invalid 2. The up slow down switch is valid before bottom slow down switch	Check the switch and its installation position
LER=10	Up slow down switch signal shaking		Check the switch and its work condition with cam
LER=11	Up slow down switch fault	1. The up slow down switch is always valid or always invalid 2. The up limit switch is valid before up slow down switch	Check the switch and its installation position
LER=12	Total floors setting is error	Total floors setting is found error when doing Hoistway learning	Check total floor setting in main board is according to quantity of leveling inductor plate or not
LER=14	up and down leveling switch position in the plate at same time		1. Check the up and down leveling switch and their wiring 2. Check the door zone plate
LER=15	Press 'esc' to cancel hoistway learning		press 'esc' key
LER=17	up and down leveling switch signal come into system at same time	1. wire of two door zone switch is parallel connected together by mistake, or 2. Bottom limit switch is installed close to 1st floor leveling position	
LER=18	Save data in error	After Hoistway learning ,then cannot save the date successfully	Contact with us

LER=19	Arrival at top limit, two leveling switch signal input at same time		1. Check the inspection speed value is less than 0.3m/s or not; 2. Check the top limit switch installation position; 3. Check the top floor leveling floor door zone plate
LER=20	Bottom limit switch installation position is too high		1. Check the inspection speed value is less than 0.3m/s or not; 2. Check the top limit switch installation position; 3. Check the top floor leveling floor door zone plate
LER=21	In hoistway learning process, when elevator run to up limit switch, bottom slowdown switch or bottom second slow down switch is still valid.		Check the switches installation position and their wiring.
LER=22	In hoistway learning process, when elevator start running from bottom limit switch, up slow down switch or up second slow down switch is still valid		Check the switches installation position and their wiring.

Notice:

When the rated speed of elevator is equal or more than 2.0m/s, we should install the second slow down switch at both up and down floor in shaft.

8.3 Driver faults detection

Driver error code and their possible reasons and solution are flowing table 8.3.

Table 8.3

Code	Display	Meaning	Probable Causes	Corrective Actions
DF	UV	DC bus overvoltage	1. Open phase of input supply; 2. Instantaneous power failure; 3. excessive voltage fluctuation of input supply; 4. loose terminals of input supply; 5. Pickup current load in a same power supply system.	1. Inspect input supply; 2. Correct the voltage to normal, 3. Reset and restart; 4. Check input wiring; 5. Change power supply system into a conformable one.
DF2	OV	DC bus overvoltage	1. too short deceleration time, 2. excessive motor regenerated energy; 3. too high supply voltage 4. doesn't connect the braking resistor	1. increase deceleration time; 2. connect brake resistor; 3. Lower voltage in the supply range.
DF3	OH	Heat sink overheated	1. excessively elevated ambient temperature, 2. damaged cooling fan, 3. Existence of heating source around.	1. make cooling unit available; 2. remove heating source around 3. Check the fan and wiring
DF4	IF	IPM fault	1. instantaneous current, 2. shorted output three phases, 3. abnormal power module supply, excessively elevated ambient temperature	contact supplier for service
DF5	OC	Over current The Inverter's output current exceeded the over current detection level.	1. Shorted Inverter output phase-to-phase, shorted motor, locked rotor, 2. too heavy load, accel/decel time too short, 3. Magnetic contactor on the Inverter output has opened or closed a special motor or a motor with a rated current larger than the Inverter's output current is used.	1. Remove the motor and run the Inverter without the motor. 2. Check the motor for a phase to phase short. 3. Verify the accel/decel times 4. Check the Inverter for a phase to phase short at the output.
DF6	CF	CPU faults	Too high level magnetic interface	seek technical support
DF7	OS	Elevator over speed. The speed feedback exceeds the allowable speed	1. The motor speed feedback exceeded the setting for a time longer than the setting 2. Overshooting/Undershooting occurs. 3. The feedback of encoder is not good	1. Check the speed setting 2. Check the P/I parameter 3. Check encoder
DF8	OE	Excessive speed deviation	1. overloaded; 2. too short of acceleration/deceleration time 3. the parameter deviation value setting is not correct 4. encoder cannot work properly	1. reduce the load; 2. Increase acceleration/deceleration time. 3. Check the parameter 4. Check the encoder
DF9	PGO	PG Disconnection	1. broken wiring of PG; 2. wrong wiring of PG; 3. defective PG card hardware	1. check wiring; 2. replace PG card

DF10	FF	Flash memory fault	When saving parameter, the data is not correct	seek technical support
DF11	BF	Base block circuit error A base block circuit error occurred at power-up.	1. Wiring for base block is incorrect 2. Setting electric level for base block is incorrect	1. Check the wiring 2. Modify the parameter
DF12	OL	Motor overload	1) power network voltage low 2) improper motor rated current 3) motor rotation blocked, or sudden load change 4) close loop vector control, PG connection reversed and long term low speed running 5) drive capacity far larger than the load	1) check power network voltage 2) reconfigure motor rated current 3) check the load, adjust torque gain 4) adjust PG signal direction 5) use a proper motor
DF13	MC	Contactor MC Not Closed	1) power network voltage low 2) contactor damaged 3) soft start resistor damaged 4) control circuit damaged	1) check power network voltage 2) replace the power circuit contactor or seek technical support 3) replace the soft start resistor or seek technical support 4) seek technical support
DF14	BR	Brake unit fault	1. defective brake cable or damaged brake elements; 2. External brake resistor disconnected or not connected.	check brake resistor and its connection
DF15	OF	Output Open-phase An open-phase occurred at the Inverter output.	There is a broken wire in the output cable. The motor winding is broken. The output terminals are loose.	Reset the fault after correcting its cause.
DF16	SCF	current flow exists due to ineffective shut down	Drive is broken	Change the driver
DF17	SRF	Slip after stop	1. loose band brake causes elevator sipping; 2. interfered or loosened encoder	1. check band brake; 2. Fasten encoder; remove interference.
DF18	UF	Signal U of encoder wire breakage	broken any U phase of encoder broken	check encoder and its wiring
DF19	VF	Signal V of encoder wire breakage	broken any V phase of encoder broken	check encoder and its wiring
DF20	WF	Signal W of encoder wire breakage	broken any W phase of encoder broken	check encoder and its wiring
DF21	DF	Parameter setting error	Parameter setting error	Check parameter setting
DF22	SDF	Inner programmer self check error	Inner data setting error	Check with supplier

8.4 Encoder phasing faults detection

Encoder phasing error code and their possible reasons and solution are flowing table 8.4, table 8.5 and table 8.7.

Incremental encoder phasing error code

Table 8.4

Code	Meaning	Probable Causes	Corrective Actions
RF1	Rotating auto tuning error	1. There are loads on motor or brake is not open 2. Motor phase is not correct 3. Encoder damaged or wiring is un correct	1. Move the loads out of sheave and open the brake 2. Correct the phase of motor 3. Check the encoder and wiring
RF2	Auto tuning data incorrect	1. The motor parameter setting error 2. Encoder damaged or its wiring is incorrect 3. High electric interface	1. Correct the motor parameter setting 2. Check encoder and its wiring 3. Decrease the level of electric interface
RF3	Z_SRCH_ERR (permanent magnet motor tuning only)	1. Encoder damaged 2. Encoder wiring is incorrect	1. Check encoder 2. Correct the wiring of encoder
RF4	Auto tuning data is not fully correct	1. Motor parameter setting incorrect 2. Encoder and its wiring is incorrect 3. Auto tuning motor with loads	1. Check motor parameter setting 2. Check encoder and its wiring 3. Move the loads away from the sheave
RF5	Auto tuning UVW repeated input	The encoder UVW have short circuit or disconnected	Check the encoder and wiring
RF6	Motor can not rotating normal	1. Motor with load 2. Motor sequence is fault	1. Check the motor is no-load or not 2. Check the motor sequence is correct or not
Others	Check the drive is fault or not	If the drive is fault, it can not tuning, and give the fault code directly	Check the drive is fault or not

Sin/Cos encoder phasing error code

Table 8.5

Code	Meaning	Probable Causes	Corrective Actions
RF232	Auto tuning motor doesn't rotate	1. Encoder connection is fault 2. There are loads on motor or brake is not open 3. Motor phase is not correct 4. Encoder damaged or wiring is un correct	1. Connect the encoder signal correctly 2. Move the loads out of sheave and open the brake 3. Correct the phase of motor and AIO 4. Check the encoder and wiring
RF233	Motor rotating in wrong direction	Motor sequences doesn't correspond with encoder	1. Adjust motor sequence 2. Adjust encoder A-, A+ or B-, B+
RF234	Encoder Z_SRCH_ERR	1. Cannot check Z_SRCH 2. There are loads on motor has wrong connect	1. Check the encoder Z_SRCH wiring 2. Motor is no-load 3. Correct the phase of motor and AIO

Static Angle error code

Table 8.7

Code	Meaning	Probable Causes	Corrective Actions
PF235	No Z_SRCH input	After motor running 7s, also can not detect Z_SRCH	If motor can running, check the encoder wiring; otherwise, find out the cause
PF236	Internal counting errs.	Internal counting errs.	1. Check the parameter input 2. Correct the phase of motor and AIO
PF237	Motor have not fixed	Motor brake is not open or brake is too loose	Check the motor brake is strictness
	Tuning result has too large errors	After running 3 laps, tuning result has large errors, then quote fault	After check the motor and encoder wiring, then try again
PF238	Detection current is too small	Load side connect is error	Correct the phase of motor and AIO
PF239	Tuning result has too large errors	After running 3 laps, tuning result has large errors, then quote fault	After check the motor and encoder wiring, then try again
PF240 ~ PF249	When the incremental PG tuning, UVW signal error	The encoder UVW have short circuit or disconnected	Check the UVW wiring with encoder

8.5 Auto tuning motor faults detection

Auto tuning motor error code and their possible reasons and solution are flowing table 8.6 and table 8.8. (Including rotating and static parameters tuning)

Motor parameters rotating tuning error

Table 8.6

Code	Meaning	Probable Causes	Corrective Actions
PF2	Doesn't set motor parameters or input wrong motor parameters	1. Doesn't set motor parameters or input wrong motor parameters 2. Load side open or contact undesirable	Setting correct parameter in dicer and verify it
PF3	Line-to-Line Resistance Fault	1. Input motor data is incorrect, 2. wire is not secured on terminal block	1. Check the input data. 2. Check the motor wiring and secured them on terminal block.
PF4	Leakage Inductance Fault	1. Input motor data is incorrect 2. Auto tuning motor with loads	1. Check the input data 2. Move the loads out of sheave
PF5	Auto tuning, motor doesn't rotate	1. Input motor data is incorrect 2. Wire is not secured on terminal block 3. PG card damaged or wiring on it is wrong	1. Check the input data 2. Secure the wire on terminal block 3. Check PG card and its wiring
PF6	Motor running speed cannot reach the full speed or rotation is wrong	1. Input motor data is incorrect 2. Motor input phase is incorrect 3. Encoder or PG card or wiring is incorrect	1. Check the input data 2. Verify motor input phase 3. Check encoder and PG card and its wiring
PF7	No-Load Current Fault	Auto tuning motor with loads	Move the loads away the sheave.

Motor parameters static tuning error

Table 8.8

Code	Meaning	Probable Causes	Corrective Actions
PF2	Initial failure	1. Motor parameters is wrong 2. Load side open or contact undesirable	Setting correct parameter in dicer and verify it
PF3	Line-to-Line Resistance Fault	1. Input motor data is incorrect, 2. Wire is not secured on terminal block	1. Check the input data. 2. Check the motor wiring and secured them on terminal block.
PF4	Leakage Inductance Fault	1. Input motor data is incorrect 2. Load side connect is wrong	1. Check the input data 2. Setting correct parameter in dicer and verify it
PF237	During the initial position, motor have not fixed	Motor brake is not open or brake is too loose, during the initial position, the traction sheave displacement.	Check the motor brake is strictness
PF238	Detection current is too small	Load side connect is error	Correct the phase of motor and AIO

9Maintenance

This chapter describes precautions of drive storage and maintenance. Section 9.3 explains quick wear parts' life and their replacements. Please read carefully this chapter and use your drive properly.

Danger

Mis-operation over a running drive may cause high-volt electric shock!
Dangerous high-volt still exists in a drive just powered off.
Only trained & authorized personnel can maintain the drive.
Before the maintenance, take off all metal objects such as wrist watch and ring. The cloths and tools must meet insulation requirements.

Attention

Before the maintenance, check the following items before you can touch the drive's internal parts or electric shock may occur.
Cut off the drive's power and wait for at least 5mins.
Do not open drive's cover until all keypad indicators are off.
The CHARGE indicator at bottom right inside the drive is off.
The voltage between power terminals (+) and (-) is below 36Vdc.
Only professionals can replace drive parts. Do not leave any alien things in the drive, or fire may occur.
After replacing the control board, set parameters properly before running the drive.

The drive and its driving mechanism need routine checking and maintenance during storage and operation.

The drive's installation and running environment must comply with the regulations specified in this manual.

In daily use, clean the drive internal parts of dust regularly, check for loosened screws, keep the daily operation data, parameter setting and modifications for reference.

Through routine maintenance, you can discover abnormalities, locate the faults and remove them in time to ensure safe operation and prolong the drive service life.

See Table 9-1 for the routine check list.

9.1 Routine Maintenance

Potential hazards exist due to aging, wear and tear of drive internal components as well as environmental influences to:

Table 9.1 Routine check list

Items	Check			Criteria
	Content	Time	Means/Method	
Working environment	1) Temperature, humidity 2) Dust, vapor, water dripping 3) Gases	At any time	1) Point thermometer, hygrometer 2) Observe & smell	1) Ambient temperature lower than 40°C or derating is a must. Humidity compliant with the related requirement. 2) No dust, no dripping, no condensation 3) No abnormal gases or smell
Drive	1) Vibration 2) Ventilation and temperature 3) Noise	At any time	1) Observe 2) Point thermometer and observe 3) Listen	1) Smooth operation without vibration 2) Normal fan operation, with normal wind speed and volume. No abnormally hot parts 3) No abnormal noise
Motor	1) Vibration 2) Heat 3) Noise	At any time	1) Watch and listen 2) Point thermometer 3) Listen	1) No abnormal vibration or sound 2) No abnormally hot parts 3) No abnormal noise
Operation state parameters	1) Input voltage 2) Output voltage 3) Output current 4) Internal temperature	At any time	1) Voltmeter 2) Rectifying voltmeter 3) Ammeter 4) Point thermometer	Refer to the related specification Refer to the related specification Refer to the related specification Temperature rise by no more than 40 C

9.2 Periodic Maintenance

In terms of application environment, you need to check the drive regularly at the interval of 3 ~ 6 months or less to ensure long and safe operation of the drive.

Note:

1. Only authorized technicians can maintain the drive.
2. No iron ware such as screw, washer, cable or tools should be left inside the drive, or the drive may be damaged.
3. Do not take the liberty of altering driver internal components or layout.
4. Do not touch directly the electrostatic sensitive IC elements on the control board.

Check points:

1. Check and secure the screws on control terminals
2. Check the power circuit terminals and wires for bad connection or overheating.
3. Is there any damage on power cables, especially scratches on cable insulation layers?
4. Is the insulation tape on power cable lugs still fine?
5. Check and clean (with vacuum cleaner) the dust off PCB and air duct.
6. Before the direct-to-ground test, remove all drive-mains & drive-motor connections and connect all the power circuit input/output terminals to the ground.
 - 1) Make sure that the 500V Meg-ohmmeter (or suitable insulation tester) is faultless.
 - 2) Insulation test of single power circuit terminal to ground is forbidden, for it may damage the drive.
 - 3) Insulation test to control terminals is also forbidden, for it may damage the drive.
 - 4) After the test, do remove all the shorting wires connected to the power circuit terminals.
7. Before the motor insulation test, you must disconnect the motor from the drive, or the drive may be damaged.

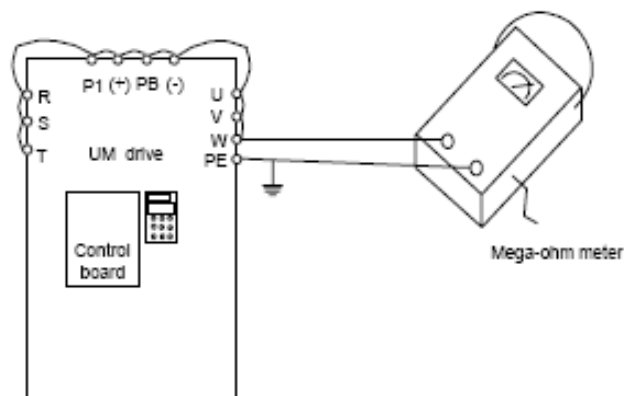


Figure 9.1 Drive insulation test



Drive's dielectric strength has been tested in the factory. Do not test again, or damage may occur.

9.3 Quick Wear Parts

The quick wear parts of drive include the cooling fan and filtering electrolyte capacitor, whose service life depends on application environment and maintenance.

Component	Service life	
Fan	30 ~ 40 thousand hs	Replace with new part.
Electrolyte capacitor	40 ~ 50 thousand hs	Replace with new part. (Determine need by inspection.)
DC bus fuse Control power fuse	10 years	Replace with new part.
PCB capacitors	5 years	Replace with new board. (Determine need by inspection.)

To ensure normal operation, you need to replace the quick wear components at regular intervals.

The new component to be used should be of the same model as the old one.



The new parts should be of the same model and electric spec. with the previous or the drive may be damaged.

The replacement interval should be determined by the actual running hour of your drive. However, if a component is faulty through checking, replace it right away.

1. Cooling fan

Possible failure causes: bearing wear-out, vane aging.

Criterion:

When drive is cut off power, check the vanes and other fan parts for cracks and any abnormalities.
When the drive is on, check the fan running state for abnormal vibration.

2. Electrolyte capacitor

Possible failure causes: electrolyte aging due to ripple current increase caused by high ambient

temperature and frequent load change

Criterion:

- 1) Whether overvoltage/current at drive's loaded startup is frequent.
- 2) Any capacitor liquid leakage.
- 3) Is the safety valve popped out?
- 4) Check the static capacitance and insulation resistance.

9.4 Storage



The power-on of a drive that has been idle for more than 2 years requires slow voltage rise with a booster, otherwise electric shock or explosion may occur.

1. Requirements on storage environment are in Table 9.2.

Table 9.2 Storage environment

Item	Requirement	Remark	
Ambient temp.	-40°C ~ +70°C	Ambient temp. should be lower than 30 C for long term storage, or capacitor performance may determinate	Free from condensing or frozen environment
Relative humidity	5 ~ 95%rh	Protective measures including plastic film coverage and use of desiccant	
Others	No direct sunlight, no dust, no corrosive/explosive gases, no oil fog, no vapor and water dripping, no vibration and little salty substance		

2. Electrolyte capacitor deteriorates in performance after long term storage, therefore should be energized regularly for preservation. If the drive is to be idle for a long time, it is recommended to energize it for more than 0.5hr every 6 months lest its components should become unusable.

9.5 Warranty

MU Power will provide warranty repair services for any of the following occurs to the drive.

The drives are guaranteed against defects for a period of 18-month from the date of delivery from the manufacturer.

After 18 months, repairs will be charged. However, even within the 18-month warranty period, repair cost will be charged in the following cases:

- 1) Damage caused by misoperation – not following manual guidance
- 2) Damage due to fire, flood or abnormal voltage
- 3) Damage due to abnormal application of the drive

Service charges will be calculated based on actual costs. If there is any service contract, the contract prevails.

10Drive EMC Installation Guidance

From such aspects as noise control, wiring requirement, grounding, peripheral equipment surge absorption, leakage current, EMC Zoning, installation precaution, use of supply filter and disposal of radiated noise, this appendix describes the elevator integrated drive controller EMC design and installation guide for reference by elevator integrated drive controller users.

10.1Noise Control

As determined by operating principle, the elevator integrated drive controller produces certain noises. The effect of noise on peripheral equipment is relevant to the noise type, transmission path as well as the design, installation, wiring and grounding of drive system.

10.1.1Noise Type

Noise types are shown by Attached figure 10.1.

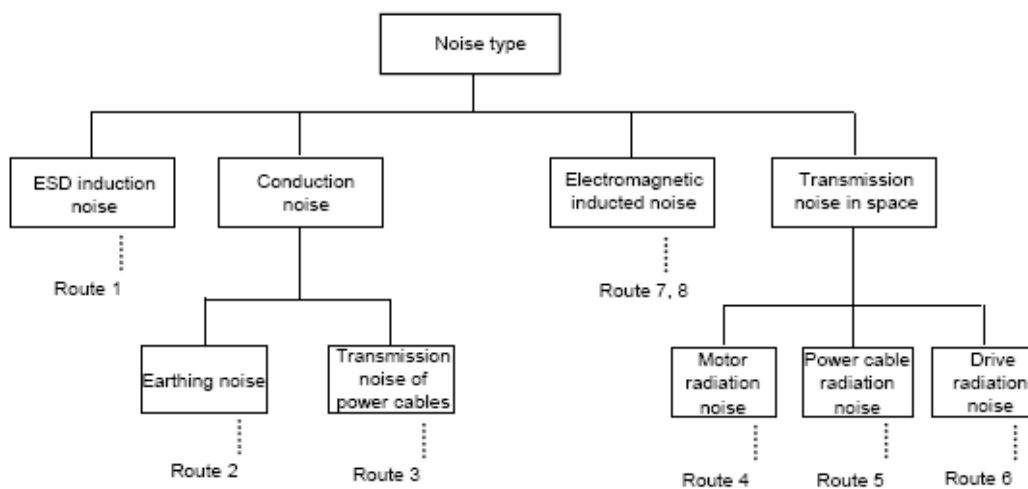


Figure 10.1 Noise type

10.1.2Transmission Path

See attached figure 10.2 for transmission path of noise.

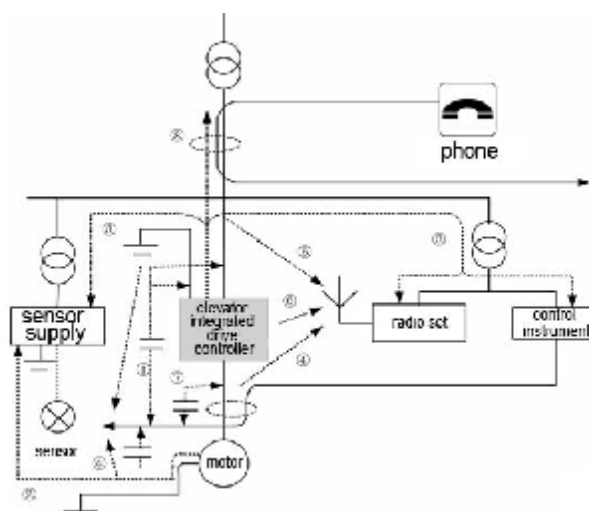


Figure 10.2Noise transmission paths

10.1.3 Basic Measures for Noise Suppression

See attached Table 10.1 for basic noise suppression measures.

Table 10.1 Basic Noise Suppression Measures.

Noise Transmission path	Possible disturbance	Suppression method
1) 7) 8)	When the signal cables are parallel to, or bound together with the power cables, the static and electromagnetic induction will cause the noise transmit through the signal cable, misoperating the related equipment.	1) Avoid laying the signal cables parallel to the power cable, or binding them together. 2) Keep the susceptible peripheral equipment away from the drive. 3) Keep the susceptible signal cables away from drive's I/O cables. 4) Use shielded cable as the signal and power cable. In addition, you can put the cable into metal tubes. Keep the tube-to-tube clearance at least 20cm.
2)	If a loop is formed between peripheral equipment and the drive cables, the grounding leakage of the drive will misoperate the equipment.	Removing the grounding of the peripheral equipment will solve the problem.
3)	When peripheral equipment shares the same power source with the drive, the noise transmitted through the power line may misoperate the peripheral equipment.	Mount a noise filter at drive input side, or isolate the peripheral equipment with an isolated transformer or power filter.
4) 5) 6)	Electronic equipment such as computers, measuring meters, radio equipment and sensors, when in the same cabinet with drive, with their wiring close to the drive, may misoperate due to radio interference.	1) Put the susceptible peripheral equipment and their signal cables away from the drive. Use shielded cable as the signal wire. Ground the shielding coat. Protect the signal cable with a metal tube and keep it off the drive's I/O cable. When crossing of the signal line and drive I/O cable is inevitable, make it orthogonal. 2) Mount radio noise filter or linear noise filter (ferrite common-mode choke) to the input/output side of the drive to suppress the radio noise. 3) The shielding coat for the drive - motor cable should be thick. Lay the cable in a tube thicker than 2mm or bury it in a cement conduit. The cable should be through a metal pipe, and has its shielding coat grounded. You may use a 4-core cable as the motor power cable. One core should be grounded at the drive side at one end, and be connected to the motor's PE at the other end.

10.2 Requirements on Cable Laying

10.2.1 Requirement on Cable Laying

As shown by figure 10.2.1(a): To prevent mutual coupling of interferences, control signal cable must be laid separately with supply cable and motor cable and be spaced as far as practicable. As shown by figure 10.2.1(b): where the control signal cable has to cross over supply cable or motor cable, they must be perpendicular with each.

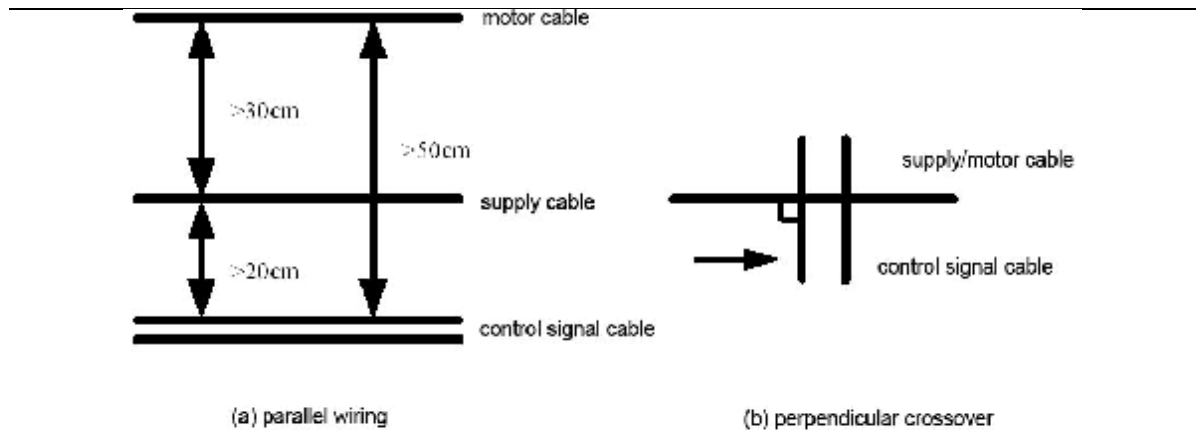


Figure 10.2 requirements on cable laying

10.2.2 Requirements on Cable Size

Greater cable size causes greater ground capacitance and greater ground leakage current. Therefore, motor cable with excessive cross-sectional area shall be used by derating so as to reduce output current (each increment of cross sectional area makes current decreasing by 5%).

10.2.3 Requirements on Shielded Cable

High frequency low-impedance shielded and armored (such as weaved copper wire mesh or, aluminum wire mesh) cable shall be used.

10.2.4 Requirements on Installation of shielded cable

Control cables are mostly shielded cables which shielding-purpose metal wire mesh must be connected to metal casing by 360° girth jointing method with cable clip at both ends, as shown by figure 10.2.2. In addition, the figure 10.2.3 shows an incorrect shield grounding method.

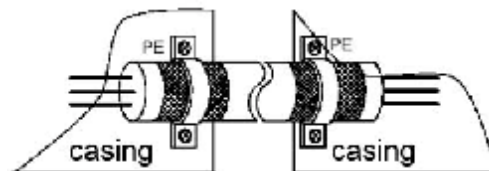


Figure 10.2.2 Correct Shield Grounding Method

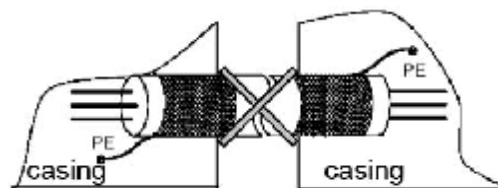


Figure 10.2.3 Incorrect Shield Grounding Method

10.3 Grounding Requirements

10.3.1 Grounding Method

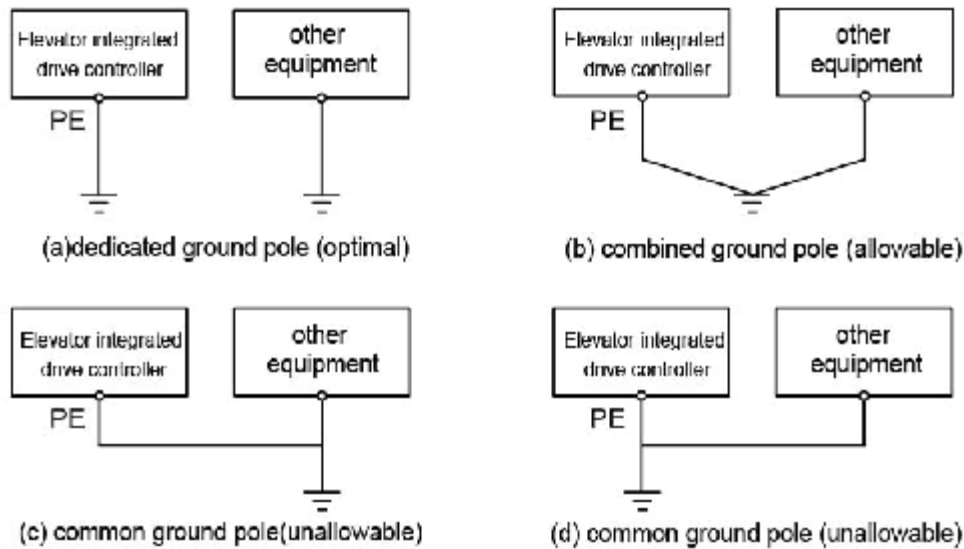


Figure 10.3.1 Dedicated Ground Pole

Among the above-shown four figures, (a) shows the optimal grounding method. Users are advised to use this method as practicably as possible.

10.3.2 Precautions for Grounding Connection

- (1) Use grounding cable of standard cross section as far as possible in order to minimize the grounding impedance; due to flat cable has smaller high-frequency impedance than round-section cable, therefore, flat cable with identical cross-sectional area will be more preferable.
- (2) The ground cable should be as short as possible and the grounding point should be close to the elevator integrated drive controller as practicably as possible.
- (3) If four-core cable is used for motor line, one of the four core conductors must be grounded in such a way that one end of it is grounded by the side of elevator integrated drive controller and another end is connected to the ground terminal of motor. The optimal grounding effect may be achieved if each of both motor and the elevator integrated drive controller has individual dedicated ground pole.
- (4) Where all ground terminals of system components are combined together, the noise source caused by ground leakage current will affect the other peripherals of elevator integrated drive controller. Therefore, in a same control system, the grounding for elevator integrated drive controller shall be separated from those for weak voltage equipment such as computer, sensor or audio devices.
- (5) To get smaller high-frequency impedance, the anchor bolts of equipment may be used as high-frequency terminal connecting rear panel of cabinet. Note to remove insulation coating around the securing point in installation.
- (6) The grounding cable must be laid far away from the I/O wirings of noise sensitive equipment. In addition, the grounding cable must be as short as possible.

10.4 Install Surge Absorption Device

Even if they are installed outside or elevator integrated drive controller, such heavy noise production devices as relay, contactor and electromagnetic brake must be additionally provided with surge suppressor. As shown in Figure 4.1.

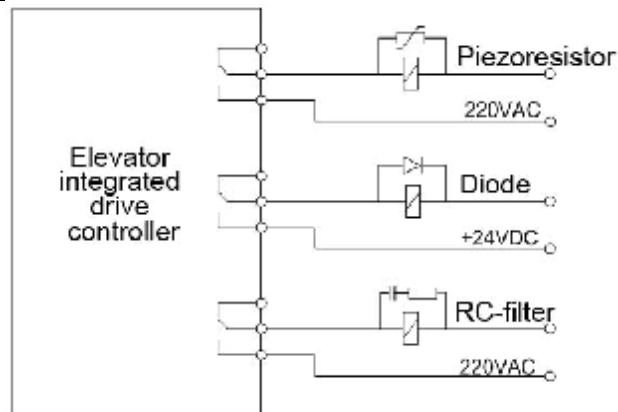


Figure 10.4 Use Requirements of Relay, Contactor and Electromagnetic Brake

10.5 Leakage Current and Countermeasures

Figure 10.5.1 shows the wiring capacitance, motor capacitance, ground leakage current and line-to-line leakage current by the input/output side of elevator integrated drive controller. The intensity of leakage current depends upon the carrier frequency and capacitance.

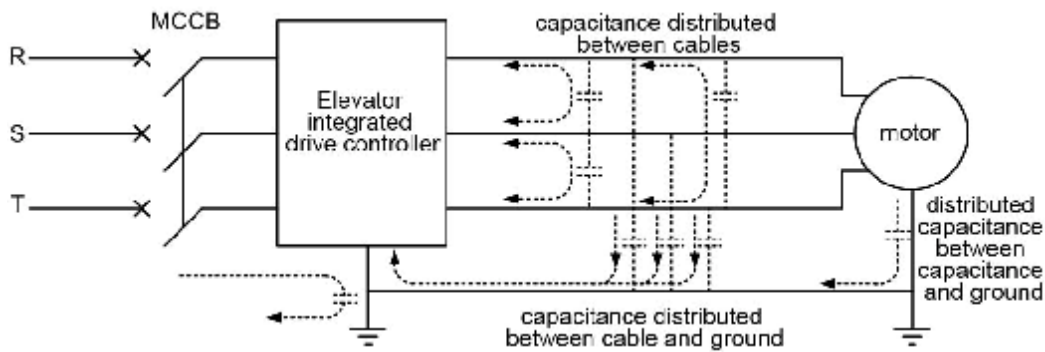


Figure 10.5.1 Paths of Leakage Current

10.5.1 Ground Leakage Current

Ground leakage current not only flows by elevator integrated drive controller but also flows by other equipment via the ground wire. As a result, it may cause malfunction of leakage protection circuit breaker, relay or other equipment. The leakage current increases with the increment of carrier frequency of elevator integrated drive controller and the extension of motor cable.

Suppression measures: reduce carrier frequency; shorten motor cable as practicably as possible; use leakage protection circuit breaker specially designed for leakage current due to higher harmonic/surge.

10.5.2 Line-to-Line Leakage Current

As for leakage current flowing by the distributed capacitance between I/O side cables of elevator integrated drive controller, its higher/sub harmonics may cause malfunction of external thermo relay. Especially when small capacity elevator integrated drive controller under 7.5Kw is used, and at the same time, long wiring (more than 50m) is provided, the increased leakage current is liable to make malfunction of external thermo relay.

Suppression Measures: reduce carrier frequency; install AC output reactor at the output side; monitor the motor temperature directly with the temperature sensor; or use electronic thermo relay, the motor overload protection integrated to elevator integrated drive controller instead of external thermo relay,

10.6 Suppression of Radiated Emission

Elevator integrated drive controller is usually installed in metal control cabinet, the external instruments and equipments are lightly affected by radiated emission of elevator integrated drive controller. Therefore, the external connecting cables are deemed as the main source of radiated emission. Due to the supply cable, motor cable as well as the control cable and keyboard cable are all necessary to be led outside of the shielding cabinet; the cable outlet must be so treated specially to avoid impairing the shielding effects.

In Figure 10.6.1: the cables in the shielded cabinet acts as an antenna, which absorbs the radiated noises in cabinet, transmits outside via cables and emits them to the space; in Figure 10.6.2: ground the cable's shielding layer at the outlet of the shielded cabinet, thus the noise radiation absorbed by the cable will flow to earth directly via the shielding cabinet so as to eliminate its effect on external environment.

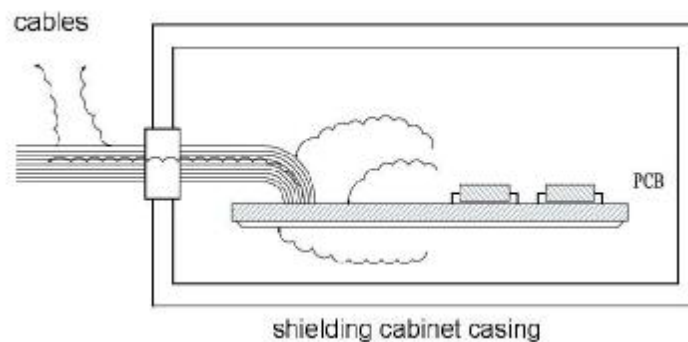


Figure 10.6.1 Radiation transmitted through cables

When using the shielding layer grounding method as shown in figure 10.6.2, the shielding layer must be grounded as close to the cabinet as possible, otherwise the sectional cable from grounding point to the outlet of cabinet will still act as an antenna coupling. The noise grounding point shall not be more than 15cm from cable outlet of cabinet, and smaller clearance will always be more preferable.

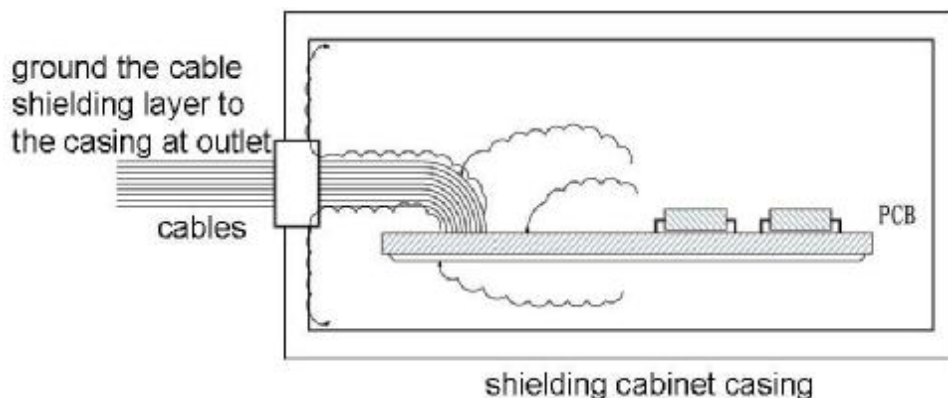


Figure 10.6.2 Suppression of radiation transmitted through cables

10.7 Guide for Use of Power line filter

Power line filter may be used by heavy-noise protection devices and noise susceptible devices.

10.7.1 Functions of Power line filter

(1) The power line filter is two-way low pass filter which only permits flowing of DC current or 50HZ operating frequency AC current but stops flowing of higher frequency electromagnetic interference current. Therefore, it can not only inhibit the equipment's electromagnetic interferences flowing into power line but also inhibit the noises in power line flowing into equipment.

(2) The power line filter may be able to make equipment meeting EMC requirements on conducted emission and transmission sensitivity. In addition, it may also suppress the radiated interference of equipment.

10.7.2 Precautions for Installation of Power Line Filter

(1) In the cabinet, the filter shall be located close to the power line inlet as practicably as possible. Additionally, the filter supply line section left in the control cabinet must be as short as possible.

(2) Where the input line and output line of filter is too closely laid, the high-frequency interference may bypass the filter, couple directly with output line and input line and as a result make the power filter failing to function.

(3) Usually there is a dedicated ground terminal on the casing of filter. However, if this ground terminal is connected by a conductor to the cabinet body, the filters will not functions due to the long conductor with great high-frequency resistance cannot plays the role of a bypass. Correct installation method shall be such: closely laid the filter casing on the conductive plane of metal cabinet so as to maximize their contact area. Note to remove the insulation coating during installation in order to ensure good electric contact.

10.8 EMC Installation Zoning

In the drive system constituted by elevator integrated drive controller and motor, the elevator integrated drive controller and peripherals such as control devices and sensors are usually installed in a same control cabinet. The interferences of control cabinet on external environment may be inhibited by taking suitable measures at the main contacts, therefore, the supply line radio noise filter and supply line AC reactor shall be installed. In order to meet the EMC directive, the interiors of control cabinet shall also be arranged in accordance with EMC requirements.

In the drive system constituted by elevator integrated drive controller and motor, the elevator integrated drive controller, brake units and contactors are all heavy noise source which may affect the normal operation of such noise sensitive peripherals as automated assembly, encoder and sensor. On the basis of electric characteristics of peripherals, they will be installed at different EMC zones in order to spatially isolate the noise sources and noise receivers. This is an effective measure for eliminating interferences.

See Figure 10.8.1 for EMC installation zoning for elevator integrated drive controller.

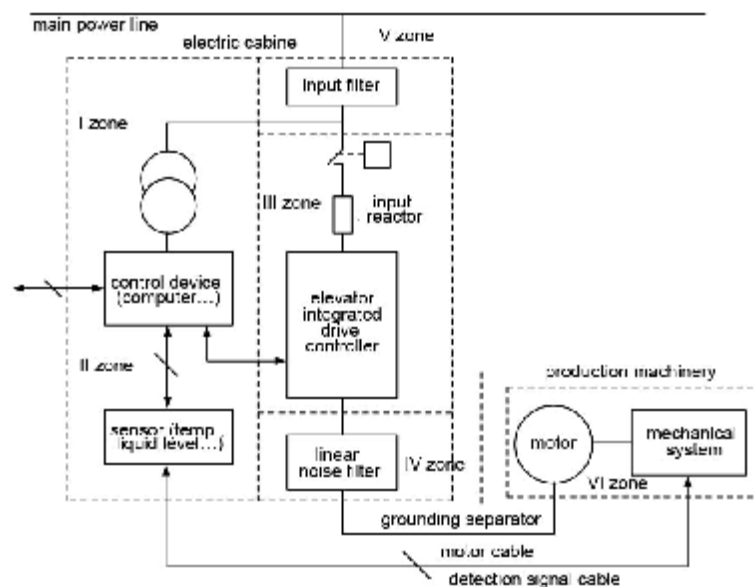


Figure 10.8.1 EMC Zone for installation of elevator integrated drive controller

Description of above shown EMC zoning for installation:

- I zone: Control supply transformer, control device and sensor
- II zone: Control signal cable interface requiring certain anti-interference capacity
- III zone: Main noise sources including input reactor, elevator integrated drive controller, brake unit, contactor and the like
- IV zone: Output noise filter and its wiring
- V zone: Power supply (includes wiring of radio noise filter)
- VI zone: Motor and its cable

These zones shall be separated by minimum 20cm in order to realize electromagnetic decoupling; for better decoupling effect, ground separator is preferred among the zones. The cables shall be laid and arranged by zone; if necessary, filters shall be installed at interfacing point between zones; all the bus cables led out from cabinet (e.g. RS485) and signal cable must be shielded.

10.9 Precautions for Electric Installation

See Figure 10.9.1 for electric installation of elevator integrated drive controller.

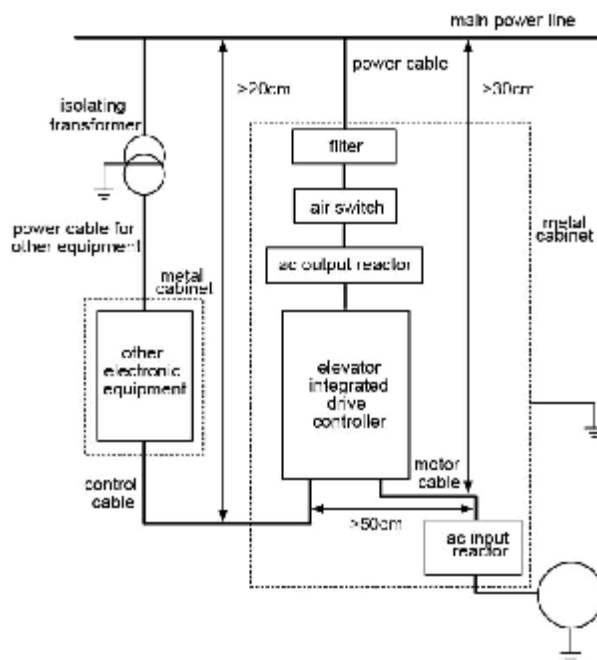


Figure 10.9.1 Electric Installation of elevator integrated drive controller In order to meet the EMC directive, the following notices must be taken:

(1) Elevator integrated drive controller shall be installed in a cabinet where the base plates of elevator integrated drive controller and such peripherals as input filter will be secured on the rear panel in order to ensure good electric contact; the elevator integrated drive controller and filters must be arranged as close as possible within 15cm in order to minimize the high-frequency impedance of their grounding line and reduces high-frequency noises.

(2) Install a wide ground terminal bank at the inlet (5cm or less from outlet) of control cabinet and then secure all the shielding layers of cabinet in/out cables onto the terminal bank by 360° girth jointing method in order to ensure good electric contact.

(3) Motor cable shall be of type shielded cable and it is more preferable to use shielded cables with such two shielding layers as spiral metal belt and metal mesh. By the elevator integrated drive controller side, the shielding layer of motor cable shall be connected, by 360° girth jointing method and with cable clip, to the rear panel of cabinet at two points: one point shall be close to elevator

integrated drive controller as far as possible within 15cm, another point is located on ground terminal bank. 360° girth jointing method shall also be used when the motor cable's shielding layer threads through motor terminal box and grounds at motor metal casing; if it is impractical, firstly strand both shielding layers as braid, then flatten it to width of more than 1/5 braid length in for connecting to motor terminal. The core conductor of motor cable and its PE soft braid lead shall be as short as possible within 5cm.

(4) Terminal control cable must be of type shielded cable, which shielding layer should be connected to terminal bank at the cable inlet of cabinet by 360° girth jointing method; the shielding layer end connecting elevator integrated drive controller may be secured to the metal casing of elevator integrated drive controller. If impractical, firstly strand both shielding layers as short and wide braid, and then flatten it to facilitate connecting to the PE terminal of elevator integrated drive controller. The uncovered section of cable core conductor and the PE soft braid lead shall be as short as possible within 15cm.

(5) Keyboard cable shall not be threaded out from the shielding cabinet.
The slot holes on shielding cabinet shall be as small as possible within 15cm in length.

11Accessories

11.1Car communication board

Car communication board BL2000-CZB shape and dimension is figure 11.1.

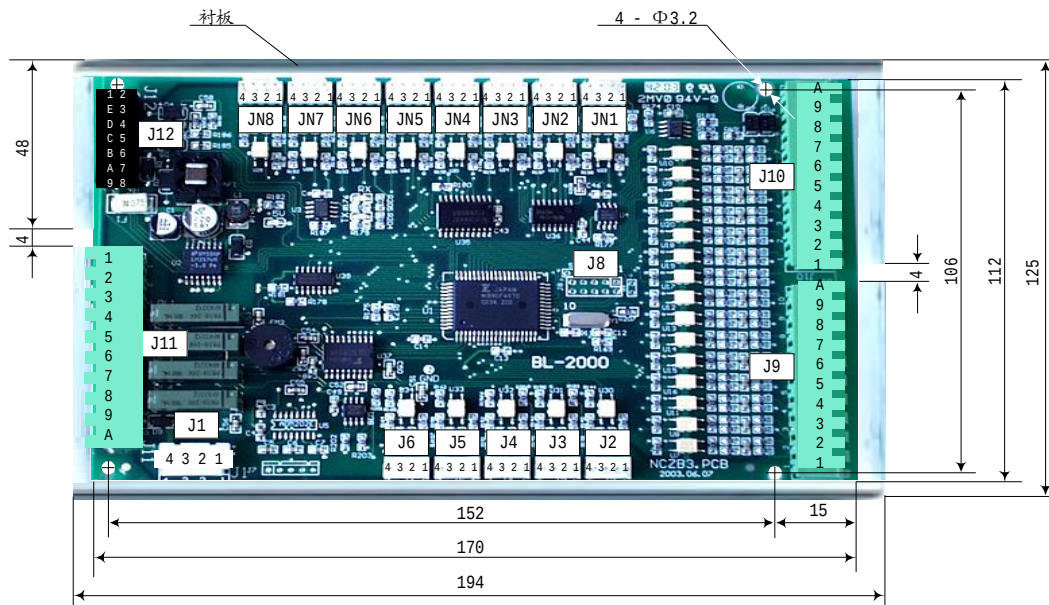


Fig11.1 Car communication board BL2000-CZB shape and dimension

11.1.1Function description

1. Car call input and registers

COP BL2000-CZB has several Opt coupler input /output interface, and it can support 8 floors car call and register. By connecting extension BL2000-CEB (it can support for another 8 floors), it mostly can support to 64 floors. The wiring of car call and register is figure 11.3.

2. Car floor number and running direction display unit

Car floor number and running direction display unit is same with hall station display unit. Its interface is on figure 11.4. In case of landing call and display unit are sue as display in car box, terminal J1 connect with COP J1, J2 not connect with J3, floor address set 0 (set method can refer to landing call and display unit).

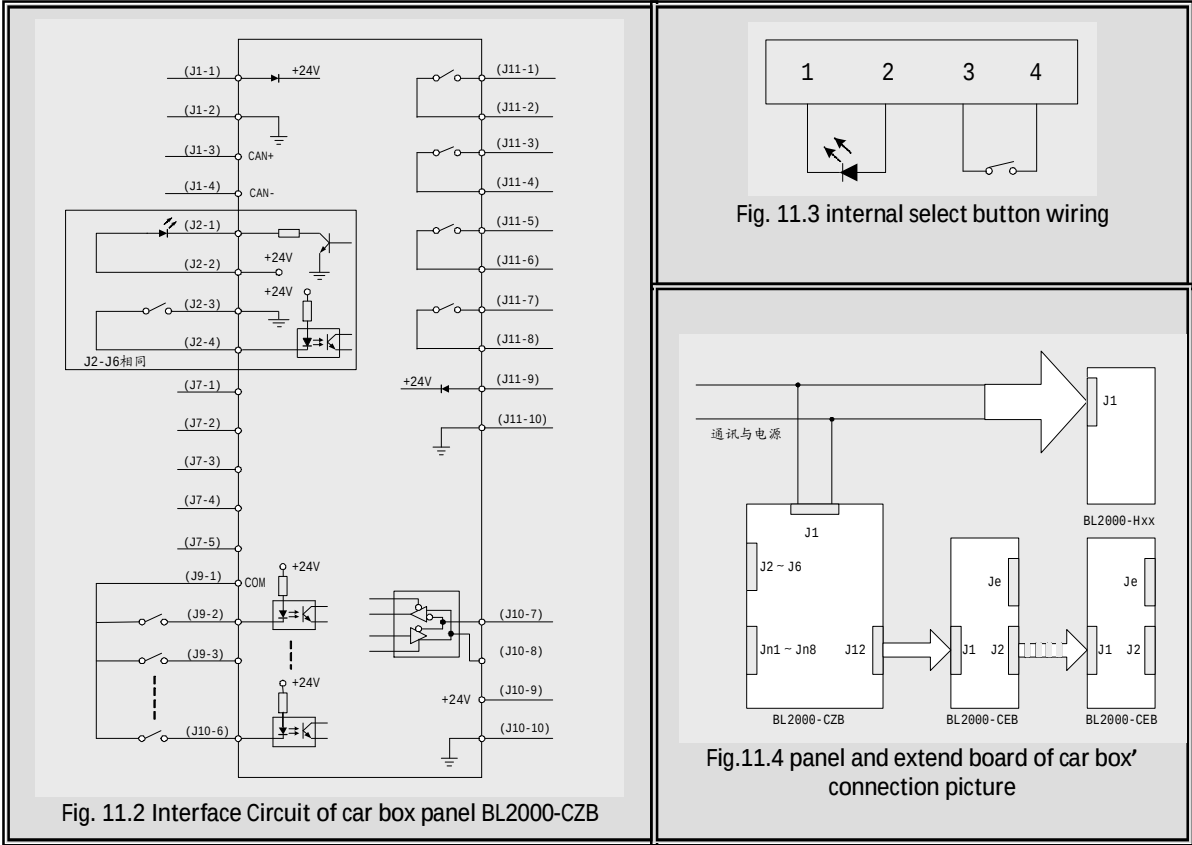
11.1.2Specification of connector

BL2000-CEB-V7 board connect or specification

1. J1 single pin bar 3.96/4P
2. J2~J6, JN1~JN8 single pin 2.54/4P
3. J7 single pin bar 2.54/5P
4. J8 double holes bar 2.54/10P
5. J9~J11 multiple wire bending socket DK5EHDR-10P, rated voltage :300V, rated current: 15A, withstand voltage: 4KV, distance: 5mm
6. J12 double pin bar 2.54/14P

11.1.3Interface Circuit

Figure11.2～11.4Interface Circuit.



11.1.4Definition and specification of terminal

Table 11.1 BL2000-CZB definition and specification of terminal

name	port	position	define	use	specification		
					form	rated load	max speed
J1		J1-1	24Vinput	power and communicate interface			
		J1-2	24Vinput				
		J1-3	CAN BUS H				
		J1-4	CAN BUS L				
J2		J2-1	Open door button 1 acknowledge	Open door button 1 input	OC door	Current limiting resistance 560Ω	
		J2-2	24Voutput				
		J2-3	24Voutput		photo coupler	8mA	
		J2-4	Open door button 1 input				
J3		J3-1	Close door button 1 acknowledge	Close door button 1 acknowledge	OC door	Current limiting resistance 560Ω	
		J3-2	24Voutput				
		J3-3	24Voutput		photo coupler	8mA	
		J3-4	Close door button 1 input				

name	port	position	define	use	specification		
					form	rated load	max speed
J4		J4-1	Close door button 2 acknowledge	Open door button 2 input(double door)	OC door	Current limiting resistance 560Ω	
		J4-2	24Voutput				
		J4-3	24Voutput		photo coupler	8mA	
		J4-4	Open door button 2 input				
J5		J5-1	Close door button 2 acknowledge	Close door button 2 acknowledge(double door)	OC door	Current limiting resistance 560Ω	
		J5-2	24Voutput				
		J5-3	24Voutput		photo coupler	8mA	
		J5-4	Close door button 2 acknowledge				
J6		J6-1	door open delay button	door open delay button and acknowledge	OC door	Current limiting resistance 560Ω	
		J6-2	24Voutput				
		J6-3	24V output		photo coupler	8mA	
		J6-4	door open delay button input				
J7		J7-1	RS232 receive	RS232 communicate	RS232 Electrical level		
		J7-2	RS232 send				
		J7-3	signal				
		J7-4	RS232 output control				
		J7-5	RS232 input control				
J8	programming connector						
J9	CMM	J9-1	Public port	input	photo coupler	8mA	500Hz
	KMV1	J9-2	open limited input				
	GMV1	J9-3	close limited input				
	KAB1	J9-4	safety touch plate 1 input				
	CZ	J9-5	Over load input				
	MZ	J9-6	full load input				
	KAB2	J9-7	safety touch plate 2 input				
	QZ	J9-8	under load input				
	KZ	J9-9	empty input				
	SZH	J9-10	Operator input				
J10	SZY	J10-1	special input	input	photo coupler	8mA	500Hz
	SZS	J10-2	direct drive input				
	ZHS	J10-3	operator direct up				
	ZHX	J10-4	operator direct down				
	KMV2	J10-5	open limited 2 input				
	GMV2	J10-6	open limited 2 input				
	RT-	J10-7	serial over load inspect communication wire RT-	SJT-150 Serial input	RS485		
	RT+	J10-8	serial over load inspect communication wire RT+				
	24V	J10-9	+24V				
	CMM	J10-10	0V				

name	port	position	define	use	specification		
					form	rated load	max speed
J11	BLV-	J11-1	Car arrival chime 1A	output	relay	DC5A24V AC5A250V	20cpm close/op en time ≦ 5/10mS
	BLV+	J11-2	Car arrival chime 1B				
	N1	J11-3	Lighting A				
	ZM	J11-4	Lighting B				
	BK1	J11-5	Stand-by 1A				
	BK2	J11-6	Stand-by 1B				
	CZD	J11-7	over load indicator A				
	CMM	J11-8	over load indicator B				
	24V	J11-9	Auxiliary power +24V input	input			
	0V	J11-10	Auxiliary power 0V input				
J12	24V	J12-1&2	Power +24V input	internal select			
	5V	J12-3&4	Power +5V input				
	0V	J12-5&6	Power 0V input				
		J12-7~ 12	Datum signal				
		J12-13 、 14	null				
JN1 ~ JN8		JNn-1	Response output	1~8 floor internal select input and acknowledge output	OC door	DC24V20m A Current limiting resistance 560Ω	
	24V	JNn-2	+24V				
	0V	JNn-3	Output				
		JNn-4	Internal select input		photo coupler	DC24V6mA	50Hz

11.2 Car Call Extend Board BL2000-CEB

Shape and flat setting size of Car Call Extend Board BL2000-CZB 11.5.

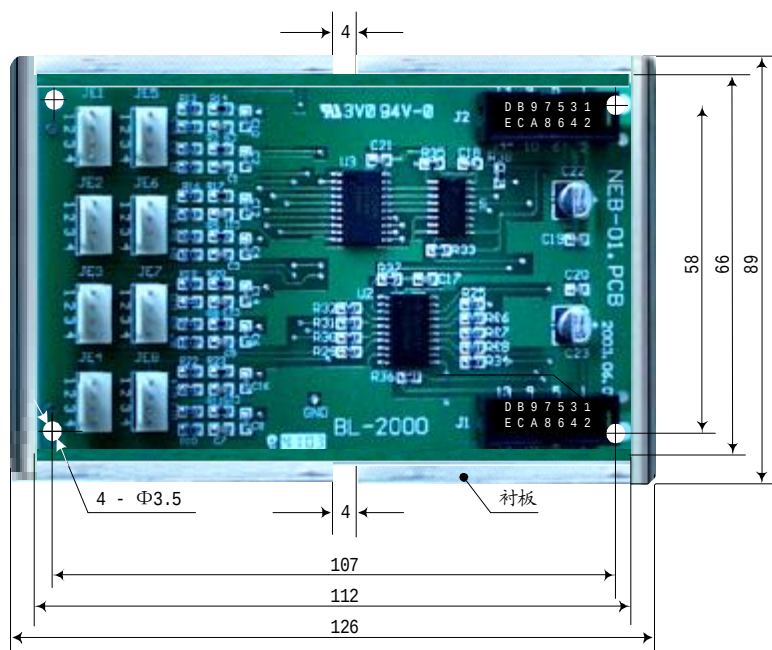


Fig. 11.5 Shape and flat setting size of Car Call Extend Board BL2000-CZB

11.2.1Function description

Prot J12 on BL2000 board is the socket for extend the car call and register. Extension BL2000-CEB can be connected to this port, and it also can be connected to next extension. Each BL2000-CEB can support 8 floors car call and register.

The max extension is 7 units.

11.2.2 Connector specification

BL2000-CEB-V1/V2 board connector specification

1. J1, J2 double pin bar 2.54/14P (DC2-14)
2. JEn1~JEn8 single pin 2.54/4P(2510-4P)

11.2.3 Interface circuit introduce

Interface circuit refers to fig. 11.6.

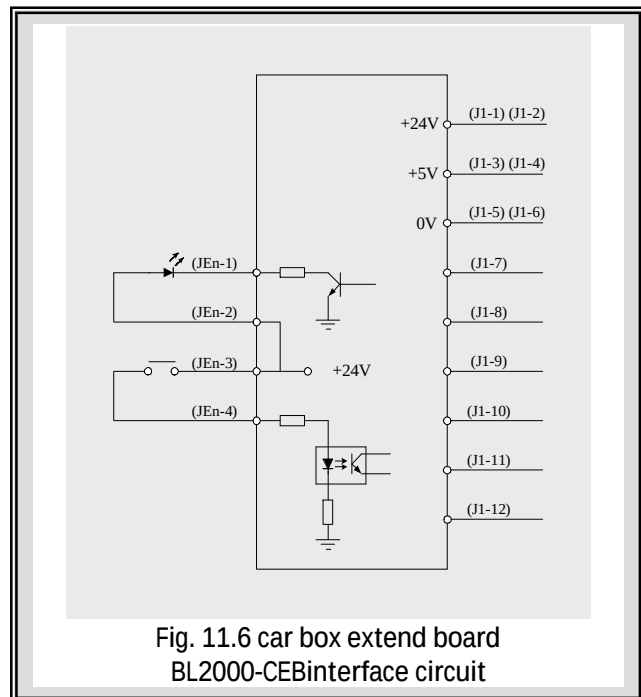


Fig. 11.6 car box extend board BL2000-CEB interface circuit

11.2.4 Terminal definition and specification

Terminal definition and specification refer to table 11.2.

Table 11.2car box extend board BL2000-CEB terminal definition and specification.

name	port	position	define	use	specification		
					form	Rated load	Max speed
J1	24V	J1-1, J1-2	Power +24V input	Previous board port			
	5V	J1-3, J1-4	Power +5V input				
	0V	J1-5, J1-6	Power 0V input				
		J1-7~J1-12	Datum signal				
		J1-13, J1-14	null				
J2			Same as J1	under board port			
JEN		JEn-1	Acknowledge output	8i+1~8i+8 floor car call button input and acknowledge	OC door	DC24V20mA Current limited resistance 560Ω	
	24V	JEn-2	+24V				
	24V	JEn-3	+24V				
		JEn-4	Internal input		Photo coupler	DC24V6mA	50Hz
remark	I is the position of extend board.						

11.3 Landing calls and display unit BL2000-Hxx

BL2000 series landing call and display unit Fig. 11.7

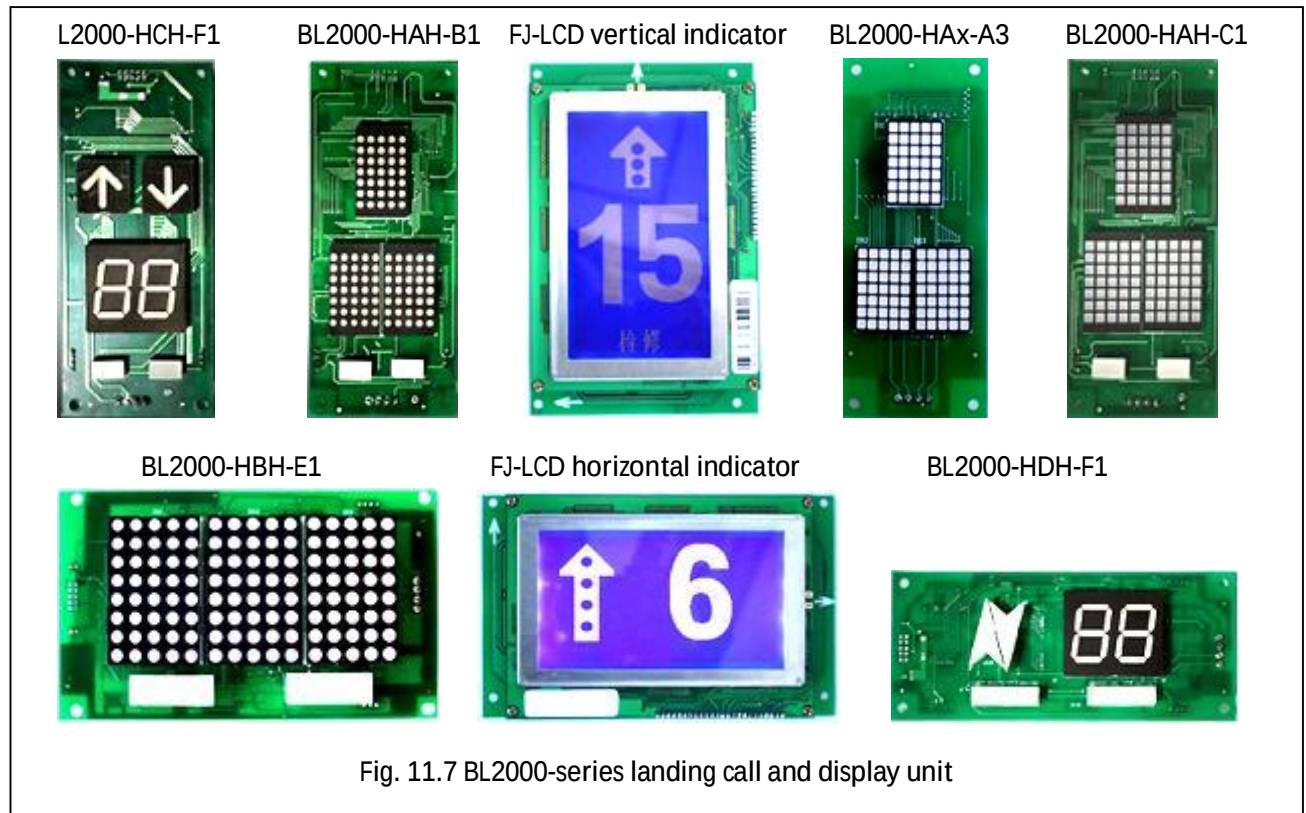


Fig. 11.7 BL2000-series landing call and display unit

11.3.1 Function description

Landing call and display board BL2000-Hxx-xn have various type. It has horizontal and vertical style.

11.3.2 Connector specification

BL2000-Hxx specifications:

1. J1 single pin bar 3.96/4P
2. J2, J3 single pin bar 2.54/4P
3. J4 double pin bar 2.54/10P
4. S1, CZ, JC Jumper2P

11.3.3

Interface introduce refer to fig. 11.8.

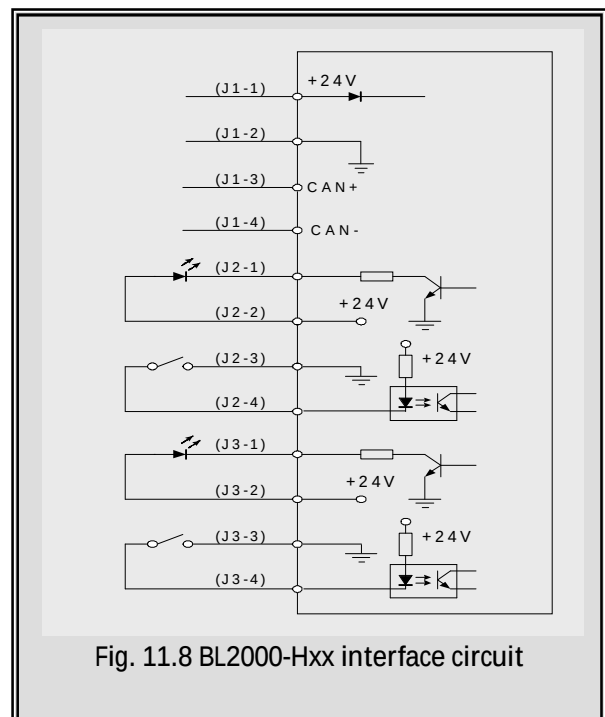


Fig. 11.8 BL2000-Hxx interface circuit

11.3.4 Terminal definition and specification

Landing call and display unit BL2000-Hxx terminal definition and specification is on table 11.3.

Table 11.3 landing call and display unit BL2000-Hxx terminal definition and specification

name	position	define	use	Specification		
				form	Rated load	Max speed
J1	J1-1	24V power input	power and communication interface		Matrix 100mA Seven sect 160mA	
	J1-2	24Vpower input				
	J1-3	CAN bus H				
	J1-4	CAN bus L				
J2	J2-1	up call acknowledge	landing call button input and acknowledge voltage output	OC door	Current limited resistance 120Ω	
	J2-2	24V output				
	J2-3	24V output				
	J2-4	Up call input		Photo coupler	8mA	50Hz
J3	J3-1	Down call acknowledge	landing call button input and acknowledge voltage output	OC door	Current limited resistance 120Ω	
	J3-2	24V output				
	J3-3	24V output				
	J3-4	down call input		Photo coupler	DC24V8mA	50Hz
J4	Program interface					
S1	serial communication terminal resistance jumper(in board)					
AN	address setting key (in board)					

11.3.5 Landing call and display address set

Hall call station is using serial communication, so each of them only can have one communication address. Hall calling and display unit setting address by observing signals and “AN” button on the call board. If set the address as 0, hall calling will set address by utterly floor (1~64), bottom floor is 1, the hypo-bottom floor is 2, the order is comply it.

The way of setting hall calling and display board address:

1. Power on, presses button AN, hall calling and display unit can show the address has been set. Press button AN 5 seconds, hall calling and display unit go into setting state.
2. At setting state, every time press the button AN, address will add 1, until to 64, then it will start again.
3. Press the button 2 seconds after finish setting, the address number will glitter and be saved.
4. When address is 1, S1 need s to short link. It means CAN bus on hall calling and display unit has connected with terminal resistance.
 - ▲ caution: Only bottom floor (address number 1) need to connect with terminal resistance.
 - ▲ caution: Address setting is various with different type.

11.3.6 Shape and platform setting size picture of landing call and display unit.

shape and platform setting size picture of landing call and display unit is on Fig 11.9~11.17.

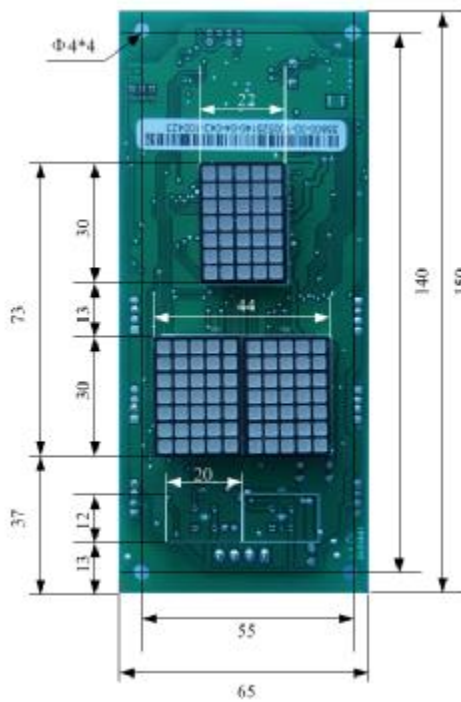


Fig. 11.9 BL2000-HAH-A8

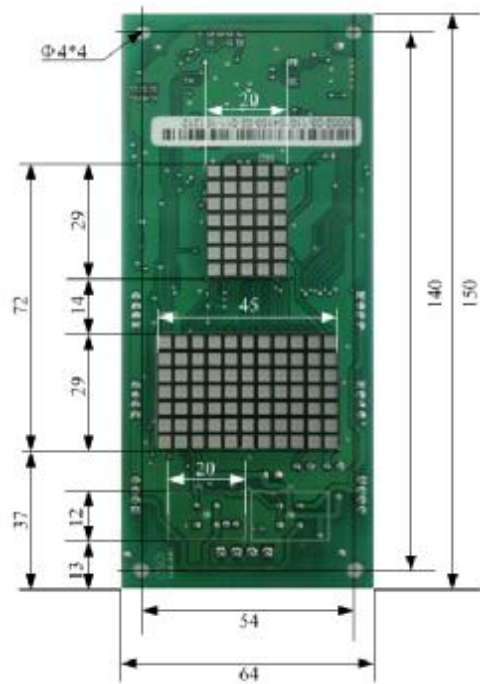


Fig. 11.10 BL2000-HAH-M1

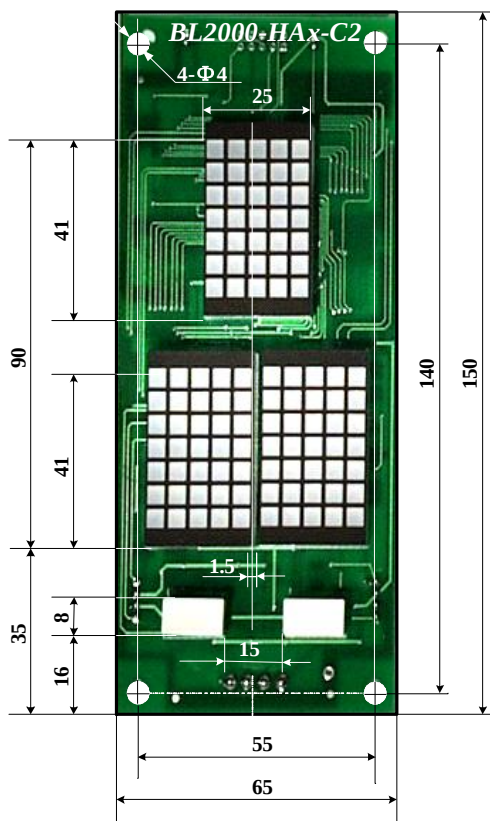


Fig. 11.11 BL2000-HAx-C2

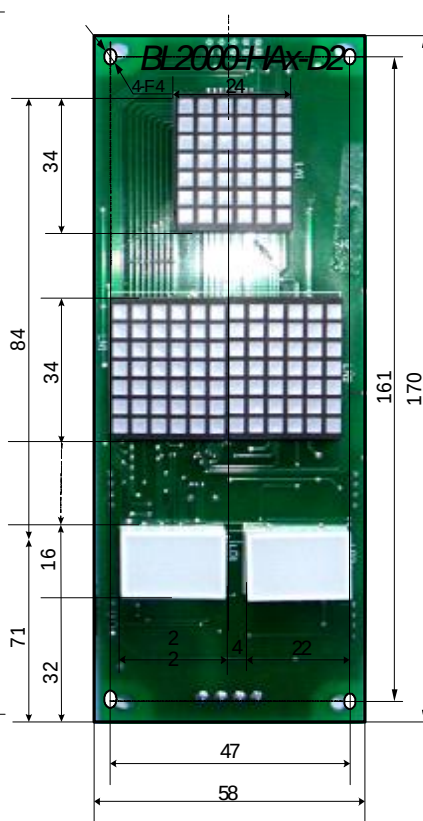


Fig. 11.12 BL2000-HAx-D2

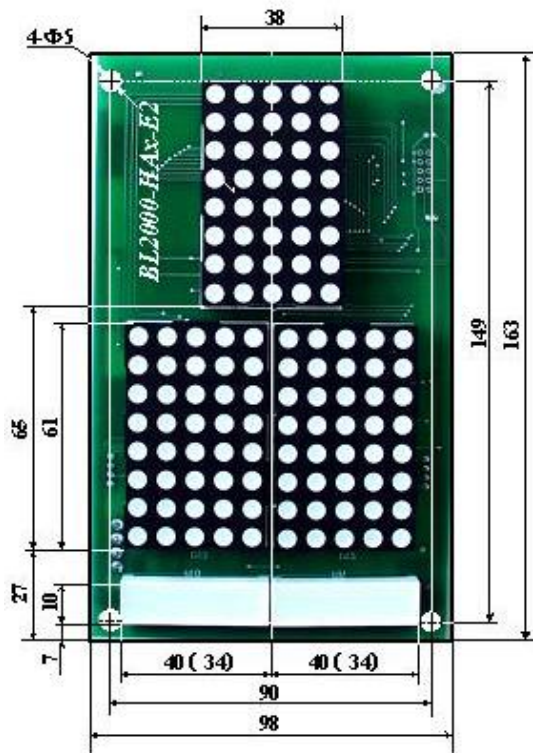


Figure 11.13 BL2000-HAx-E2

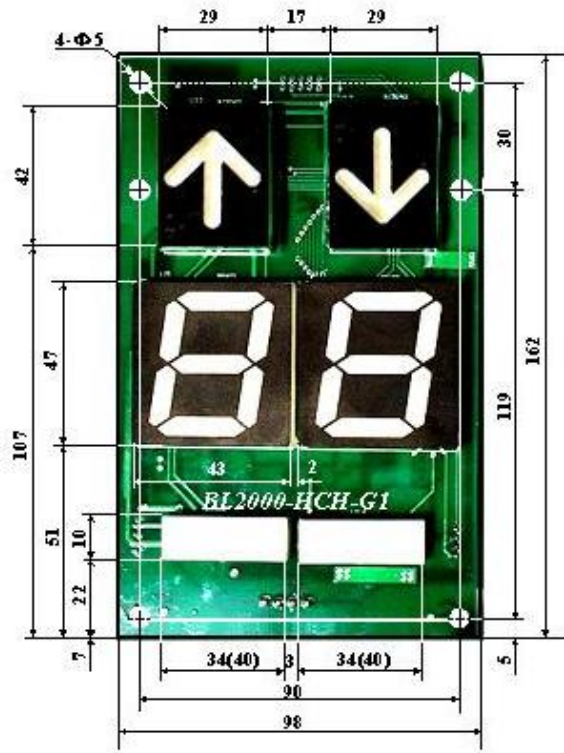


Figure 11.14 BL2000-HCH-G1

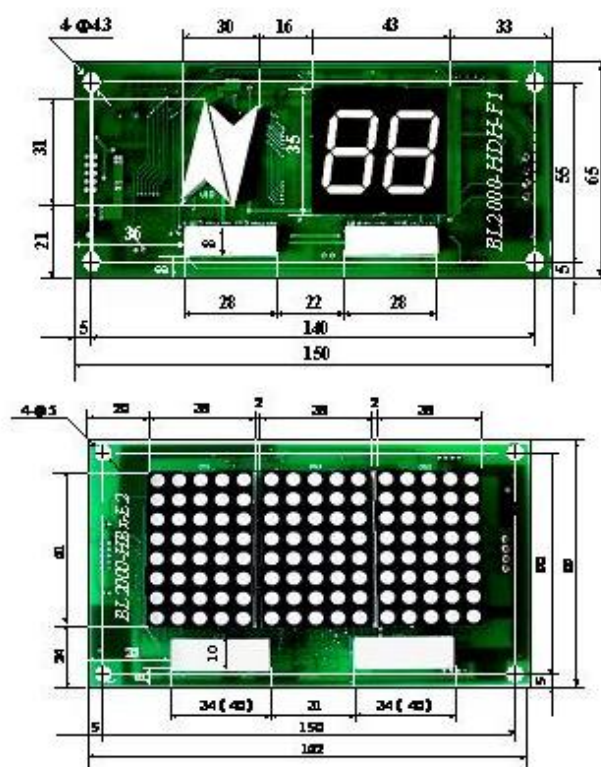
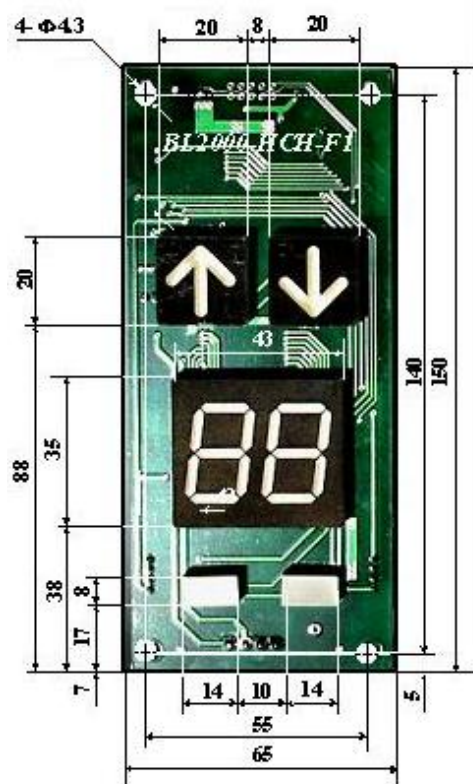


Figure 11.15 BL2000-HCH-F1

Figure 11.16\17 BL2000-HBx-E2、HDH-F1

1.3.7 Landing calls and display unit specification list

Landing call and display unit specification is on table 11.4.

Table 11.4 landing call and display unit BL2000-Hxx specification.

model	Display module specification						
	model	arrange	shape	structur e	color	Length ×wide × high mm	lightness
BL2000-HAH-A8	1357AH	vertical	Square dot	Dot matrix		30×22×10	High
BL2000-HAS-A3	1357AS						Super
BL2000-HAR-A3	1357ASR						UltraHigh
BL2000-HAH-M1	1012A	vertical	Square dot	Dot matrix		30×20×8	High
BL2000-HAS-B2	1057AS		Circle dot			39×23×8	Super
BL2000-HAR-B2	1057ASR		Circle dot			39×23×8	UltraHigh
BL2000-HAH-C2	1257AH		Square dot	Dot matrix		42×25×7	High
BL2000-HAS-C2	1257AS						Super
BL2000-HAR-C2	1257ASR						UltraHigh
BL2000-HAH-D2	1067BH		Square dot	Dot matrix		34×24×6	High
BL2000-HAH-E2	2058AS		Circle dot	Dot matrix		61×38×8	High
BL2000-HAS-E2	2058AH						Super
BL2000-HAR-E2	2058ASR						UltraHigh
BL2000-HBH-C1	1257AH	vertical	Square dot	Dot matrix		42×25×7	High
BL2000-HBS-C1	1257AS						Super
BL2000-HBR-C1	1257ASR						UltraHigh
BL2000-HBH-E2	2058AH	horizont	Circle	Dot		61×38×8	High

BL2000-HBS-E2	2058AS	al	dot	matrix			Super
BL2000-HBR-E2	2058ASR						UltraHigh
BL2000-HCH-F1	BS252	vertical	segmented	Seven segment		35×42×7	Arrow dimension 22×20
BL2000-HCH-G1	BS402					46×44×10	
BL2000-HDH-F1	BS252	horizontal				35×42×7	

11.4 BL2000-HQK for Group control hall call

BL2000-HQK for group control hall calls Appearance and dimension as fig 11.18.

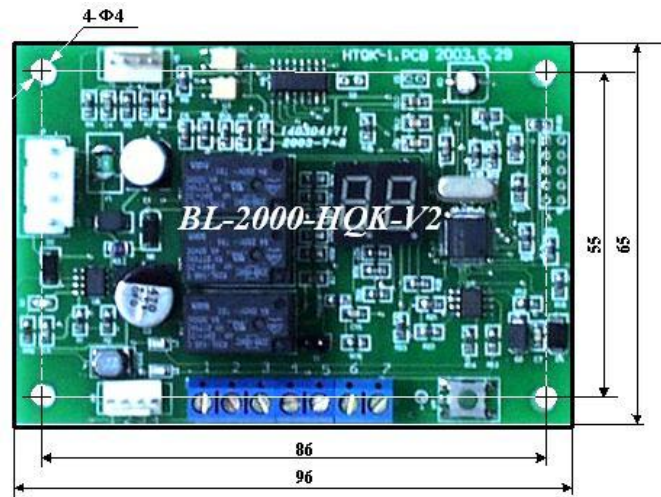


Fig. 11.18 BL2000-HQK

11.4.1 Function description

BL2000-HQK is only use for group control hall call function without.

11.4.2 Connector specification

1. BL2000-HQK-V1/V2's connector specification
2. J1 single pin bar 3.96/4P
3. SH, XH single pin bar 2.54/4P
4. J4 double pin bar 2.54/10P
5. J5 connecting terminal DG301-7P
6. S1 jumper 2P

11.4.3 Interface Circuit illustration

Interface circuit reference fig. 11.19

11.4.4

BL2000-HQK terminal definition and specification as table 11.5

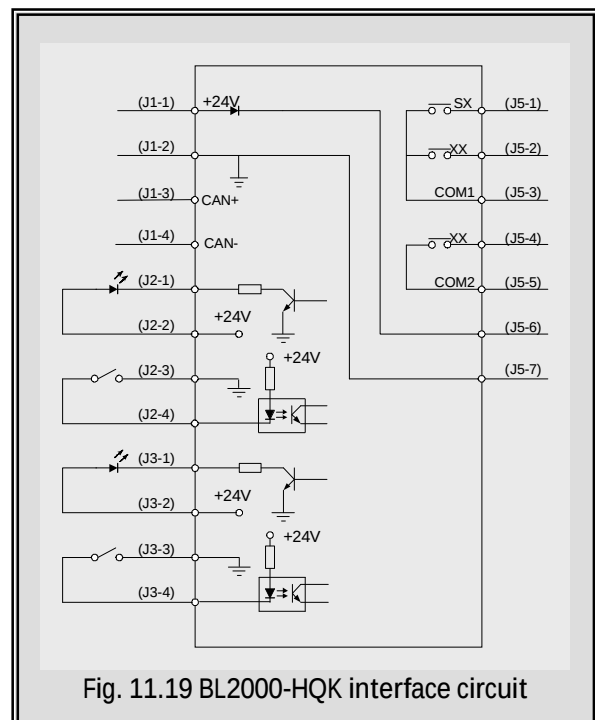


Fig. 11.19 BL2000-HQK interface circuit

Table 11.5 BL2000-HQK terminal definition and specification

Name	locati on	definition	usage	interface technique specification		
				Interface form	Rated capacity	Highest speed
J1	J1-1	24V power input	Power and communication			
	J1-2	24V power input				
	J1-3	CAN bus H				
	J1-4	CAN bus L				
J2	J2-1	Up call respond	ascending & hall call button Input and answering voltage output	OC door	current-limiting resistance 120Ω	
	J2-2	24V output				
	J2-3	24Vground		curtain	8mA	50Hz
	J2-4	Up call input				
J3	J3-1	Down call Respond	Descending & hall call button Input and answering voltage output	OC door	current-limiting resistance 120Ω	
	J3-2	24V output				
	J3-3	24Vground		curtain	DC24V8mA	50Hz
	J3-4	Down call Input				
J4	programming interface					
J5	J5-1	Up lamp output	output	relay	DC5A24V AC5A250V	20cpm off/on time ≤5/10mS
	J5-2	Down Lamp output				
	J5-3	up & down lamp output public terminal				
	J5-4	arrival gong output				
	J5-5	arrival gong output public terminal				
	J5-6	24Voutput				
	J5-7	24Vgroup				
S1	series connected communication terminal resistance wire jumper(in the board)					
AN	address setup keyboard(in the board)					

11.5 Group control computer board BL2000-QKB-V1

BL2000-QKB-V1 Shape and plane installation size as picture 11.20

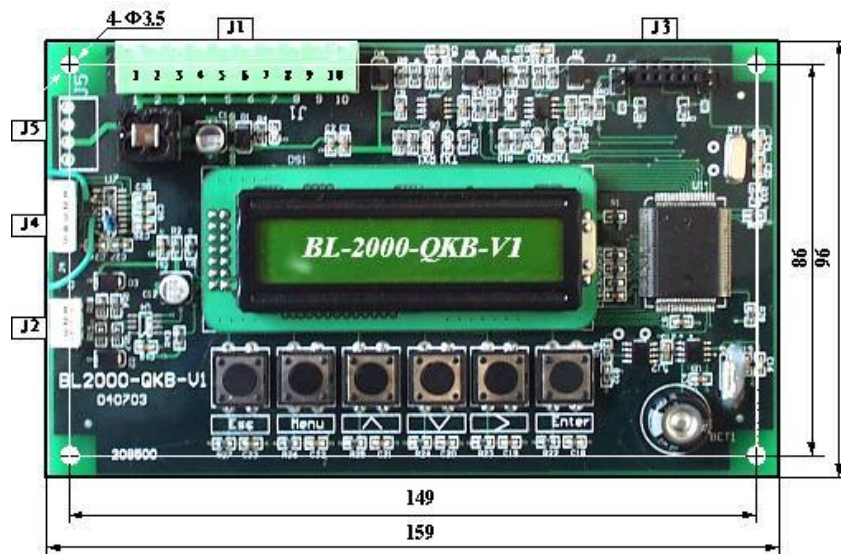


Fig1.20 BL2000-QKB-V1 appearance and plane dimension

11.5.1 Function description

1. Group control system is composed of group control computer board "BL2000-QKB-V1" and a all-in-one board which distributes in every elevator group control computer collect hall call, car call and postural information of every elevator through CAN universal serial bus communication way, after these information are processed by intelligent GROUP CONTROL COMPUTER BOARD, the board will deliver distribution order to every elevator, this will make group control for 8 elevators under 64 floors come true.

2. Four running modes

- ◆ Up peak mode---within the settled time, all elevators will supply the hall call service as per the up call the highest preference on the basic floor
- ◆ Down peak mode---within the settled time, one elevator will supply the up call service at first; the rest will supply down call service partition and make the down call get response in time.
- ◆ Balance Mode-----hall call will do optimizing assignment to the elevator, according to the hall call shortest time principle; it will do the response of hall call order.
- ◆ Spare mode----within 3 minutes at the balance mode without hall call and car call, the elevator will wait for the order from the first floor of each floor so that it can respond at the time of hall car as soon as possible.

When the elevator is at the status of breakdown, diver, inspection, haul, fire protect, Specialized status, this elevator will be excluded the group control.

As to each single elevator running function and setup in group control system, please refer the relative chapter of this manual.

11.5.2 Application scope

1. It can be used in 3-8 elevators' group control
2. It can be used in the speed scale of 0.5-4m/s
3. Usable floors: below 64 floors
4. It applies to passenger elevator and residential elevator

11.5.3 Connector assembly specification instrument

- 1.J1 multiple wire caliper socket DK5EHDRC-10P; rated voltage: 300V, rated current: 15A, withstand voltage: 4KV, pitch: 5mm。
- 2. J2connector 2.54/3P
- 3. J3 twin housing 2.54/10P
- 4. J4connector 2.54/5P
- 5. J5 single sheath plug-in board 3.96/4P

11.5.4 Interface circuit instrument

Interface circuit reference as fig. 11.21

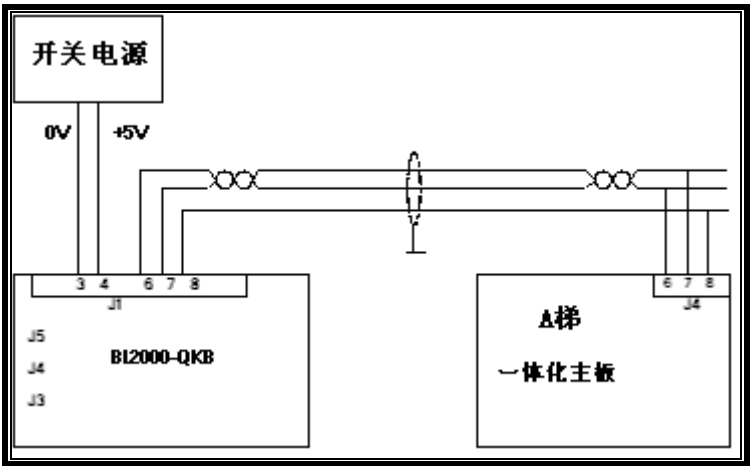


Fig. 11.21 group control computer board BL2000-QKB-V1 interface circuit picture

11.5.5 Terminal definition and specification

Group control computer board BL2000-QKB-V1 terminal definition and specification as table11.6

Table11.6 Group control computer board BL2000-QKB-V1 terminal definition and specification

Name	Terminal No.	location	definition	usage	technique specification		
					interface form	rated capacity	highest speed
J1	GND3	J1-1	0V	power and communication			
		J1-2					
	GND3	J1-3	0V				
	5V IN	J1-4	5V input			200mA	
		J1-5					
	TXA+	J1-6	group communication				
	TXA-	J1-7					
	GND3	J1-8	0V				
		J1-9	Standby CAN communication TXA+				
		J1-10	Standby CAN communication TXA-				
J2	DA+	J2-1		RS485			
	DA-	J2-2					
	GND	J2-3					
J3	programming interface						

J4	TX	J4-1	communication SND	RS323			
	RX	J4-2	communication Rec.				
	IN	J4-3	control input				
	OUT	J4-4	control output				
J5	TXA+	J5-1	group control communication				
	TXA-	J5-2					
	GND3	J5-3	0V				
		J5-4					

11.6 SJT-YY Emergency leveling device

11.6.1 Brief

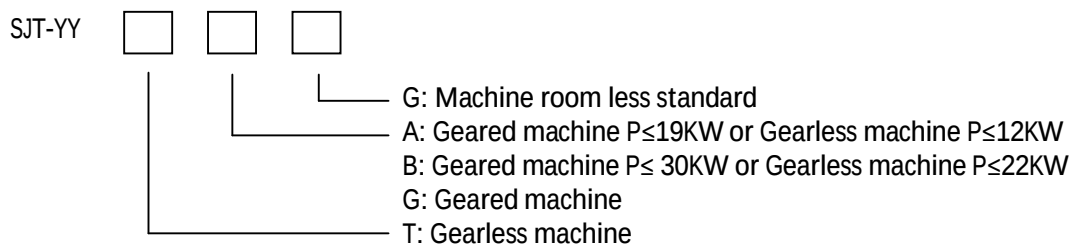
SJT-YY is a same type of safety equipment aiming at the emergent stop of the elevator which is caused by power failure or power malfunction. In the emergent rescue, the equipment is powered by storage battery. The energy stored in the battery is reversed into the electricity needed by the elevator. The elevator will be started and the cage will be pulled to the next floor, then the car door and landing door will be opened and the passenger can leave safely without being stuck in the elevator.

11.6.2 Application scope

This type of ELD is suitable for any type of elevator which driven by VVVF both geared and gearless machines.

11.6.3 Specification

Explanation for type:



11.6.4 Descriptions of principle and function

While ELD is working, it can provide the power supply and control signals to the inverter and mother board of the main control panel through its inside control circuit, and work with them together to control the elevator moving to the nearest floor at a low speed and opening the door.

The specific function described as follows:

- 1). There is a reliable electric interlocking circuit between ELD and the main control system of elevator. The elevator should be recovered to normal state once the power supply recovers.
- 2). ELD will run must at same time under following conditions.
 - a. Power supply is failure;
 - b. The elevator is on non-inspection state;
 - c. The elevator is not at door zone
 - d. The elevator's safety and door interlock circuit are normal
- 3). If the conditions above are fit except "c", then ELD will only let elevator to open the door automatically.
- 4). While ELD running, its running direction can be confirmed according to the load situation of elevator.
- 5). ELD can exactly landing and the leveling precision is within $\pm 15\text{mm}$.
- 6). ELD has the time limiter function for the protection of the longest one times running.
- 7). ELD has the function of automatic, constant voltage and current charging for the batteries.

11.6.5 Installation and commissioning

The ELD must work with YIDA elevator controller, the wiring drawing show on fig.11.22 and fig.11.23.

Port definition

mark	means	pin	instruction
PS1 PS2	Input power	CN1-1 CN1-12	PS1 & PS2 are power supply input of ELD, they should come from one of the 3 phases and nature line of main power supply, with which the voltage is AC220V±15%, 50Hz, they are used for charging ELD and also for judging the power supply failure or not, 2mm ² wire should be better.
Y5	Power failure signal output	CN1-6	Separate neutral which comes from ELD DC192V between comes from controller for brake (only use for synchronous motor)
BS+ BS-	Power Output	TB1-1 TB1-2	BS+ & BS- are power supply output with the voltage DC192V for inverter, 4mm ² wires should be better.
BK+ BK-	Power Output	CN1-3 CN1-14	BK+ & BK- are power supply output with the voltage DC110V for brake coil, 0.75mm ² wire should be better.
CP1 CP2	Power Output	CN1-4 CN1-15	CP1 & CP2 are control power supply output with the voltage AC220V 50Hz, Max. 600W for control circuit and door operator, 0.75mm ² wire should be better.
IL1 IL2	Power Interlocking Input	CN1-7 CN1-17	IL1 & IL2 are normal closed input contact for the electric interlocking between ELD and main control system.
IL3 IL4	Power Interlocking output	CN1-5 CN1-16	IL3 & IL4 are normal closed output contact for the electric interlocking between ELD and main control system.
IL5 IL6	Power Interlocking output	CN1-8 CN1-18	IL5 & IL6 are normal closed output contact for the electric interlocking between ELD and main control system.
Y4 COM4	Output Signal	CN1-19 CN1-9	For jumping out the normal open contact of phase protective relay while power supply failure
Y3 COM3	Output Signal	CN1-20 CN1-10	Input signal for the 32-bit microprocessor of main control system
Y2 Y6 COM2	Output Signal	CN1-21 CN1-13 CN1-11	Input signal for the inverter of main control system

Wiring and commissioning

- Switch off F1 (breaker) and F2 (switch) (F1 & F2 are switches for stopping ELD which are "off" when ELD leaves factory);
- Switch off the main power supply;
- Make wiring according to electric drawing and the table above;
- Check the wiring seriously, incorrect wiring will cause ELD and main control system broken down;
- After checking, if no problem, switch on F1 and F2 of ELD, press the button of UPS, and then switch on the main power supply;
- If elevator can run normally, please set the relative battery running parameters in inverter;
- Level the elevator on one floor; switch off the main power supply, the door should be opened

automatically by ELD;

h. Recover the main power supply and move the elevator to the middle between two floors, switch off the power supply again, the elevator should move toward light load direction, level and open the door once all conditions of emergency running are accordant.

11.6.6 Troubleshooting

No	Phenomena	Analysis
1	The power supply is deviant while link the ELD to main control system	Wiring of electric interlocking is not correct
2	The inverter has no display when power supply is off	a. The output voltage of BS+ & BS- is less than DC192V; b. Breaker F1 of ELD is off; c. Wiring of batteries is not reliable and firm
3	The mother board has no display when power supply is off	a. The switch F2 of ELD is off; b. The Switch of UPS is off;
4	The brake does not work when ELD is running	The voltage of BK+ and BK- is incorrect
5	ELD does not work when power supply is failure	a. Safety circuit or door interlock circuit is not in normal state b. The elevator is on inspection mode c. The contact used for jumping out the “no” contact of phase protective relay is failure d. The output signal from ELD to inverter or mother board is failure

11.6.7 Main technical data

- 1). Input power supply: AC220V+15 % 50Hz
- 2). Output power supply: DC192V
DC110V $I_{ed} \leq 3A$
DC220V 50Hz $I_{ed} \leq 3A$
- 3). Environmental temperature: 0℃~40℃
- 4). Relative humidity: 20%~90%, no dew
- 5). Leveling precision: $\pm 15mm$
- 6). the maximum applicable motor output
A Type Asynchronous motor $\leq 19KW$ synchronous $\leq 12KW$
B Type Asynchronous $\leq 30KW$ synchronous $\leq 22KW$
- 7). Maximum running time: $\leq 2min$
- 8). Exterior Diagrams
YA: 570×240×450
YB: 840×240×450
YG: 570×165×720

11.6.8 Maintenances

- 1). Batteries should work in such an environment that is clean, dry and good airiness, it strictly prohibits short, touching organic compound liquor and fire; and also prohibits to convert the batteries during transportation, moreover avoids to impact and acutely vibrate; keep good earth;
- 2). Check the voltage of batteries (it should be larger than DC192V between F1-1 and BS-) and try to run ELD once a quarter.

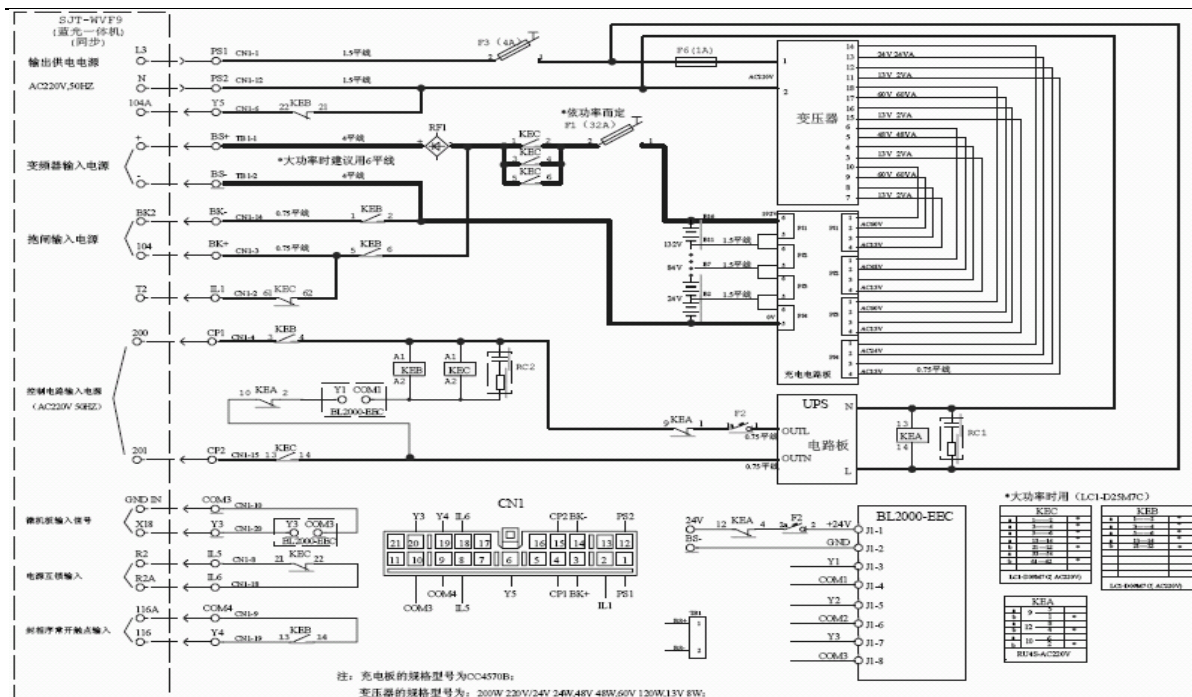


Fig.11.22 drawing of SJT-YY with BU-3 for synchronous motor

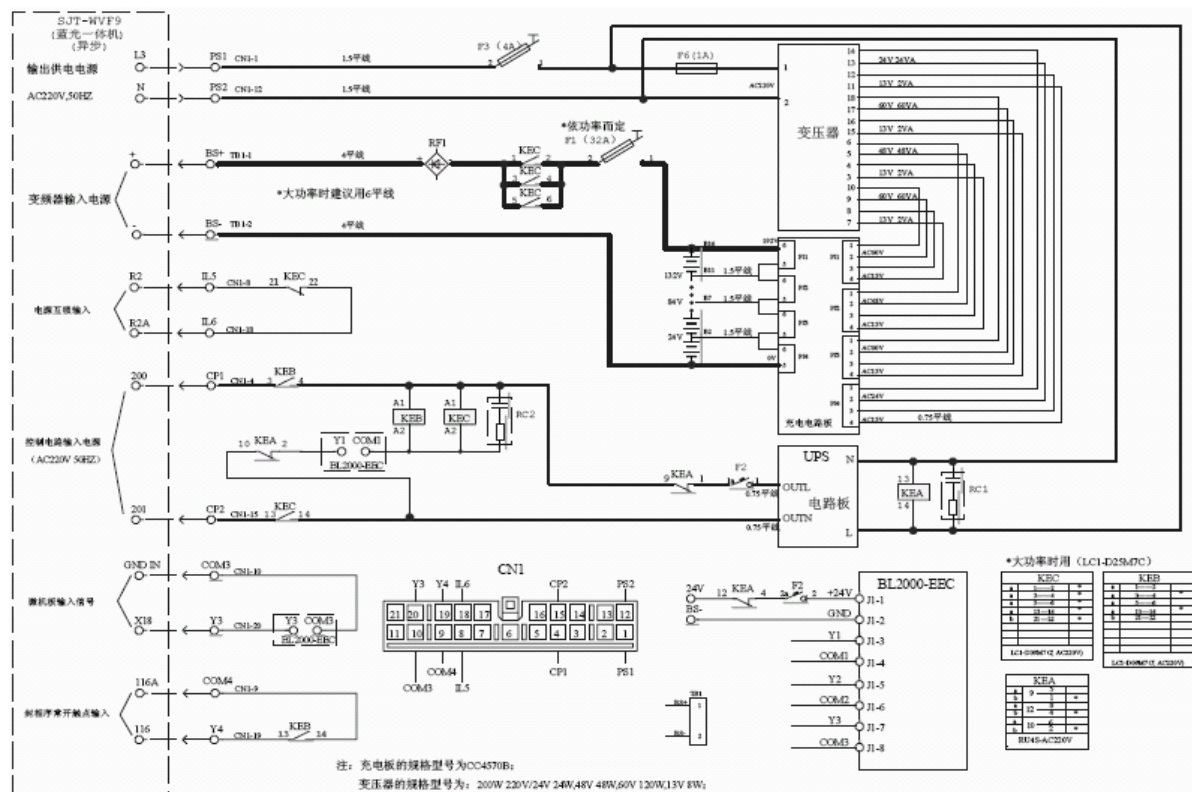


Fig.11.23 drawing of SJT-YY with BU-3 for asynchronous motor

Appendix 1 Shaft information

A1.1 leveling switches and flag installation

Elevator needs two leveling switches on top of car and flag for each floor mounted on the guide rail by flag frame. Leveling switch adopt photo electronic switch or magnetic switch.

Normally the flags length is 200mm, but it is suggested that flags length is 250mm when elevator's speed greater than 2m/s.

Both leveling switches are installed at car top; flags are installed in shaft, which position refers to the following fig. A1

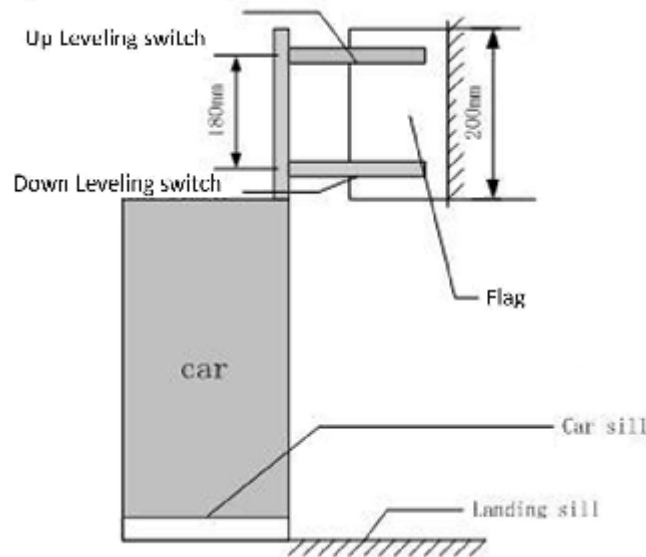


Fig.A1.1 Leveling Switch and flag

A1.2 Top/Bottom slow down Switch Installation

Elevator needs one top slow down switch and one bottom slow down switch (speed<1.75m/s) and cam installed on the car frame.

Top/bottom slow down switch is installed in shaft which is mounted on the guide rail by frame. The distance is 2.5m (<1.75m/s) from car sill to top/bottom landing door sill.

The number of slow down switch should be increased for elevator speed is greater than 2.0m/s. refer to table:

Table A1.2 Terminal Switch Installation Position

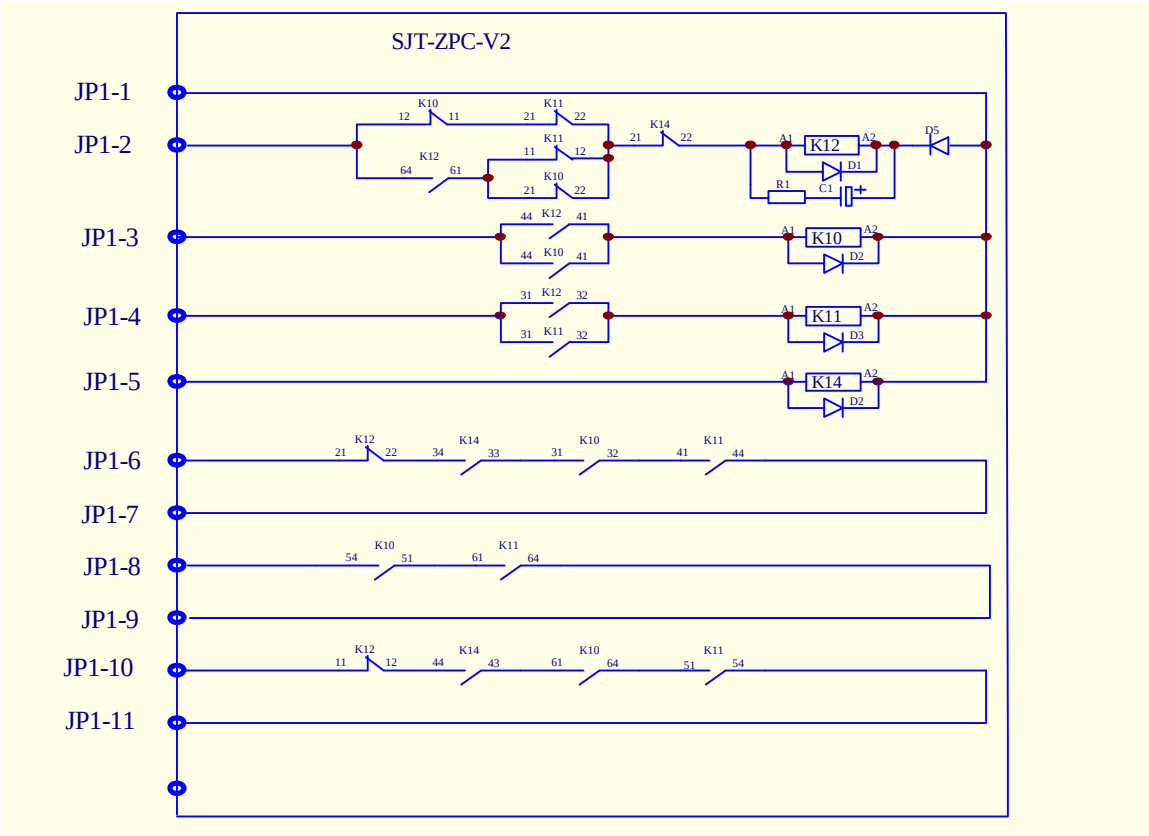
	Position in shaft of slow down switch					
	0.5m/s	1.0m/s	1.5m/s to 1.75m/s	2.0m/s	2.5m/s	4.0m/s
Up/bottom slow down 1	1m	1.3m	2.5m	2.5m	2m	2.5m
Up/bottom slow down 2				4m	6.25m(4m)	*8m
Up/bottom slow down 3						

Appendix 2 Advance door opening /relabeling zone

A2.1 safety circuit board (SJT-ZPC-V2) eletric drawing

As shown in Fig.A2.1

Fig.A2.1 safety circuit board (SJT-ZPC-V2)



A2.2 safety circuit board interface define

As shown in table A2.1

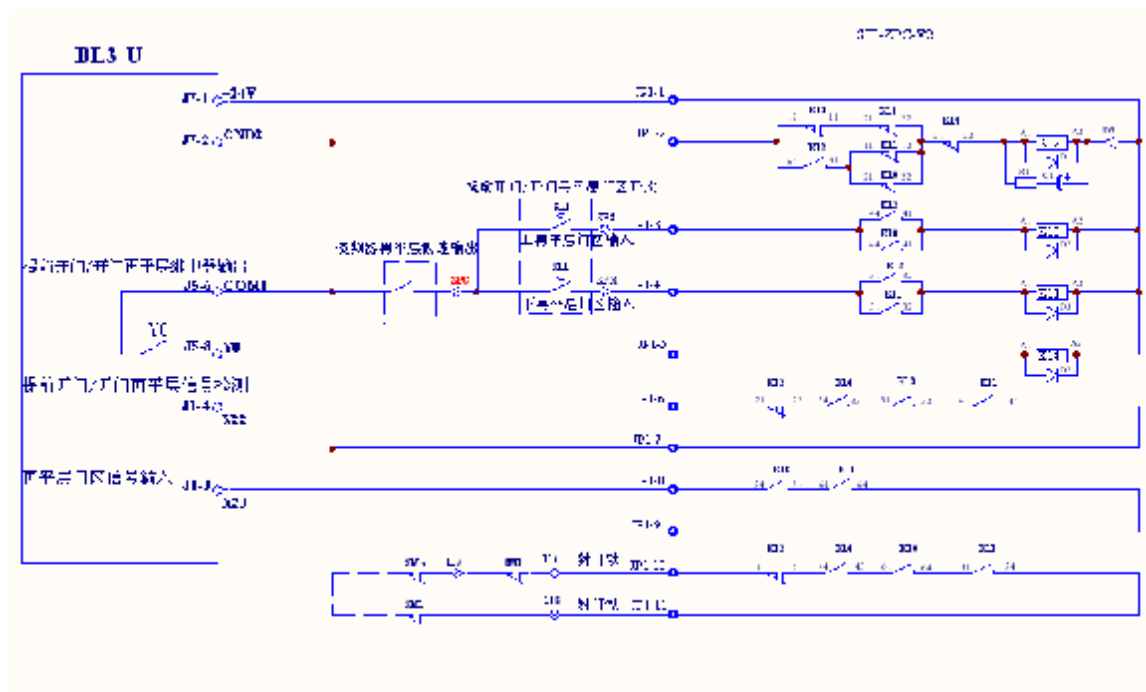
Fig F2.1 safety circuit board interface define

Name	Position	Define
JP1	JP1-1	24V+
	JP1-2	0V
	JP1-3	Up re-leveling door zone
	JP1-4	Down re-leveling door zone
	JP1-5	Control system re-leveling condition required
	JP1-6	Re-leveling condition required output
	JP1-7	Re-leveling condition required output common terminal
	JP1-8	Re-leveling door zone output
	JP1-9	Re-leveling door zone output common terminal
	JP1-10	Sealing door lock output+
	JP1-11	Sealing door lock output -

A2.3 advance door opening/re-leveling function wiring drawing

As shown in F2.2

Fig A2.2 Advance door opening/ re-leveling function wiring drawing



A2.4 installation for re-leveling door zone switch and door zone switch

While using the function of advance door open and re-leveling, besides the up and down leveling switch, two re-leveling switches should also be installed. The installation of them is shown in Fig A2.3.

mq1: up leveling switch; mq2: down leveling switch; sl1: up re-leveling switch; sl2: down re-leveling switch; All leveling switches should be installed in order. Otherwise, the direction of the re-leveling running will be reverse.

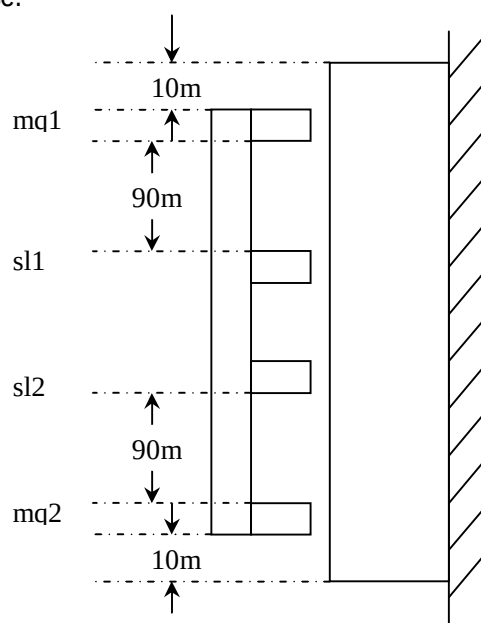


Fig A2.3 installation for re-leveling switch and leveling switch



If SI1 and SI2 use magnetic inductor, the distance between them should be more than 5m. Otherwise, the two inductors will not work normally for the interference of each other.

A2.5 setting related parameters for main control board

A2.5.1 setting advance door opening/ re-leveling function for special function of the special parameter.

As shown in Table A2.2

Table A2.2 Setting and choosing advance opening/ re-leveling function

Function No.	Function instruction
*FU4-06-19	ON: start re-levelling function; OFF: no re-levelling function
*FU4-06-20	ON: start re-levelling function; OFF: no re-levelling function

A2.5.2 setting advance door opening/ re-leveling function for running parameter

There are three related advance door opening/re-leveling functions:

1. advance door opening speed (F1-07) : The speed that elevator advance open door after the elevator's normal running into re-leveling door zone
2. re-leveling protection speed (F1-08) : if the speed is above the value during advance door opening or re-leveling running, the elevator stop running.
3. Re-leveling running speed (F1-09) : Be settled

Appendix 3 Parameter Table

The setting for the controller parameter and factory value is shown in Table A3.1

Table A3.1 The setting for the controller parameter and factory value

Appendix 3 Parameter Table

The setting for the controller parameter and factory value is shown in Table A3.1 Table A3.1 The setting for the controller parameter and factory value

Parameter No.	Name	Factory value	Setting value
U0-00	Lower Limit	0	
U0-01	Upper Limit	0	
U0-02	Lower Slowdown 1	0	
U0-03	Lower Slowdown 2	0	
U0-04	Upper Slowdown 1	0	
U0-05	Upper Slowdown 2	0	
U0-06	Floor Data 1	0	
U0-07	Floor Data 2	0	
U0-08~68	Floor Data 3~63	0	
U0-69	Floor Data 64	0	
U1-00	Input Data	--	
U1-01	Input Bin	--	
U1-02	Input App	--	
U2-00	Output Data	--	
U3-00	Car Input Data	--	
U4-00	Run Times	0	
U4-01	Run Hours	0	
U4-04	CAN1SendAPP	--	
U4-05	CAN1ReceiveAPP	--	
U4-06	Interfer Apprais	--	
U4-07	Encoder Apprais	--	
U4-09	Loading Data		
U5-00	Ctrsoftware NO	--	
U5-01	DriveCodeVer	--	
U5-02	CpldEdition	--	
U6-00	Power	--	
U6-01	Ref Speed	--	
U6-02	Feedback Speed	--	
U6-03	Load	--	
U6-04	DC Voltage	--	
U6-05	Output current	--	
U6-06	temperature	--	
U6-07	Output torque	--	

F0-00	Total floor	6	
F0-01	Homing Floor	1	
F0-02	Fire floor	1	
F0-03	Parking floor	1	
F0-04	VIP floor	1	
F0-05	1 st floor indicator setting	1	

Parameter No.	Name	Factory value	Setting value
F0-xx	Xxth floor indicator setting	1~63	
F0-68	64 th floor indicator setting	64	
F1-00	Car speed	1.60m/s	
F1-01	Motor speed	1450r	
F1-03	Insp speed	0.3m/s	
F1-04	Start speed	0m/s	
F1-05	Leveling Speed	0.3m/s	
F1-06	Least Speed	0.96m/s	
F1-07	Open Door Speed	0.25m/s	
F1-08	Relevelst Speed	0.30m/s	
F1-09	Relvelrun speed	0.05m/s	
F1-10	Acceleration B1	0.7m/s ²	
F1-11	Deceleration B2	0.7 m/s ²	
F1-12	S Curve P1	0.6 m/s ³	
F1-13	S Curve P2	0.6 m/s ³	
F1-14	S Curve P3	0.6 m/s ³	
F1-15	S Curve P4	0.6 m/s ³	
F1-16	Zero Speed	1RPM	
F1-17	Leveling Adj	50mm	
F1-18	Load Adj	0	
F1-21	Drive mode	0	
F1-22	Two Door Mode	0	
F1-23	Fire Mode	0	
F1-24	Parallel No.	0	
F1-25	Twins Control	0	
F1-26	Group Control	0	
F1-27	Far monitor	0	
F1-28	Auto Parking	0	
F1-29	Load Enable	0	
F1-30	Open Delay Time	0	
F1-31	Brake Feedback	0	
F1-32	Solution Ladder Password	0	
F2-00	Brake ON Time	0.5s	
F2-01	Brake OFF Time	0.5s	
F2-02	Insp Brake Time	0.5s	
F2-04	Zero Time	0.3ms	

F2-05	Open Door time	3s	
F2-06	Open Delay Time	60s	
F2-07	Homing Time	60s	
F2-08	Door Run Time	5s	
F2-09	Beep Delay Time	0.15s	
F2-10	Enable Delay	0s	
F2-11	Lamp Off Time	15min	
F2-12	Over Time	45s	
F2-13	Smooth Start Time	0	
F2-14	Start time	00:00	
F2-15	Stop time	00:00	
F3-00	Input type	--	
F3-01	car input type	--	
F4-00	Set Stop Floor1	0xFFFFFFFF	
F4-01	Set Stop Floor2	0xFFFFFFFF	
F4-02	TIM Stop Floor1	0x00000000	
F4-03	TIM Stop Floor2	0x00000000	
F4-04	Door Select A1	0xFFFFFFFF	
F4-05	Door Select B1	0xFFFFFFFF	
F4-06	Function Select	0	
F5-00	Motor type	--	
F5-01	Poles	--	
F5-02	Rated Freq	--	
F5-03	Motor rated power	--	
F5-04	rated speed	--	
F5-05	V IN	--	
F5-06	L_phase	--	
F5-07	R_phase	--	
F5-08	Rated FLA	--	

Parameter No.	Name	Factory value	Setting value
F5-09	non-load current	--	
F5-10	slip frequency	2HZ	
F6-00	Carrier Freq	10KHZ	
F6-02	Speed Zoom	100%	
F6-03	DirSel	0	
F6-04	Kp	220	
F6-05	KI	110	
F7-00	PIMulEnable	0	
F7-01	PI1 Range	0	
F7-02	PI2 Range	0	
F7-03	PI3 Range	0	
F7-04	PI4 Range	0	
F7-05	Kp1	300	
F7-06	Kx1	200	
F7-07	Kp2	300	
F7-08	Kx2	200	
F7-09	Kp3	300	

F7-10	Kx3	200	
F7-11	Kp4	300	
F7-12	Kx4	200	
F8-00	Encoder PPR	8192	
F8-01	PG Ratio	4	
F8-02	PG Type	0	
F9-00	Max Torq Comp	0%	
F9-01	SPD SourceSel	2	
F9-02	Tigh Torq	0	
F9-03	Spderr Set	10%	
F9-04	Spderr Time	700ms	
F9-05	Num of Restarts	10	
F9-06	Time of Restarts	30S	
F9-07	ZeroTime	6000ms	
F9-08	PowerGetEnable	0	
F9-09	Cur Limit	100A	
F9-10	Cur Time	20ms	
F9-11	Load Comp Enable	0	
F9-12	Uart Send Time	10ms	
F9-13	Load Source Sel	0	
F9-14	Fan Mode	1	
F9-15	Alarm Temperature	70°C	
F9-16	Fan Hold Time	60s	
F9-17	UP Comp Pro	100%	
F9-18	DOWN Comp Pro	100%	
F9-19	UP Comp Bias	0	
F9-20	DOWN Comp Bias	0	

Parameter No.	Name	Factory value	Setting value
F9-21	FULL Comp Pro	100%	
FA-00	Strat KP	180	
FA-01	Stra tKI	550	
FA-08	PLKP	1500	
FA-09	PL Time	80ms	
FA-10	PL Rate	1ms	
FA-11	Integral Gain	100	
FA-12	Flux Gain	0	
A0-00	Language Sel	0	
A0-01	User Password	000000	
A0-02	Factory password	111111	
A0-03	Set Quick Menu	000000	

Appendix4 UPS Emergency Function

BL3-U series are designed to cooperate with the time series of elevator emergency leveling device to offer customers the reliable elevator emergency plan.

F4.1 Name standards of UPS Emergency Device

Suitable range of the motor power:

SJT-YU-A: below 7.5W

SJT-YU-B: 7.5~15W

SJT-YU-C: 15~22KW

Series' name: YU (PS: UPS control)

F4.2 technical standards of UPS Emergency Device

1. Power supply: $AC220V \pm 15\%$, $50HZ \pm 10\%$.
2. Output power: $AC220V \pm 10\%$, $50HZ \pm 2\%$.
3. Ambient temperature: $0^{\circ}C \sim 40^{\circ}C$
4. Relative humidity: 20~90% No dewing.
5. Leveling accuracy: $\pm 15mm$
6. Suitable motor power: A: below 7.5w
B: 7.5~15KW
C: 15~22KW
7. Maximum runtime $\leq 2min$
8. Size of cabinet: SJT-YU-A/B/C: 604*247*556 (reference dimension)
(PS: With the different motor power of UPS' appearance change, the size of emergency cabinet may change. If you need the actual size, please choose the motor power and contact the factory.)

F4.3 Recommendations of UPS Emergency Device

F4-06-22 Set to ON, emergency power supply is chosen 380V

Set to OFF, emergency input power supply is chosen 220V

The effect of F4-06-22 parameter is that when the emergency power supply is 220V, the voltage of ec-bus will be down, the production will be stoppage with lower voltage; when the emergency level input of all-in-one machine master leaf X18 is effective, the elevator will in leveling running mode, at that time if the set of parameter is OFF, the all-in-one machine will screen the stoppage and begin to rescue operation.

If emergency power supply is 380V, or output of UPS is 380V, and then please set the parameter to ON, or when the X18 input signal is effective, the transformer can't detect the decline of ec-bus, it can not begin the rescue operation.

Appendix5 Functional specification of separation design of Star shaft seal contactor and running contactor

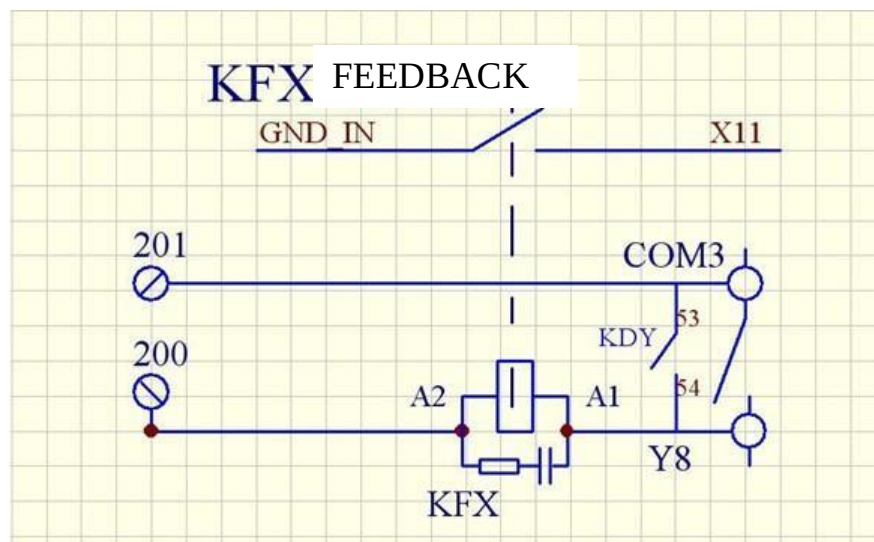
This function is used in the cabinet running contactor and star shaft seal contactor, to lower the cost, some customers ask to have a star shaft seal contactor alone, not choose the star shaft seal contactor of our standard layout, this moment please use this function.

When star shaft seal contactor and running contactor are separated, please set the all-in-one machine's parameter F4-06-29(FU-29) to ON, then the master leaf will automatically increase the output (Y8) and input (X11) of star shaft seal contactor. When running contactor acts, star shaft seal will operated too. If you use this function, the feedback contact X11 of master leaf doesn't receive the feedback signal of star shaft seal's action, the fault ER33 will prompt the error of the feedback.

Pressing on the running contactor, star shaft seal contactor will get rid of the star shaft seal after detecting the action of running contactor, to prevent the short out.

The conditions of the feedback contact are detected by the master leaf. If the action is not fit, it will appear the fault ER-33, to prompt star shaft seal contactor doesn't work, or contact can't be broken.

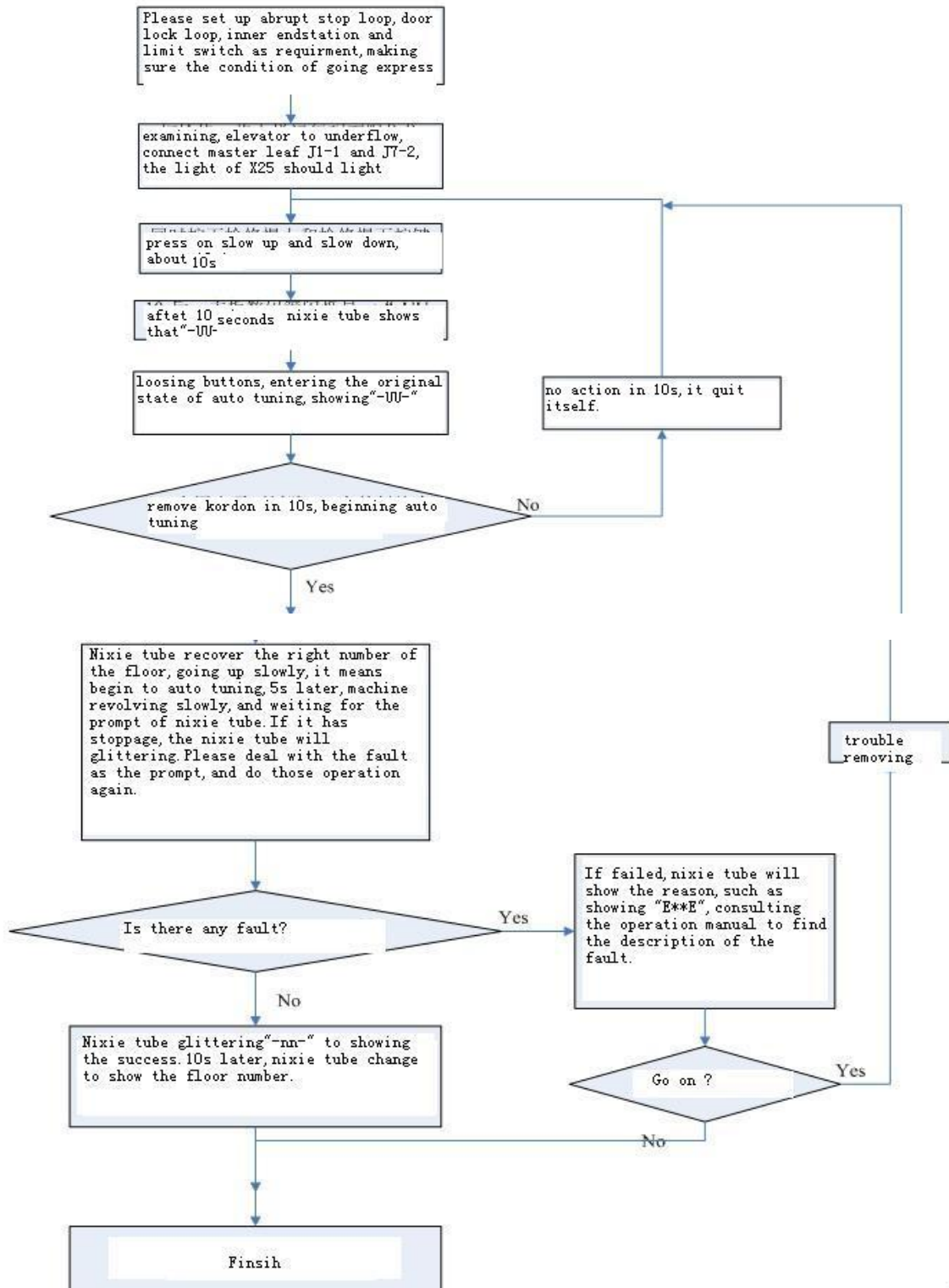
Star shaft seal contactor (closed contact) cancels the action delaying 300ms (after delaying it renew the closed condition). So that it can make sure running contactor close 300ms earlier, protecting the machine by star shaft seal, preventing the short out.



PS: The specific wiring way is subject to blue-ray drawings confirmation. This picture only shows the connection type of Y8 output and X11 input.

Appendix6 All-in-one machine auto tuning procedure of simple hoistway

PS: This function only use in edition 6017 or upper.

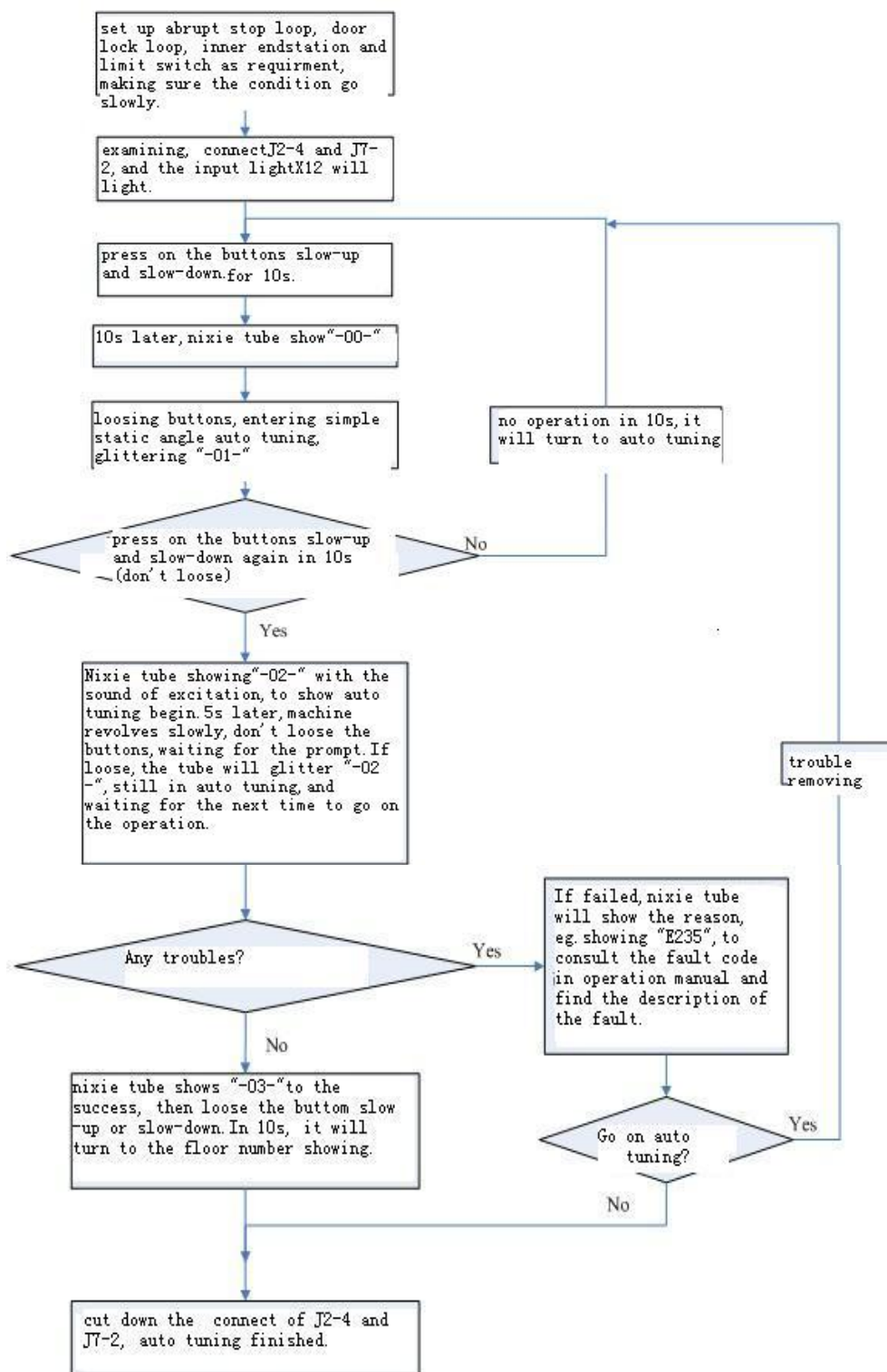


Attention: Making sure door, end station, limit, and signal is connected exactly, before it.

Appendix7 All-in-one machine simple static angle auto tuning

PS: 1.This function only used in MU-V5 master leaf

2. Before the operation, please make sure the software is 6014 or upper, and the drive software in 52 or upper.



Attention: All-in-one machine must be set up to match up with the basic parameter (F5) and PG parameter (F8), and change the mode to static auto tuning.